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FOODSHARE: CONNECTING SURPLUS TO NEED, REDUCING FOOD WASTE

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FoodShare: Connecting Surplus To Need, Reducing Food Waste

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SUPERVISOR APPROVAL

FOODSHARE: CONNECTING SURPLUS TO NEED, REDUCING FOOD WASTE WITHIN UITM JASIN

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This proposal was prepared under the supervision of the project supervisor, Puan Nur Suhailayani Binti Suhaimi. It was submitted to the College of Computing, Informatics and Media and was accepted in partial fulfilment of the requirements for the degree of Bachelor of Information Systems (Hons.) Information System Engineering.

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JUNE 13, 2024

STUDENT DECLARATION

I certify that this proposal and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.

……

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ABSTRACT

The FoodShare: Connecting Surplus to Need, Reducing Food Waste System, is a web-based platform designed to streamline the redistribution of surplus food within UiTM Jasin. The project addresses the challenges faced by cafeteria owners and students, including difficulties in sharing leftover food information, ineffective distribution, and the lack of an event booking feature. By leveraging a centralized platform, the system promotes sustainability and mitigates food waste while supporting food-insecure students. The project adopts the Waterfall methodology, encompassing three key phases: Requirement Gathering and Analysis, Design, and Implementation. Interviews with stakeholders informed system requirements, which were translated into Use Case Diagrams, Domain Class Diagrams, and System Sequence Diagrams during the design phase. Implementation involved creating a web-based platform using Java and Postgress, featuring real-time notifications, food requests, and an event booking module. The findings indicate that FoodShare significantly improves the efficiency of surplus food distribution, reduces reliance on manual processes, and enhances communication between stakeholders. The system ensures transparency and equitable access to surplus food while addressing the unique needs of UiTM Jasin's community. Future enhancements include advanced notification filters, real-time food tracking, and integration with external food donation platforms to expand its reach. These recommendations aim to further reduce food waste and promote a sustainable food-sharing ecosystem.

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LIST OF ABBREVIATIONS

SDD	Software Design Document
SDLC	Software Development Life Cycle
SMS	Short Message Service
SRS	System Requirement Specifications
SDD	System Design Document
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

The system is entitled FoodShare: Connecting Surplus to Need, Reducing Food Waste is a management system. This management system is targeted specifically to UiTM Jasin's students and cafeteria owners. This chapter provides additional details and an overview of the project. The project's background and problem statement are presented at the beginning of this chapter, followed by the aim and objectives. This chapter also includes the project significance, project scope, outline of the thesis, and summary.

1.1 Background of Study

This project's stakeholders include café owners at UiTM Jasin and students. There are 20 cafes in total in UiTM Jasin but only 18 cafeterias are operating, which allow students to purchase their food and beverages whenever they feel the need to do so. While some of the food vendors offer food from morning until dinner, others only offer lunch. Some of the food stalls cover the entire day. Their business hours typically begin at ten o'clock in the morning and continue until ten o'clock at night. Following their preferences, students have access to a wide variety of options, such as *Nasi Lemak, Nasi Ayam, Nasi Campur*, and many others. In addition, there is a wide variety of beverages available for students to select from, depending on their preferences. Table 1.1 shows the list of the cafeteria names with their number of cafe.

Table 1.1 List of cafeteria names and number

Cafe Number	Cafe Name
1	Cabar Niaga Trading
2	Sarapan Dan Makan Tengahari
3	Kedai Minuman
4	HS Gerak Wawasan
5	Gerai 05
6	JNZ Beringin Enterprise
7	Selera Pantai Timur

Table 1.1 (Cont.)

8	H&K Food Cafe
9	Molek Gemilang
10	Gerai 10
11	Ads Oasis Enterprise
12	Zumo Cafe
13	Noordin Bin Saad
14	Afiq Kitchen
15	Nasi Ayam Ala Cina
16	Roti Canai
17	Che Ramli Bin Che Ismail
18	SNZ Bumi Enterprise
19	Norys Enterprise
20	Gerai Minuman

According to the current procedure outlined by Puan Amalina, the owner of Cafe 6, any leftover food from their menu is first offered to the students who are present in the cafeteria. If the students do not want the food, the cafe staff will ask help from the students to broadcast the availability of the leftovers in the college's Telegram group since they do not have access to it. This can allow those who are interested to collect it at the cafe lobby.

The process described does not happen every day. It only occurs when there is leftover food from the café's menu or surplus meals from college events or seminars. On regular days, the cafe does not have a need to distribute leftover food in the manner described. The procedure takes place when there is an excess of food that needs to be shared with students, which typically happens after a meal period or during events. Therefore, it is not a daily occurrence but rather happens as needed, depending on the situation.

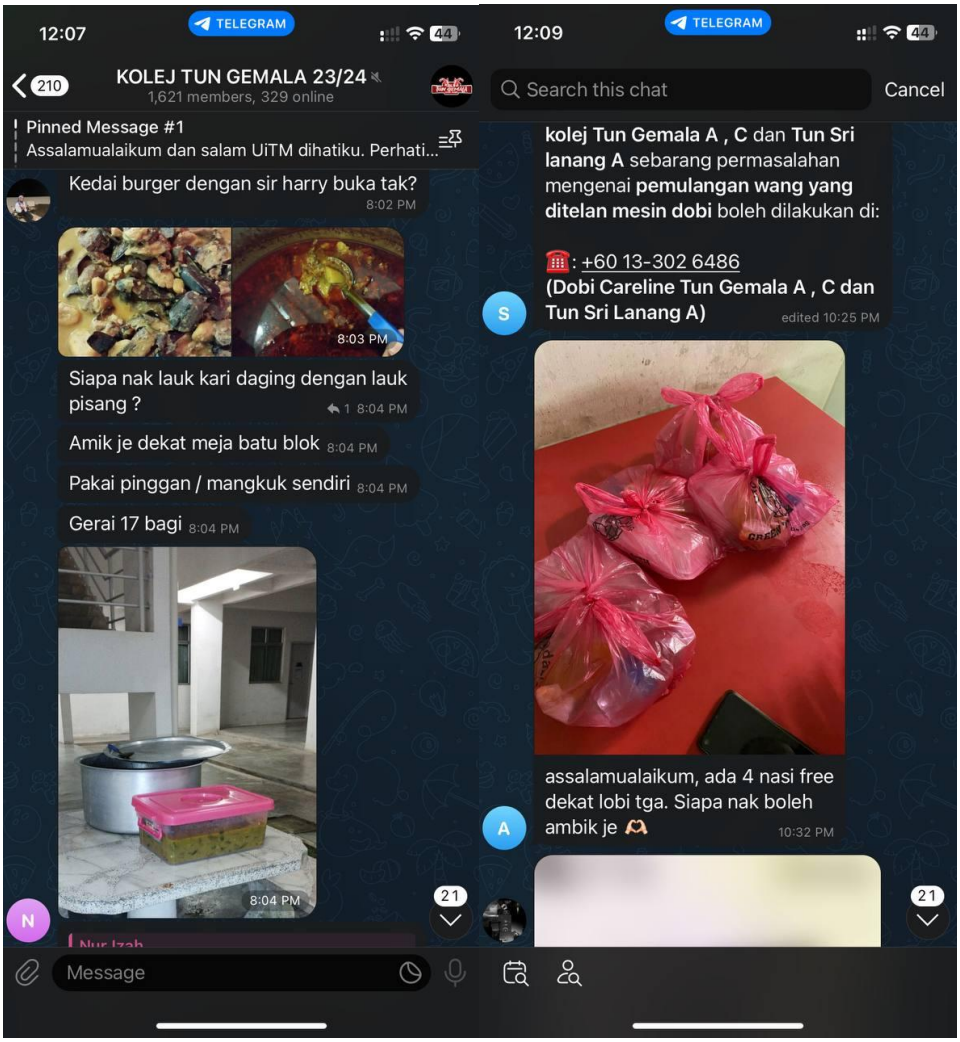


Figure 1.1 Current info distribution via Telegram application

Figure 1.1 highlights the challenges of informing students about leftover meals. Many students may not regularly check the college group, as it often focuses on other topics, causing them to miss notifications about leftovers from stall 17. The figure also shows the ineffective distribution of leftover meals, with cafeteria owners not knowing who will take the food or who has already taken it. Additionally, the lack of status about food availability can lead to food waste if no one claims the leftovers.

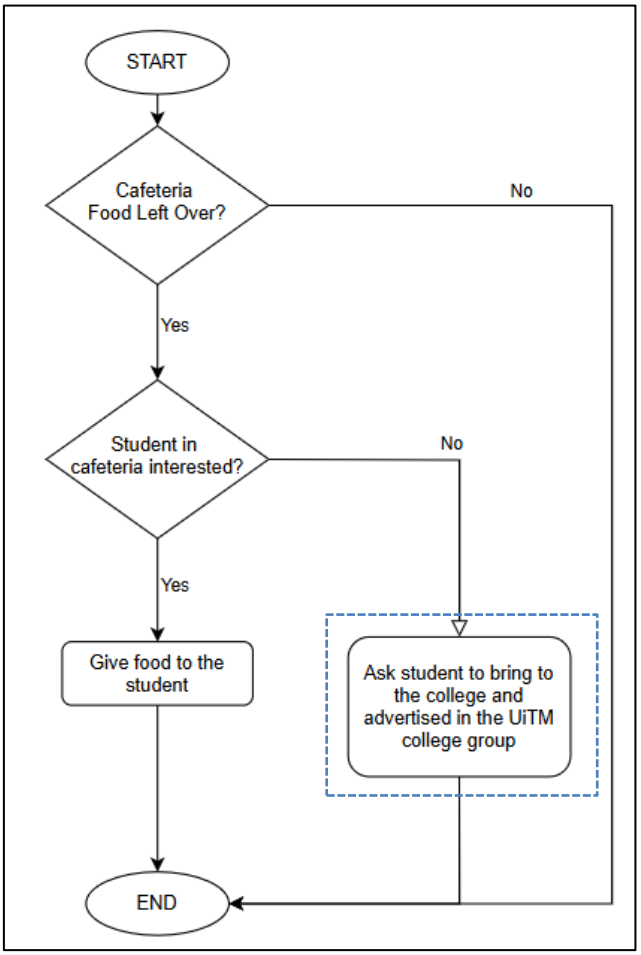


Figure 1.2 Current business process of handling leftover food

This step in the flowchart highlights one of the problems in the current process of managing leftover food. When no students in the cafeteria are interested in the food, the staff relies on a student to take the food to the college and advertise it in the UiTM Telegram group. This introduces a potential issue as it depends on the availability and willingness of a student to help with this task.

If no student is available or willing to assist, the leftover food may go to waste, which is inefficient and problematic. Additionally, relying on a student to broadcast the availability of the food in the Telegram group may lead to delays or confusion, as it is not a direct and structured part of the process. This could cause some students to miss out on the opportunity to collect the leftover food, further exacerbating the problem of food wastage.

1.2 Problem Statement

Generally, managing leftover food could be tedious since it involves multiple parties, from the cafeteria owner to students, before the food reaches those who are in need. Three main problems that occur during the process are difficulties in sharing the leftover information with the students, ineffective distribution of leftover meals among the students, and lack of booking food features for the event. Further elaboration on each of these problems are as follows:

a) Difficulties in sharing the leftover information with students

Currently there is no centralized platform to display the specific information on the extra foods (Puan Amalina, personal communication, April 55, 2024).. Despite having surplus food available, cafeteria owners face significant challenges in efficiently distributing leftover meals. The majority of students already have consumed their meals by the time the surplus is identified. This results in food wastage and inefficiencies in reaching those in need. Consequently, the owners must rely on random students to deliver the surplus food to designated locations within the college and publish it in the college telegram channel, which adds complexity and tedious to the distribution process. The owner of Cafe which is Puan Aida voice out similar concerns, agreeing with Puan Amalina that the current system is ineffective (Puan Aida, personal communication, April 17, 2024).. She notes that the manual approach of relying on students for food transport and posting in the Telegram group is not only inefficient but also adds unnecessary steps to what should be a simple process. Both cafe owners believe that a more streamlined, centralized system could significantly reduce food wastage and improve the overall management of surplus food distribution within the college. They suggest that with a proper platform, leftover meals could be allocated more quickly and to a larger number of students, ensuring better resource utilization and reducing food waste.

b) Ineffective distribution of leftover meals among the students

According to Puan Siti, after the owner of the cafeteria number 4 has already distributed the food to the students, the students who are in charge of the situation are responsible for ensuring that the food is distributed to those students who are in need of it if the students do not want the food (Puan Siti, personal communication, April 15, 2024).. By publishing it in the Telegram channel group, those who are in need can take it to the specific location that has been stated by the student in charge. This can be a problem if there is a student who has already taken the food but did not inform in the telegram channel due to information overload. The individual who already goes to the particular locations that the person in charge has stated, but another individual may have already taken the food. There is also a possibility that the same group of students only received the food; from multiple cafes.

The absence of an organized system to manage and redistribute surplus food creates a gap between food waste and students in need. Similar to the challenges mentioned by Puan Siti, colleges also struggle with effectively communicating the availability of leftover food. In some cases, food may be picked up by students without others knowing, or certain groups of students may repeatedly benefit, leading to unequal access. Furthermore, relying on informal processes like word-of-mouth or group messages can easily lead to miscommunication and waste when students are unaware of available leftovers (Diekmann & Germelmann, 2021).

1.3 Project Aim

This project aims to develop a platform for cafeteria owners to distribute their leftover food to students. The main objectives of this project are:

a) To gather data and analyze requirements of FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin.
b) To design the system of FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin.

c) To develop a system of FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin.

1.4 Project Scope

This project scope of the demographic area covered in this project is within UiTM Jasin. The system was developed using Java. To eliminate the problem of a lack of proper channels for sharing food, this system helps cafeteria owners share their leftover food and not rely on their specific channel. Therefore, the database platform implemented is Postgress.

1.5 Project Significance

Several benefit was discovered for the specific users:

a) Significance to cafeteria owners

   i. The significance of this project to the cafeteria owner is to make a proper channel to share their leftover with the students who need food.
   ii. By implementing this system, the cafeteria owner can ensure that food is not wasted, but instead reaches students who may be struggling with food insecurity.

b) Significance to student

   i. The significance of this project to the students is to reduce information overload in Telegram.
   ii. By having this system, students will get instant updates regarding the leftover information by getting the notification whenever the leftover is available.

1.6 Outline of the Thesis

This thesis is divided into five chapters: introduction, literature review, methodology, results and findings, and conclusion and recommendation.

a) This thesis began with Chapter 1: an explanation of the project details, and the introduction. The following describes the outlines of the remains 4 chapters.
b) An overview of some studies on FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin is provided in Chapter 2.
c) The methodologies and phase of development for this project, including research and activities, are included in Chapter 3. The methodology study designs that have been employed to learn more about this project are included in this chapter.
d) The discussion centres on the findings and insights gained throughout the project, based on the methods that have been selected earlier and will be elaborated in Chapter 4. It provides a detailed analysis of the results and explains how our approach led to these outcomes.
e) The final part reflects on the challenges and limitations encountered during the project. It also offers recommendations for future improvements, suggesting ways to enhance the project if it were to be undertaken again will be discussed in Chapter 5.

1.7 Summary

This chapter summarized by outlining the study's history and stating FoodShare's issue statement, which is "Connecting Surplus to Need, Reducing Food Waste." In addition, an explanation of the project's goal and its accomplished objectives has been provided. Moreover, the project's scope is discussed, and this chapter concludes with an explanation of the thesis outline.

CHAPTER 2

LITERATURE REVIEW

This chapter discusses the literature review related to the development of the FoodShare: Connecting Surplus to Need, Reducing Food Waste system. All the information gathered in this chapter is based on articles, books, journals, and other reliable materials related to this project.

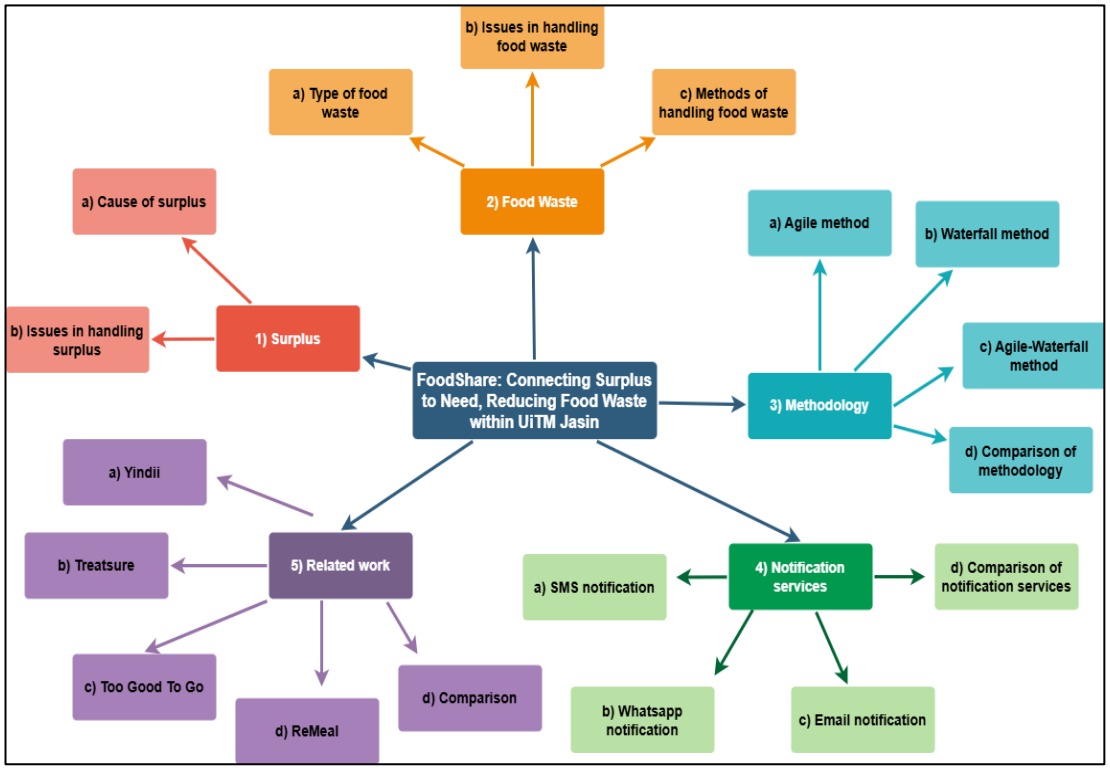


Figure 2.1 Overview of literature review

Figure 2.1 illustrate the overview of the topic that has been reviewed in this chapter 2. The research topics are related to the project and consist of 5 subtopics: surplus, food waste, the methodology used to develop FoodShare system, notification services, and related work. Finally, a summary of this chapter is also elaborated.

2.1 Surplus

Teigiserova, Hamelin and Thomsen (2019) defined, "Surplus food represents the leftover edible food fit for human consumption." Meanwhile, the definition of surplus food according to Huang, Liu and Hsu (2020) is that surplus food arises from factors like overproduction, excessive imports, weather impacts, and market prices, leading to the discarding of uneaten fruits and vegetables before reaching suppliers, processors, restaurants, or consumers. This statement is supported by Lai (2022), who also claimed that food surplus happens when there is more food available than needed, occurring at all stages from farms to homes, resulting in unsold edible items at supermarkets or restaurants or excess food piling up at farms and storage facilities, ultimately leading to food waste.

2.1.1 Cause of Surplus Food

There are a few causes that can lead to surplus food, which are overproduction, food imperfections, and retail:

a) Overproduction

Plazzotta, Manzocco, and Nicoli in 2017 mentioned that in developing nations, waste primarily arises during agricultural production, post-harvest handling, and distribution due to seasonal surpluses and a lack of effective preservation methods for perishable crops, while agricultural waste remains significant even in developed countries. According to Lai (2022), in Hong Kong, local bakeries frequently make more food than they can sell, leading to leftover items that are discarded if unsold by the end of the day, which adds to the 40% of surplus food in landfills in Hong Kong. This statement is supported by ReFED (2022), which said that even though there are efforts like gleaning and farm-to-food-bank programs to collect this unharvested food, most of it remains in the fields and gets plowed under. This is also agreed by Bartezzaghi, Cattani, Garrone, Melacini, and Perego (2022), where weather changes, pests, diseases, new pesticide regulations, poor communication by retailers, shifting consumer preferences, market demand fluctuations with low prices, and difficulties

finding buyers can all lead to produce being left unharvested in fields during cultivation and harvesting, contributing to surplus food.

b) Food imperfections

According to Lai, in 2022, food sellers often end up with extra surplus because of customer demand, but supermarkets throw away imperfect yet edible produce to keep their image or because customers prefer nicer-looking items. Meanwhile, strict quality standards frequently lead to the disposal of perfectly edible food that does not meet specific aesthetic criteria (Deliverect | What Is Food Waste? Causes, Impacts, and Solutions for a Sustainable Future, 2023). Additionally, Plazzotta et al. in 2017 stated that issues that arise during the packaging and processing stage include excessive inventory, challenges with operational and logistical performance, failure to meet quality standards or adhere to retailers' shelf-life policies, and contractual obligations regarding order cancellations and product returns with distributors and retailers. This is also agreed by ReFED (2022), where customer expectations regarding freshness can prompt businesses to discard perfectly safe and edible food due to the perception that it has exceeded its optimal condition.

c) Retail

One part of the excess food is the outcome of issues particular to the retail stage. Among these issues are poor demand forecasting and ordering, which results in an overstock because of forecasting issues; store personnel mishandling; defective packaging; temperature control problems; inefficient shelf and display management; and failure to meet internal quality standards and grading (Plazzotta et al., 2017). This strains purchasing and inventory management procedures, making it challenging to change stocking procedures or product sizes—especially when these procedures are intimately associated with a retailer's brand identification. In the meantime, inventory management and purchasing procedures are under strain to satisfy consumer demand for a wide variety of premium food goods, claims ReFED (2022). Retailers must continually adjust to prevent waste because of the complexity added by changes in customer behavior and preferences.

2.1.2 Issues in Handling Surplus

There are many issues of handling food surplus. According to Thapa Karki, Bennett, and Mishra in 2020, challenges in handling food surplus include a lack of a legislative framework for food donations and organizational sustainability issues, requiring coordinated efforts between actors. Moreover, Hastorf and Foxhall (2017) mentioned that handling food surplus involves creating and reflecting social and political relations within a group, with individuals storing food within a network of interactions and rules for access. Also, there is a model for managing food surplus, encompassing business, environmental, and social players, and it demonstrates its use in identifying food waste reduction strategies, which indicates that there are many ways to handle food surplus (Garrone, Melacini & Perego, 2014). The key problem in handling food surplus is the lack of analysis on critical success factors for redistribution, such as selling in secondary markets, donation to non-profit organizations, and marketing and sponsoring actions (Sert, Garrone, Melacini & Peregi, 2015).

2.2 Food Waste

Food waste is defined as reducing the amount or quality of food typically associated with retail, food services, human behaviour, and the phases of final consumption (Moraes, Lermen and Echeveste,2021). According to Garske, Heyl, Ekardt, Weber and Gradzka (2020), food waste highlights the distributive problems associated with an unsustainable global food and agriculture system. This is also agreed by Huang et al. (2020), who claimed the worrisome nations worldwide issues related to food waste. Huang et al. (2020) also mentioned confusing terms like "food surplus" and "food waste" with other related proper nouns. However, Lins, Zandonadi, Raposo and Ginani (2021) stated that "Food loss" and "surplus food" are included in the definition of "food waste".

2.2.1 Types of Food Waste

Kim, Lee, Andrew Lin, Hong, Kwon and Lee (2020) determined that food waste can be divided into four categories: animal byproducts, mixed catering waste, organic crop residues, and inedible food waste. In the meantime, Teigiserova et al. (2019) mentioned, food waste can be categorized into six categories, which consist of edible food, naturally inedible items, unavoidable waste with potential optimization, inedibility due to external factors (e.g., weather, pest damage), inedibility due to perishability or resource constraints and loss or unaccounted amounts. Different from Li, Jiang, and Qing in 2023, there are five categories of food waste: whole foods, fruits and vegetables, poultry, meat, eggs, and dairy products, staple foods such as rice and noodles, snacks, and candy (not including the above foods). Additionally, this is supported by Silvennoinen, Nisonen, and Katajajuuri (2022), who claimed that the categories of food waste consist of vegetables, potatoes, fruit and berries, bread, meat, and fish.

2.2.2 Issues in Handling Food Waste

Handling food waste is not an easy task. According to Lins et al. (2021), understanding the economic aspect is essential for promoting the reduction of waste generation in the food production process, as the structure and administration of the business must balance the needs of the other two pillars with the equipment and tools required to generate financial returns. This is also supported by Çosan and Yakut Aymankuy (2023), who highlighted that handling food waste involves issues such as fertilizer and water usage, soil occupation, transportation and cooking time, and environmental pollution. Meanwhile, Makanjuola, Arowosola, and Du (2020) mentioned that collecting, storing, and preparing food according to cultural norms and turning it into useful byproducts can be challenging. This problem is exacerbated by insufficient care and attention given to food from the farm to the store, leading to poor quality and waste.

In addition, Chen, Zhao, Qiu, Zhu, Liu, Yin, Shen and Fheng (2021)

emphasized the economic challenges associated with food waste treatment facilities, noting that their viability often depends on government subsidies. Factors such as location, processing techniques, and urban income levels play a significant role in determining the success of such facilities. Similarly, Niyonsaba, Höhler, Kooistra,Van Der Fels-Klerx, and Meuwissen (2021) explored the use of insect farming as an innovative method for food waste management. They noted that while insect farming holds promise as a sustainable solution, profitability remains a challenge due to high operational costs and the need for market development. Both studies underscore the importance of understanding economic and systemic factors to address food waste effectively.

2.2.3 Methods of Handling Food Waste

Food waste should be handled as much as possible through reuse and recycling; dumping and energy production should only be used as a last resort (Dhir, Talwar, Kaur & Malibari, 2020). According to Dhir et al. (2020), packing leftovers for taking away or donating unconsumed food are standard approaches for handling consumable waste. This is agreed by Cook, Goodwin, Porter, and Collins (2022), who propose the food recovery hierarchy, which prioritizes using excess food for human consumption, repurposing it as animal feed, recovering energy via anaerobic digestion, and composting to managing food waste effectively based on its social, economic, and environmental impacts.

2.3 Methodology of Software Development

The Methodology for software development outlines a structured approach aimed at effectively navigating through all the stages involved in the process to ensure successful outcomes (Soobia, Jhanjhi, Naqvi & Humayun, 2019). According to Gbaranwi, Precious, Ojekudo and Akpofure (2021), software development methodologies are essential for efficiently managing tasks to ensure timely completion and high-quality outcomes, with each approach having advantages and disadvantages. On the other hand, according to Badkar

(2023), the term "software development methodology" refers to a framework that emphasizes planning and structure when creating information systems.

2.3.1 Agile Method

Badkar (2023) claimed that agile methodology focuses on user satisfaction and communication through short sprints and regular feedback loops to drive software improvements. Meanwhile, Ríos and Pedreira-Souto in 2019 stated that agile approach empowers flexibility to adapt to changes throughout the life cycle, prioritizing user involvement and software functionality over extensive documentation while fostering support for modifications as needed. On the other hand, Chaudhari and Joshi (2021) mentioned that agile is a software development framework that employs 1-4 week iterations to deliver higher quality software aligned with customer needs.

There are a few characteristics to identify the agile methodology. According to Ríos and Pedreira-Souto (2019), agile methodology emphasizes adaptability to changing requirements, continuous team communication, end-user involvement, minimal yet precise documentation, and utilization of UML artifacts for design and documentation. Meanwhile, V2Soft (2016) characterized Agile by transparency, iterative releases, early detection and resolution of issues, and early and predictable delivery. As a result, Badkar stated in 2023 that Agile is most suitable for projects with evolving requirements, particularly in emerging software domains, enabling developers to incorporate new concepts based on market demands, provided they operate autonomously and prioritize rapid progress.

2.3.2 Waterfall Method

According to Alzeyani and Szabó (2022), the Waterfall software development model will organize the process into sequential phases, emphasizing the completion of each phase before moving to the next, thereby treating the project as a unified entity. This statement is supported by Yahya and Maidin (2022), who claimed that the waterfall model is a conventional Software Development Lifecycle (SDLC) that is characterized by its sequential phase

development. The waterfall paradigm comprises six phases: analysis, design, development, testing, implementation, and maintenance.

There are a few characteristics that can be determined as waterfall methodologies. The Waterfall model mandates a structured, sequential completion of project steps, with each phase being immutable once finalized and commencing with clearly defined requirements that remain unchanged throughout development Andrei (2016). According to Mantik, Nurcahya, Nurfauziah, Dwiatmodjo, Raya, No, Wetan and Timur (2022), each stage of the waterfall methodology, from analysis to maintenance, involves a thorough examination and adherence to a structured progression, with changes in data or functionality impacting subsequent phases, ensuring the prioritization of functional requirements during implementation and maintenance.

2.3.3 Agile-Waterfall Method

The Agile-Waterfall method, though problematic for purists, effectively combines the strengths of both methods and is suitable for developing software or products that integrate hardware and software (Lucid Content Team, 2019). Reiff and Schlegel (2022) mentioned that the Agile-Waterfall method follows the agile methodologies; each iteration involves defining requirements, seeking customer feedback, conducting tests, and making corrections for continuous improvement. Other than that, Hassani, El Bouzekri El Idrissi and Abouabdellah (2018) stated that a project can be initially defined and planned using an agile approach, then later developed and implemented using traditional procedures.

Taller (2021) stated that the Agile-Waterfall method is used to shorten design, analysis, and planning phases while defining project parameters like budget and delivery time, ensuring compliance with standards, enhancing collaboration, and being a useful initial step when adopting Agile culture is challenging. Meanwhile, Żurawiecki (2024) stated that the key advantage of the Agile-Waterfall method is its customizability, allowing a mix of Waterfall and Agile elements without strict rules, providing complete control to incorporate Agile elements based on team and stakeholder readiness and enabling adjustments over time to improve workflow.

2.3.4 Comparison of Methodology

Choosing between the dynamic Agile, Waterfall, and Agile-Waterfall methodologies is crucial in software development, as they encompass distinct characteristics and approaches. Table 2.1 describes the characteristics represented by each methodology.

Table 2.1 Comparison between Characteristics Of Methodologies

Characteristic	Agile	Waterfall	Agile-Waterfall
Planning Horizon	Short-term planning focused on iteration objectives	Long-term planning focused on the entire project life cycle	Combines long-term planning (Waterfall) with short-term iterations (Agile), allowing for an overall roadmap with detailed planning for each iteration.
Activities Details	Unpredictable, nonlinear, and unmeasurable detailing with incremental and constant deliveries, as well as lean and informal documentation	Predictable, linear, and measurable detailing with integral or phased deliveries and sophisticated, formal documentation	Detailed documentation and upfront design for initial phases (Waterfall), with flexibility to adapt and refine activities in later phases based on iterative feedback (Agile).
Project Scope	Based on the vision of what must be built.	Based on the detailed specifications of what will be built,	Broad, well-defined scope set at the beginning (Waterfall), with flexibility to modify scope within iterations to incorporate changes and new insights (Agile).
Scope Conformance	Changes are identified and planning is adjusted for each interaction	Deviations are identified and activities adjusted to maintain planning	Adherence to the overall project scope defined initially (Waterfall), while allowing iterative adjustments to scope within each iteration to better meet project goals (Agile).
Control and Monitoring	Made from physical and/or virtual visual artifacts and devices, such as posters, murals and pictures, with short and frequent team meetings	Carried out via performance indicators, time lines, formal documentation, performance reviews, and audits, with extensive and infrequent team meetings	Regular monitoring against the overall plan (Waterfall), combined with iterative reviews and continuous integration to ensure alignment with project goals and stakeholder feedback (Agile).

Table 2.1 (Cont.)

Management Style	Flexible, variable, and adaptive.	Mechanic, formal, and bureaucratic.	Structured management for planning and milestones (Waterfall), followed by collaborative teamwork for iterative development and improvement (Agile).
Project Planning	Lean planning for the cycle of iterations with constant re-evaluations and refinements	Sophisticated planning for the entire project life cycle	Initial high-level planning to define project frames like budget and timeline (Waterfall), followed by iterative, adaptive planning to adjust based on feedback and progress (Agile).

(*Source :* Azenha, Reis and Fluery, 2020)

The table compares three project management methodologies across various characteristics. Agile emphasizes short-term planning and iterative development, Waterfall focuses on long-term planning and structured phases, while Agile-Waterfall combines elements of both methodologies to provide flexibility and adaptability throughout the project lifecycle. Each methodology differs in its approach to project planning, activities detailing, project scope, scope conformance, control and monitoring, and management style.

2.4 Notification Services

Notifications are communications that enhance the mobile broadcast service by allowing the transmission of information regarding upcoming or imminent events Hornsby, Bouzazizi and Defee (2010). Meanwhile, a notification service is a system that informs a significant number of individuals of crucial or urgent occurrences, prompting them to take action (Li, Mallows, & Landwehr, 2012).

2.4.1 SMS Notification

Short Message Service (SMS) communication has grown significantly in popularity in the last 10 years, both at home and at business (Gao, Yang and Wang, 2010). According to Ali, Alam and Mustafa (2016), SMS was originally created for person-to-person messaging, and it has been widely utilized in the

wireless realm, leveraging computer technology to transmit various types of information, such as news updates, financial alerts, and notifications. This makes SMS more useful.

According to research by Mohd Ekhsan, Hamid and Mazlan in 2020, SMS offers advantages over other social media platforms like WhatsApp, Telegram, and Twitter due to its high "open rate" and independence from internet connectivity, which is particularly beneficial for individuals in rural areas or those unable to afford internet access. This website, STRINGFELLOW (2023), stated that SMS messages boast a 209 % higher response rate than phone, email, and social media marketing, as they are directly delivered to personal devices, unlike emails, which can be easily deleted or end up in spam folders. Moreover, Mackeviciute, in 2024, mentioned that speedy delivery and immediate visibility characterize SMS, as opposed to emails, which can easily be overlooked, with 60 % of text messages being opened within five minutes, resulting in heightened engagement and maximizing the effectiveness of marketing efforts. The immediacy of SMS also means that time-sensitive information reaches the target audience almost instantly, which is crucial for urgent communications and real-time marketing campaigns. Furthermore, the simplicity and directness of SMS communication make it a powerful tool for delivering clear and concise messages without the distractions commonly found on other platforms.

2.4.2 Email Notification

According to Iqbal and Horvitz (2010), email notifications are used to provide passive awareness of new email arrival while a user is attending to a primary task. Email notifications serve to inform subscribers about product changes, new features, and scheduled maintenance timings, enabling better task planning for prospects and customers and distinguishing themselves from marketing emails or newsletters by offering valuable anticipated information (Uplers, 2023). Indirectly, it enhances the overall user experience by providing timely and relevant updates.

Email notifications offer numerous advantages, making them a powerful communication tool for businesses. They provide a high return on

investment with minimal costs, allowing companies to reach a broader audience effectively (Beginner, 2022). Unlike other channels, emails are accessible to almost all customers and are ideal for conveying detailed or longer content. Their customization options enable tailored messages that resonate with specific audience segments, enhancing engagement (Uplers, 2023). Moreover, with no strict time constraints, emails can be read at the recipient's convenience, ensuring flexibility. Uplers in 2023 also justified the high open rate of email notifications also plays a crucial role in keeping users informed, fostering customer retention through regular updates, and promoting new features or products efficiently.

2.4.3 Whatsapp Notification

WhatsApp notifications serve as a powerful communication tool in various contexts, including educational settings and organizational notifications. The application allows users to send messages instantly to individuals or groups, ensuring that information is quickly shared. This makes WhatsApp an effective notification service because of its widespread use, ease of access, and real-time delivery capabilities (Sharma, Mavuduru, Singh, & Mandal, 2019)Surveys show that WhatsApp is popular among businesses for sending alerts, promotions, and transactional information since 90% of messages sent to the application are opened by users due to the easy-to-use format of the application and deregulated accessibility in any country. The setting of multimedia including images, videos and documents adds to the richness and interest of these notifications (Sayudin, Kartono & Aang, 2024). This makes WhatsApp an effective means for communication that is briefly and engagingly informing the user.

According to Sayudin et al. (2024), some of the benefits of the WhatsApp notification include; Real-time notification, Customized and bi-directional communication. These features enable organizations to provide relevant and accurate responses to customers hence enhancing their satisfaction and hence their loyalty. Moreover, knowing that WhatsApp is an end-to-end encrypted app, users feel safe in sharing material that may be personal, like a financial statement report or appointment calendar (Zarouali, Brosius,

Helberger & De Vreese, 2021).

2.4.4 Comparison of Notification Services

Based on the analysis outlined in Table 2.2, an assessment was undertaken on three forms of notifications: SMS, email, and push notifications. Below are the comparisons of each characteristic based on the specified parameters.

Table 2.2 Comparison between Characteristics Of Notification Services

Feature	WhatsApp	SMS	Email
Delivery Speed	Instant, with internet connection	Instant, no internet needed	Fast, but can be delayed by spam filters
Read Rate	Extremely high, thanks to app popularity	High, due to direct to phone delivery	Lower due to cluttered inboxes
User Engagement	High, with interactive features	Moderate, text-only	Moderate, depends on content
Message Length	Long (Up to 4096 characters)	Limited (160 characters)	Unlimited
Multimedia Support	Rich (Images, Videos, Documents, etc.)	None	Rich (Images, Videos, Documents, etc.)
Personalization	High, appears as individual message	Low, appears generic	High, can be personalized
Delivery Guarantee	Dependent on app usage and internet	High, as it doesn't need internet	Usually high, but can land in spam
User Base	Requires WhatsApp installation	Universal, no app needed	Requires email account
Automation	Possible with WhatsApp Business API	Widely supported	Widely supported
Cost-Effectiveness	Free with internet, charges for Business API	Carrier fees apply	Generally free, charges for premium tools
Interactivity	Very interactive (buttons, quick replies)	Not interactive	Interactive with HTML and links

(*Source* : Soares, 2023)

These two comparisons illustrate specific advantages and disadvantages of WhatsApp Broadcast, SMS, and Email as the means of communication. WhatsApp delivers messages instantly provided internet is available, boasts near 100% read rates, and has high interaction rate because of interactive features such as quick replies and buttons. It allows for use of multimedia, long message up to 4096 characters and allows personal communication which makes it perfect for businesses who wish to engage their customers. Compared to email, SMS sends messages instantly and works without the Internet, has high read rates, but low participation is due to the lack of graphical and much longer texts than 160 characters. It does not have

multimedia messaging and it is not customizable to a certain extent especially in terms of the interface however it only requires a basic interface to work and is reliable for simple messaging mostly on any mobile phone. Email, however, enables one to send as many messages as possible with a wide reach with no limit on the word limit and comes with the possibility of customized marketing tools. However it has lower read rates because the mail inboxes are cluttered and most emails end up in the spam folder, that it is cheaper and comes with high analytical support. Commitment, satisfaction, and security are strong suits of WhatsApp while Short Message Service (SMS) is best for its simplicity and without app and internet, and Email service is appropriate for detailed messages and marketing. WhatsApp channel is for engagement, the SMS for convenience, while the Email channel is for depth of content.

2.5 Related Work

Within the framework of the project, related research carefully analyzes the platforms or systems now in use to obtain new perspectives on food sharing. Understanding the capabilities these platforms provide and pointing up places where our system might be improved are the main objectives. Platforms that reflect different food-sharing strategies, such as Yindii, Graze Market, and Too Good to Go and ReMeal, are the main focus of this study. Examining these sites closely is intended to help one understand their importance and applicability in the food-sharing industry. Making use of this knowledge, our goal is to improve our own ideas and methods to create a more successful and efficient food-sharing arrangement.

2.5.1 Yindii

Yindii plays a crucial role in promoting sustainability by connecting consumers with surplus food from cafes, restaurants, and grocery stores at discounted prices. The platform's user-friendly design ensures that customers can easily navigate through a diverse selection of excess food items, making it convenient to purchase affordable and high-quality goods. By collaborating with businesses that aim to minimize food waste, Yindii helps reduce environmental

impact while supporting a circular economy. Additionally, it empowers users to make more sustainable choices in their daily consumption, contributing to global efforts to combat food waste and promote responsible resource management.

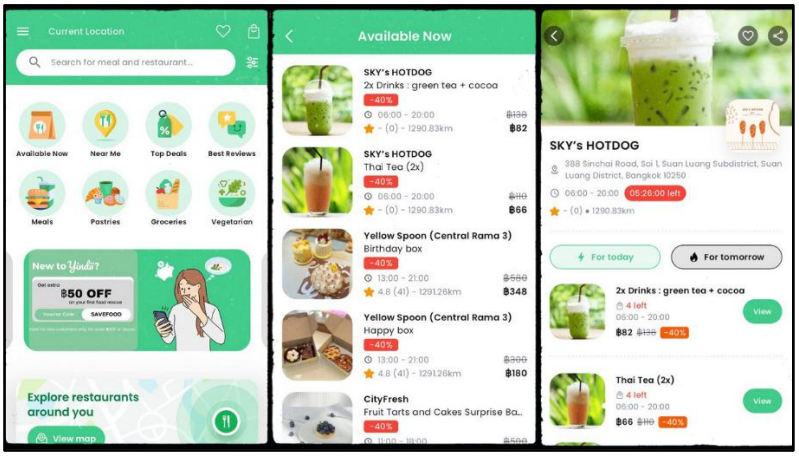


Figure 2.2 Interface of Yiindi application

Figure 2.2 shows the interface of the Yiindi application. The additional food items are prominently displayed on the UI, frequently with eye-catching graphics and detailed descriptions that entice customers to explore further. Users can easily navigate the platform by sorting options based on food type, cuisine, or region to receive personalized recommendations. In addition, the payment procedure has been streamlined to guarantee a prompt and seamless buying encounter for customers wishing to buy cheap excess food products. By providing a user-friendly platform with eye-catching displays and simple navigation for dispersing excess food, Yindii helps minimize food waste and offers consumers cost-effective options.

2.5.2 Treatsure

Treatsure also promotes sustainability by encouraging users to adopt environmentally-friendly practices, such as reducing food waste and supporting local businesses. The platform offers a convenient and user-friendly interface that allows customers to explore a variety of surplus food options from participating vendors. Additionally, Treatsure fosters community engagement by creating partnerships with restaurants, cafes, and food establishments that share the goal of minimizing waste. The application not

only benefits consumers by providing cost-effective meals but also supports businesses in managing their surplus inventory efficiently. By bridging the gap between food suppliers and consumers, Treatsure contributes to a more sustainable food ecosystem.

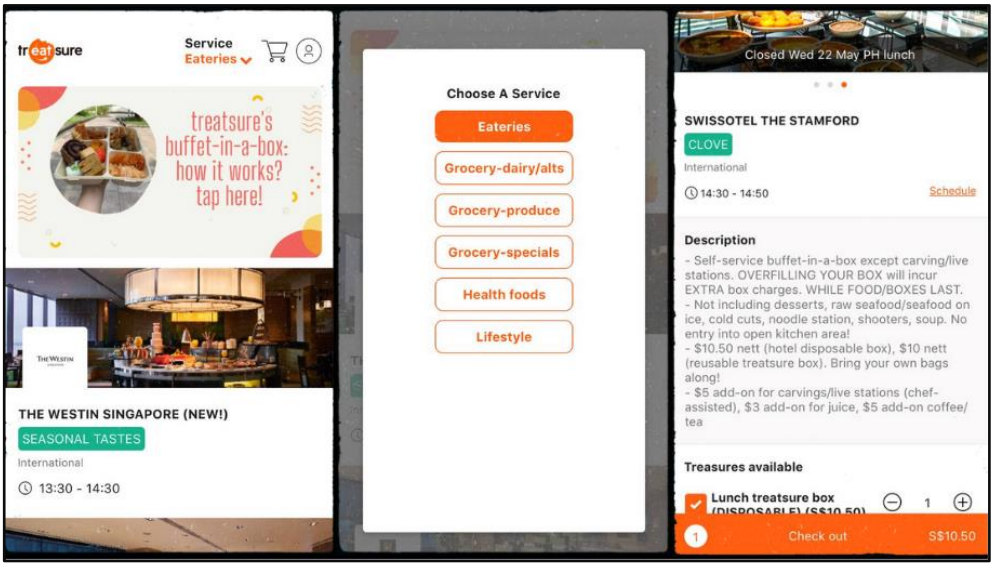


Figure 2.3 Interface of Treatsure application

The display of the Treatsure application's user interface is shown in Figure 2.3. The approach that Treatsure takes to the design and layout of its user interface most likely places a stronger premium on usability and simplicity in order to improve the speed and ease with which users can browse and make purchases. It is possible that the website will generate a sense of urgency in customers by utilizing tools such as countdown timers. This will encourage customers to make purchases from the website as soon as possible, so saving food from being wasted. The mission of Treatsure is to develop a community of conscientious consumers that are committed to sustainable consumption practices in addition to tackling the issue of food waste. This will be accomplished via the utilization of intelligent design decisions and engaging techniques.

2.5.3 Too Good To Go

Too Good To Go is a strategic partner that reduces food waste for restaurants, bakeries, supermarkets, and hotels by selling excess food at a discount before

it is no longer salvageable. By serving as a middleman between these businesses and customers, the portal gives users the chance to peruse surplus food listings from a variety of companies.

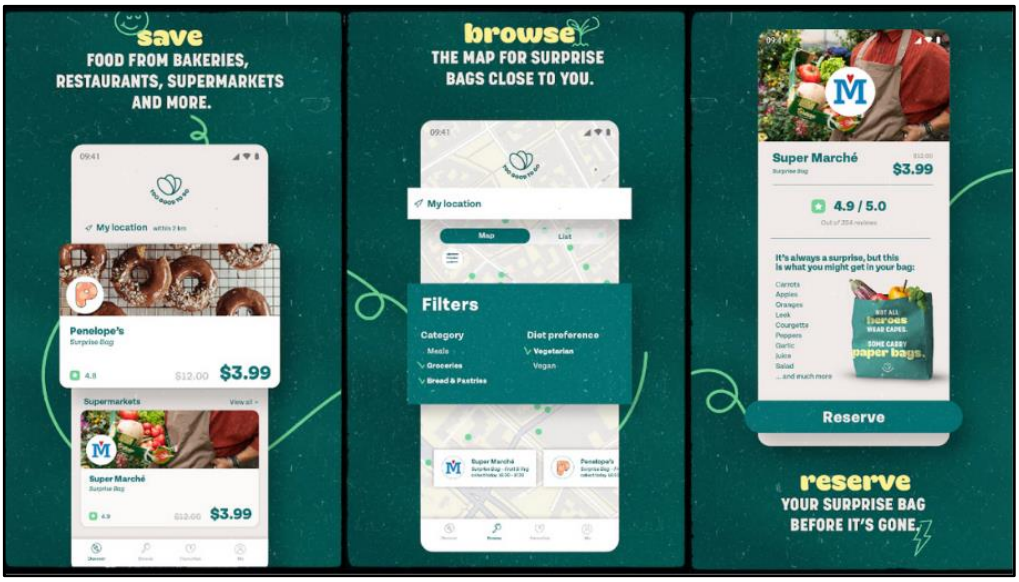


Figure 2.4 Interface of Too Good To Go application

The interface, as shown in Figure 2.4, has been carefully designed to display listings for excess food from different kinds of businesses, such as restaurants and supermarkets, making it simple for people to look through the possibilities. Users can choose to filter postings for surplus food according to criteria like geography, cuisine, or type of business, which gives them the opportunity to find products that suit their requirements and tastes. Through the provision of detailed facts like discounted rates, pickup times, and accessible quantities, each ad empowers consumers to make well-informed judgments when perusing the platform. In addition, interactive elements like ratings, maps, and user reviews improve the user experience overall by making navigating easier and supporting decision-making. By means of its intuitive design and extensive functionalities, Too Good To Go efficiently links surplus food supplies with consumers, thereby mitigating food waste and advancing sustainability within the food business.

2.5.4 ReMeal

ReMeal is an application that aims to reduce food waste in cafes, bakeries,

supermarkets, and restaurants, and to offer such food to consumers at a lower price. With its roots in Malaysia, ReMeal introduced a convenient application that consists of a list of available products and buys food that has been in excess at different stores.

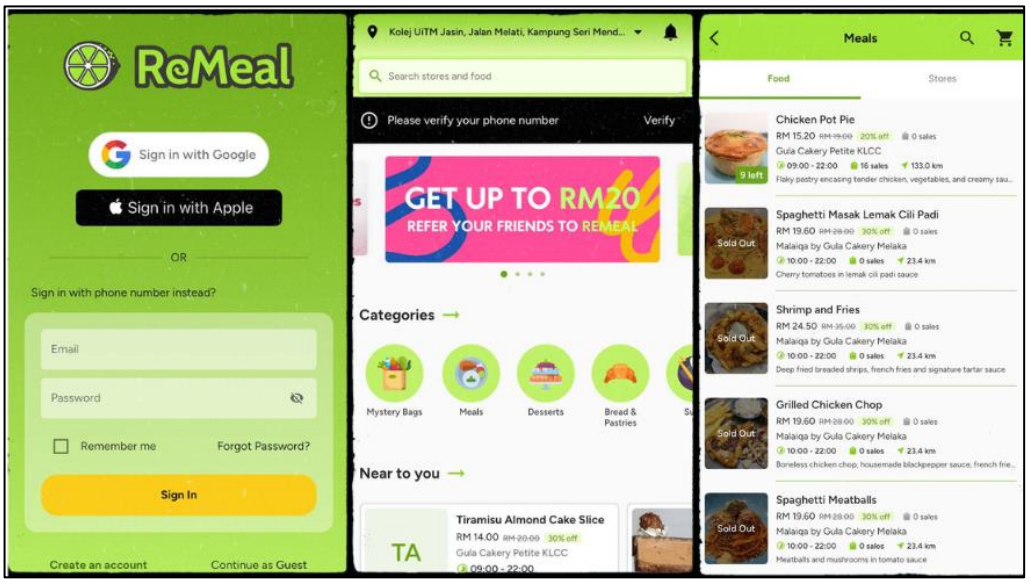


Figure 2.5 Interface of ReMeal application

Figure 2.5 shows that the ReMeal application shares key characteristics such as ease of navigation, easy to use and clean UI. When the app is launched, whereby the users end up being presented with a selected list of surplus food items a number of which are accompanied by beautiful pictures and very detailed descriptions. It enables users to sort the platform's restaurants by food kind or cuisine, geographic region, and it gives out relevant suggestions according to the input. This design helps one to browse and search for the available inexpensive foods with a lot of convenience.

As a part of the ordering process optimization, ReMeal has the fast and protected payment acceptance mechanism. Individuals can be able to make payment directly via the app hence enhancing the overall transaction. Customers are directed and instructed to go to the stores and pick their orders for consumption and once ordered they are provided with an opportunity to make more economical orders since most of them may consider a second option of food which would otherwise go waste.

2.5.5 Comparison Related Work

Table 2.3 concisely compares the key features and functionalities of Yindii, Treatsure, and Too Good To Go, focusing on their respective functions, interface/design, navigation, and additional features.

Table 2.3 Comparison between the features of related work

Feature	Yindii	Treatsure	Too Good To Go	ReMeal
Interface/Design	Showcases surplus food items prominently, with appealing visuals and descriptions	Likely prioritizes simplicity and user-friendliness to facilitate quick and seamless browsing and purchasing experience	Designed to highlight surplus food listings from different types of businesses, ranging from restaurants to supermarkets	User-friendly interface with appealing images and detailed descriptions of surplus food items
Navigation	Users can easily navigate through categories such as food type, cuisine, or location	Users can navigate for quick and efficient browsing	Users can explore surplus food listings based on location, cuisine, or business type	Sorting options based on food type, cuisine, or location for personalized recommendations
Additional Features	Streamlined checkout process	Countdown timers may create urgency to motivate timely purchases and prevent food waste.	Interactive features such as maps, ratings, and user feedback contribute to a user-friendly experience	Secure checkout system, referral program to promote sustainable consumption
Function	Offers surplus food at discounted prices	Present surplus food from restaurants and cafes at discounted rates	Partners with various businesses to sell surplus food at discounted prices before it goes to waste	Connects users with surplus food from various establishments at discounted prices

The table presents a comparison of four platforms which are Yindii, Treatsure, Too Good To Go, and ReMeal, by analyzing their interface, navigation, extra options, and capabilities. Yindii focuses on the aesthetic appeal of its designs, where fresh, leftover food is showcased, and the simple and easy-to-use menu options help sort the products based on food type and store or restaurant and a clean check-out method within sold and bulk products. Treatsure is designed to be simple and easy to use to allow users efficiently browse through products, and include countdown timers to ensure consumers make their purchase within the shortest time possible to avoid foods expiring. Too Good To Go has an extensive homepage that displays listings from various companies, including

restaurants and supermarkets, clickable features that include maps, rating, and consumers' reviews. Convenience is seen in a tabular form of description and illustrations, customize filter options, check out process and ReMeal referral program to encourage use of sustainable products. All of the four platforms offer a similar function to link users with food sellers, buyers with unsold food to be sold at cheaper prices; however, each one of them comes with features and advantages to facilitate the use and embrace sustainability.

2.6 Discussion

The FoodShare system is designed to efficiently connect surplus food sources with those in need while minimizing food waste. The literature review highlights key themes, including surplus food causes, food waste management, software development methodologies, notification services, and related platforms.

A crucial element of this system is the handling of surplus food, which arises from overproduction, food imperfections, and retail inefficiencies. Overproduction in bakeries and farms leads to large amounts of food waste, while strict aesthetic standards at supermarkets result in the disposal of perfectly edible food. Retailers also struggle with demand forecasting, which contributes to surplus accumulation.

To manage food waste effectively, different methodologies have been considered. The Waterfall methodology was identified as the most suitable for the FoodShare system due to its structured and predictable nature. Unlike Agile, which relies on flexible iterations, Waterfall ensures that all requirements are well-defined before moving forward, minimizing risks for stakeholders like cafeteria owners.

For communication and user engagement, various notification methods such as SMS, email, and WhatsApp notifications were compared. WhatsApp notifications emerged as the most effective due to their immediacy, multimedia capabilities, and high engagement rates. Unlike SMS, which has character limitations and costs per message, WhatsApp provides an interactive platform at a lower cost.

Existing food-sharing platforms, such as Yindii, Treatsure, Too Good

To Go, and ReMeal, were analyzed to identify best practices for FoodShare. While Yindii and Treatsure focus on offering surplus food at discounted rates, Too Good To Go and ReMeal incorporate user-friendly navigation and real-time surplus tracking. By integrating these features, FoodShare aims to create a more efficient and sustainable food redistribution system.

In conclusion, FoodShare: Connecting Surplus To Need Reducing Food Waste leverages structured development, effective notification services, and insights from existing platforms to address food waste issues. By ensuring systematic food surplus management and timely communication, this system enhances accessibility while reducing waste, benefiting both food providers and consumers.

2.7 Summary

This chapter discusses and explains the study of the literature review associated with the system, which is the FoodShare system. This chapter examines the causes and handling issues of food surplus and the types and methods of handling food waste. This chapter also describes the different methodologies, including Agile, Waterfall, and Agile-Waterfal, to ensure which one is more effective in the process of development. It also evaluates various notification services like SMS, Whatsapp notifications, and email to enhance communication. Last but not least, by analyzing existing platforms such as Yindii, Too Good to Go, ReMeal and Treatsure, FoodShare seeks to integrate best practices and innovative features to create a sustainable and efficient food-sharing system.

CHAPTER 3

METHODOLOGY

This chapter describes the project's methodology that was used. Furthermore covered in this chapter are prerequisites for system development, the relevant process, and every phase of the project's development. The goals of the project were met and obtained using the technique implementation. Moreover, it produced a summary of each phase and activity according to the selected system development life cycle (SDLC) model. This chapter used the Waterfall Method as the project methodology.

3.1 Overview of Methodology

A set of accepted guidelines and practices used to carry out a task or carry out a study is referred to as methodology. Methodology, in its broadest sense, refers to a collection of guidelines or rules that specify how information about a certain topic is obtained and verified. Different approaches might be used to offer direction on the completion of required tasks. The Software Development Life Cycle (SDLC) is an all-inclusive process that includes effective information management, methodical development, and strategic planning. Recognising that each model has a unique set of benefits and limitations is essential.

3.2 Waterfall Methodology

The Waterfall methodology takes a project management and software development strategy that is linear and systematic. One of the original software engineering approaches consists of discrete stages that are sequentially followed by one another. Every phase has to be finished before the one after it starts, and once one is finished, it can usually not be returned. The Waterfall methodology for the FoodShare: Connecting Surplus to Need Reducing Food Waste project is divided into three main phases: Requirement Gathering and Analysis, Design, and Implementation, as shown in Figure 3.1.

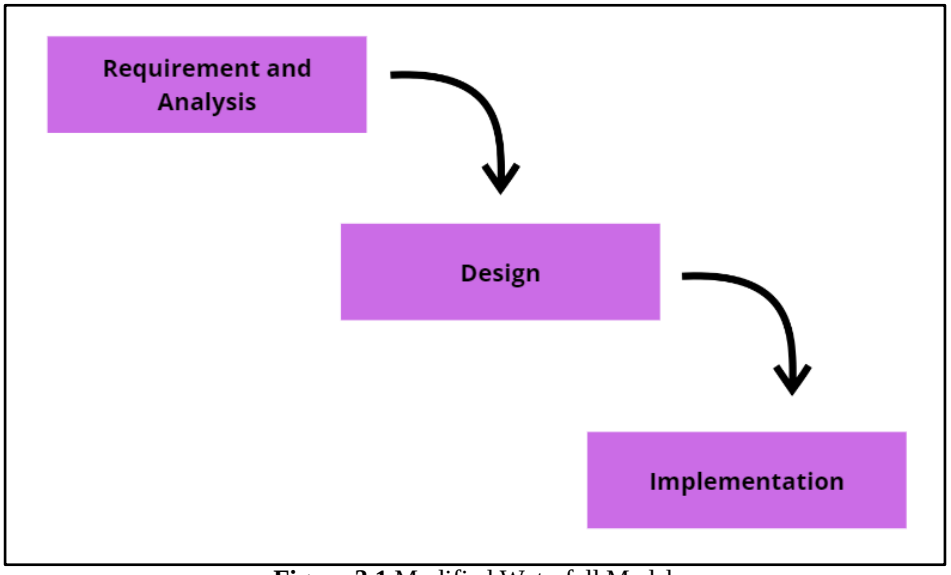


Figure 3.1 Modified Waterfall Model

3.3 Phases

Based on Table 3.1, the Waterfall methodology for the FoodShare: Connecting Surplus to Need Reducing Food Waste project is divided into three main phases: Requirement Gathering and Analysis, Design, and Implementation. There are several tasks and phases in the methodological table to follow so that it can meet the objectives in Chapter 1.

Table 3.1 The project framework using the SDLC approach

Phase	Activities	Deliverables	Objectives
Requirement gathering and analysis	a) Prepare the interview questions. b) Interview session with stakeholders. c) Analyze list of requirements and construct the diagrams d) Document the analyzed requirements.	a) Interview questions. b) Interview data c) Analyzed Requirement diagrams: i. Use case Diagram ii. Domain Class iii. Diagram Activity Diagram iv. Sequence Diagram d) Software Requirement Specification (SRS).	To gather and analyze FoodShare: Connecting Surplus to Need, Reducing Food Waste

Table 3.1 (Cont.)

Design	a) Design a the system b) Design the user interface of the system c) Design the database d) Document the system design	a) Design diagrams: i. Mulilayer Sequence Diagram (MLSD) ii. Design Class Diagram b) Storyboard c) Entity Relationship Diagram (ERD) d) System Design Document(SDD)	To design FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin based on the requirement
Implementation	a) Develop the web system with notification	a) FoodShare: Connecting Surplus to Need, Reducing Food Waste System	To develop a system of FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin.

Phase 1: Requirement Gathering and Analysis

This phase is where the requirements are gathered from the cafeteria owner to get information about the current business process and what should be included in the FoodShare. In order to get the requirements, interview questions must be prepared beforehand for the cafeteria owners to answer later. Interviews were conducted through WhatsApp to understand more about the current distribution process of leftover food.

The phase starts by gathering information from the cafeteria owners. It will create several diagrams, including a use case diagram to show user interactions, a system sequence diagram to detail event sequences, and a class diagram to outline the FoodShare structure. Finally, the Software Requirements Specification (SRS) document was prepared to comprehensively document all the FoodShare requirements, ensuring clear communication and understanding to smooth the development process of FoodShare.

Phase 2: Design

In the second step, the design phase, the system is created based on the requirements. The goal is to design the FoodShare system, including the user interface. An interface was created to guide the system flow and meet the cafeteria owners' requirements, ensuring user satisfaction. This interface includes features like Whatsapp notifications for surplus availability and reservations for surplus. Java was used to develop the interface, and Draw.io was used for diagrams like the Design Class Diagram. The next step is these diagrams were developed, including domain class and design class diagrams. Each diagram is important before moving on. The final task is to document the entire system design in a Software Design Document (SDD) refer on Appendix C.

Phase 3: Implementation

During the implementation phase, the FoodShare System was developed using Java programming language with the help of Visual Studio, a software tool used for building web applications. This phase is the most time-consuming and costly part of the development process, as it involves translating the project requirements obtained from cafeteria owners into a functioning system. The main task in this phase is to develop the system based on the documentation prepared in the previous phase. Additionally, a database was created using Oracle software which is a crucial step in saving and managing data for the system. Some hardware and software are required for implementing the system, as outlined in Table 3.3 and Table 3.4.

Table 3.3 Hardware Specification

No	Hardware	Specification
1.	Laptop	Brand: Acer Aspire 3 Processor: Intel(R) Core(TM) i5-1035G1 CPU @ 1.00GHz 1.19 GHz Installed RAM: 12.00 GB System Type: 64-bit operating system, x64-based processor OS Name: Windows 11 Home Single Language

2.	Phone	Brand: Apple Model: Iphone Xr
3.	Tablet	Brand: Apple Model: Ipad 8th Gen

Three key hardware components support the development of FoodShare: the main workstation for creating reports, configuring the system, and carrying out resource-intensive tasks in the reliable Windows 11 Home Single Language environment is the Acer Aspire 3 laptop, which has an Intel Core i5 processor and 12GB of RAM. As a vital communication center, the phone allows for continuous connection for handling incoming calls and receiving supervisory messages. With its bigger screen for comfortable note-taking and document viewing during consultation or brainstorming sessions, the tablet complements these gadgets and offers flexibility and convenience. Although less often used than a laptop or phone, tablets are a useful backup that may cover simple chores like research or report writing when the main workstation is away, guaranteeing efficiency throughout various work settings.

Table 3.4 Software Specifications

No	Software	Specification
1.	Microsoft® Word for Microsoft 365 MSO (Version 2304 Build 16.0.16924.20150) 64-bit	Report and Documentation
2.	Google Chrome	Internet Browser
3.	Draw.io	Illustrate Diagram
4.	Visual Studio	Code editor
5.	Postgress	Store and retrieve data
6.	GitHub	Repository system

The software requirements describe several software instruments that have been utilised in the expert FoodShare development process. Reports and documentation were produced using Microsoft Word, a well-known word-processing program that is part of Microsoft 365. Meanwhile, the internet browser of choice for project research and information collecting is Google Chrome. The diagram was illustrated using Draw.io. On the other hand, the programming environment chosen for coding chores is Eclipse IDE. The

database system Oracle will handle data storage and retrieval. Furthermore, GitHub will deploy The system as a repository system to the cloud. The development process is facilitated by these software solutions used together, from data management and coding to report writing.

3.4 Summary

In this chapter, the methods used in this project to construct the FoodShare: Connecting Surplus to Need Reducing Food Waste system are explained. Each point clarified the different activities carried out at every stage of the Software Development Life Cycle (SDLC) paradigm.

CHAPTER 4

RESULT AND FINDINGS

This chapter details the results and findings of the FoodShare system, a solution designed specifically for cafeteria owners at UiTM Jasin to share surplus food with students, thereby reducing waste. It covers all three phases of the project which are requirements gathering and analysis, design, and implementation. This chapter brings information on how the project has fared and its Ability to deliver as per the expectations of the stakeholders by presenting a brief on the development aspect of the project.

4.1 Findings Requirement Gathering and Analysis Phase

During the initial phase of gathering requirements, information was collected from stakeholders through interviews. The primary aim was to gain insights into the FoodShare system, understand the current process for distributing leftover food to students, and identify key components necessary for developing the FoodShare: Connecting Surplus to Need, Reducing Food Waste System.

The main features and functionalities required for the system were identified through an analysis of the data collected from the interviews. This phase was crucial for ensuring that the system would meet the demands and expectations of the students and cafeteria owners. Several deliverables were produced during the requirements analysis phase, including a use case diagram, domain class diagram, class activity diagram, and sequence diagram.

4.1.1 Prepare the interview questions

In this phase, it is crucial to prepare detailed interview questions to gather the necessary insights from stakeholders. These questions will cover both functional and non-functional requirements for the FoodShare system, which aims to manage surplus food distribution and improve communication between

cafeteria owners and students. The functional questions will focus on understanding core tasks such as how food surplus should be logged and how notifications about available food should be managed. Non-functional questions will seek clarification on system performance, security, accessibility, and overall user experience. Additionally, stakeholder-specific questions will be developed to understand the unique needs of cafeteria owners, students, and other key users. Preparing these interview questions will ensure that all relevant requirements are captured, providing a solid foundation for the design and implementation phases of the system.

4.1.2 Result of Interview

In the requirement gathering and analysis phase, an interview with the cafeteria owner Puan Amalina which from Cafe 6 and Puan Siti from Cafe 4. The purpose of this interview was to understand and clarify the requirement of the system. During the interview, several questions were asked, and answers were recorded. Table 4.1 shows the results of this interview session.

Table 4.1 The analysis from the interview session with Puan Amalina and Puan Siti

`	Question	Answer
1.	How do you identify when there are leftover foods to distribute?	The cafeteria staff recognizes when there are leftover foods that need distribution, usually when some food cannot be saved for the next day.
2.	What process do you follow to offer leftovers to students?	The cafeteria staff asks random students if they want to take the leftover food.
3.	Who is the main users for this system?	Cafeteria owners and students can use this account. They can sign up to the system and can update their information.
4.	Are there any difficulties in determining where the food goes after students take it from the cafeteria?	The staff faces difficulties in knowing the status of the food once it is taken by the students and brought to the college. They also felt bad about asking students to share the food in the telegram group.

Based on the interview session, Puan Amalina and Puan Siti explained that the purpose of their current process is to distribute leftover food to students. They usually offer food to students by asking if they want it, but face challenges when no students are available. In such cases, they either bring the food home to distribute it to the mosque or consume it themselves. They also encounter difficulties in tracking the status of the food once it's taken by the students, as

the communication method is not always effective.

4.1.3 Analyzed Requirements from Interview Session

All data was compiled for analysis and to create diagrams based on the information obtained from the interview sessions. This included the System Sequence Diagram (SSD), Domain Class Diagram (DCD), and Use Case Diagram (UCD). The results from the requirements gathered during the interviews are presented in Table 4.2.

Table 4.2 Requirement Gathering Result

`	**Answer**	**Findings**	**Actor**
1.	The cafeteria staff recognizes when there are leftover foods that need distribution, usually when some food cannot be saved for the next day.	A system that can track leftover food, recognize food that needs distribution, and ensure it is managed properly. Use case invoved: -Add leftover information include notify students -Update leftover Information -Delete leftover Information -View leftover information	Cafeteria owner Student
2.	The cafeteria staff asks random students if they want to take the leftover food.	The system should enable students to request leftover food and notify the cafeteria owner. Use case involved: -Request leftover include notify cafeteria owner -Accept leftover include notify students - View leftover request	Cafeteria owner Student
3.	Cafeteria owners and students can use this account. They can sign up to the system and can update their information.	Both Cafeteria Owner and students can create or update their accounts. Use case involved - Create account - Update account - View account	Cafeteria owner Student
4.	The staff faces difficulties in knowing the status of the food once it is taken by the students and brought to the college. They also felt bad about asking students to share the food in the telegram group.	The system should allow tracking of the leftover food Use case involved: -Food pickup status update include notify cafeteria owner	Student

Table 4.2 presents 11 use cases derived from stakeholder interviews. Each use case was crafted to connect the specific issues stakeholders identified to

targeted solutions for the system's implementation. They are designed with careful attention to address specific challenges and complexities highlighted during these discussions. This structured approach not only aims to meet stakeholder needs but also supports development by having an adaptable system which capable of handling the various issues raised.

4.1.4 Use Case Diagram

Based on the analyzed requirements, the use case diagram has been created. So, the figure 4.1 has shown the use case diagram for this project.

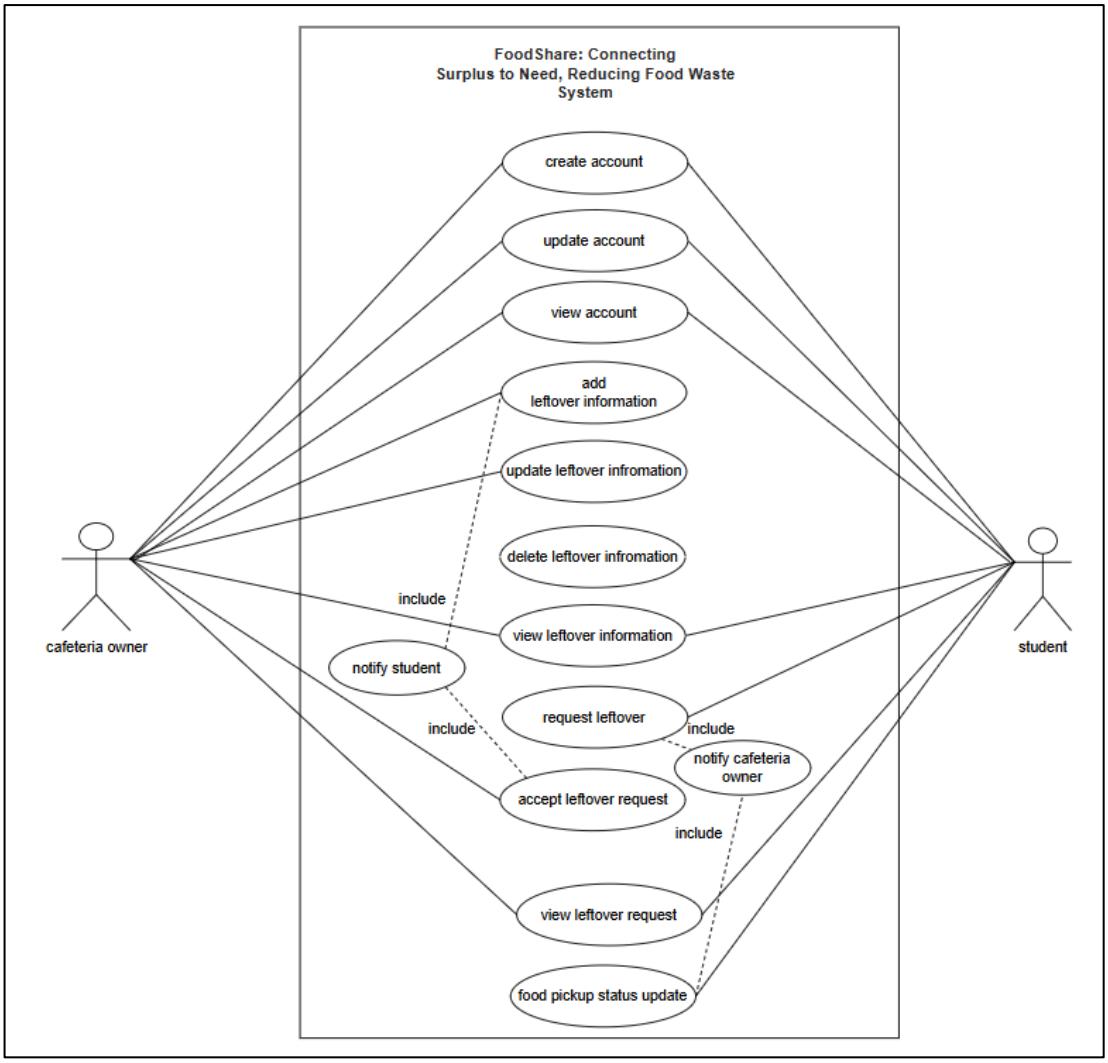


Figure 4.1 Use case diagram for FoodShare System

4.1.5 Domain class diagram

The class diagram for the FoodShare system (Figure 4.2) provides an overview of the key entities, their attributes, and relationships, highlighting how the

system facilitates surplus food sharing to reduce waste. It includes five main classes: Cafeteria Owner, Student, Leftover, Request, and Pickup, each representing critical system components.

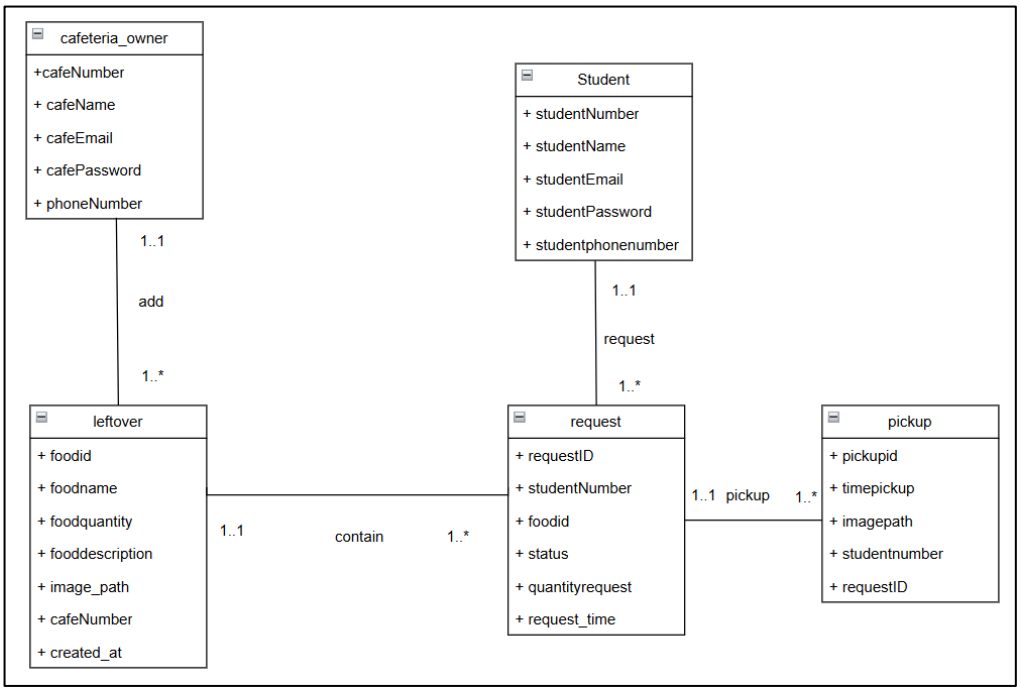


Figure 4.2 Domain class diagram for FoodShare System

4.1.6 Activity Diagram

During the analysis phase, an activity diagram is a valuable tool used to visually represent the flow of actions, tasks, and processes within a system or a specific procedure. It provides a high-level overview of how different activities are connected, showing the sequence and dependencies between them.

Figure 4.2 presents the step-by-step process for a student to log in and complete the registration, as well as the interactions between the three main entities: the Cafeteria Owner, the System, and the Student. This system is designed to efficiently manage leftover food information and student requests.

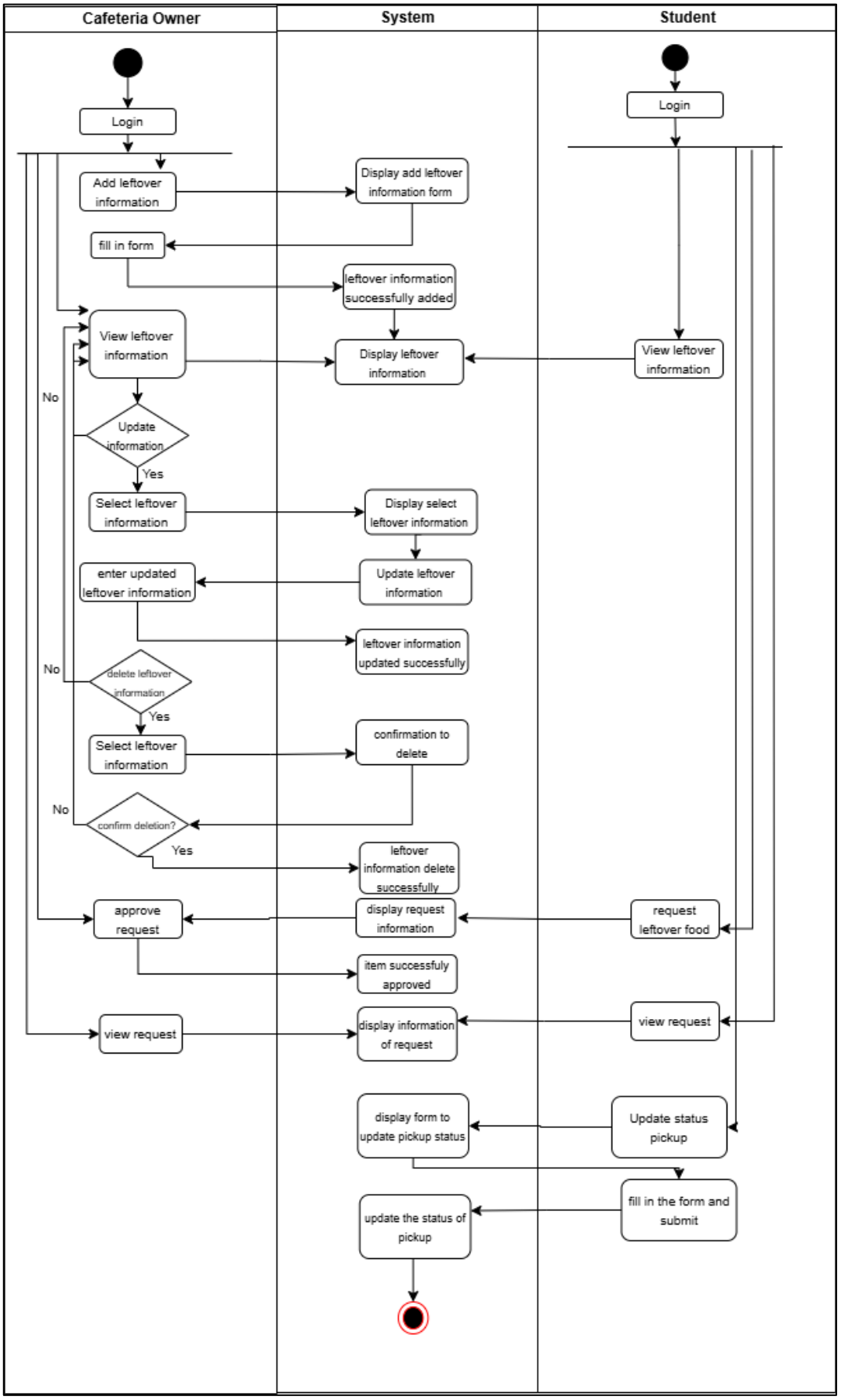


Figure 4.3 Activity diagram for FoodShare system

On the student side, the process begins with logging into the system, followed by viewing available leftover food information. Students can request leftover food by submitting a request through the system, which then displays the request details. Additionally, students can create new orders by filling out and submitting a form provided by the system. The system processes these orders and allows students to view the status of their requests.

The system serves as an intermediary, ensuring smooth communication and processing between the Cafeteria Owner and the Student. It facilitates the display and management of forms, updates, and requests, while also confirming successful operations such as adding, updating, and deleting information. This structured process enhances the efficiency of managing leftover food information and responding to student requests.

4.1.7 System Sequence Diagram

A system sequence diagram (SSD) illustrates and documents how a proposed system interacts with external entities, including users and other systems.

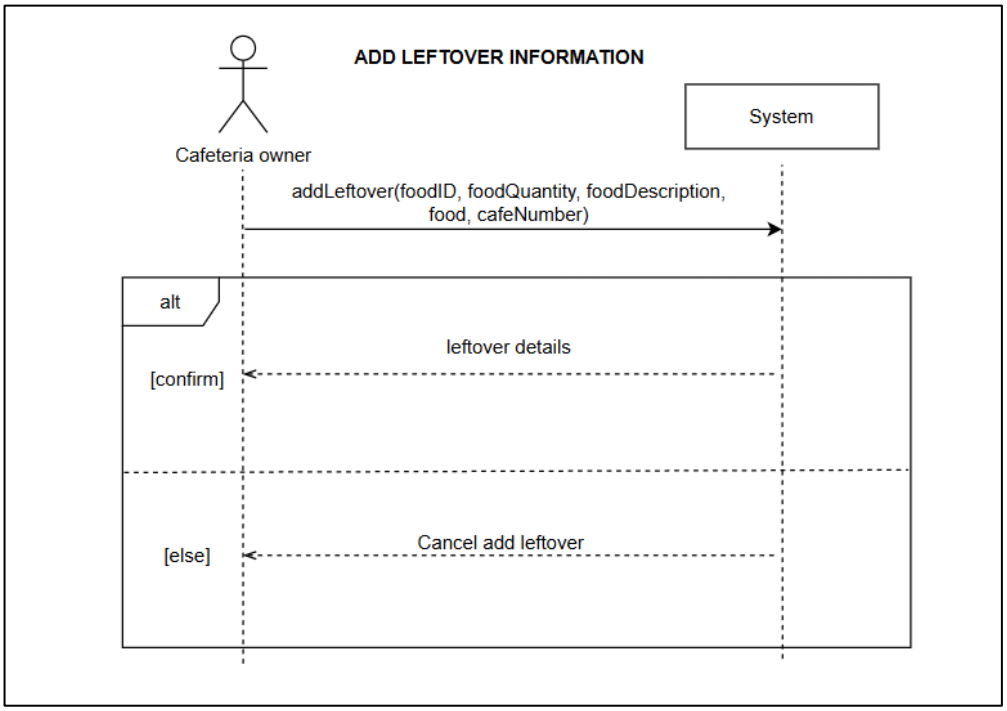


Figure 4.4 SSD for add leftover information

Figure 4.4 illustrates the System Sequence Diagram (SSD) for the Add Leftover Information use case, showing how the cafeteria owner interacts with

the system to input leftover food details such as foodID, foodQuantity, foodDescription, and cafeNumber. The system processes the input and provides two possible outcomes: if the details are confirmed, the leftover information is successfully added; otherwise, the operation is canceled. Additional SSD diagrams for other use cases are available in the System Requirements Specification (SRS) document in Appendix B.

4.2 Findings on Design Phase

The design phase is a crucial step in the Waterfall model that comes after the analysis phase and before implementation. The system's direction is greatly influenced by the design phase making certain that the software is created in an organised and systematic approach.

4.2.1 Storyboard of the user interface

A system storyboard visually represents the user interface, interactions, and functionality of a software system or application. It acts as a guide for developers and designers, detailing the steps required to complete tasks and providing a clear overview of user interactions with the system. The process begins with planning and drafting the user interface layout, followed by iterative refinements based on user feedback and usability testing. Figures 4.5 and 4.6 illustrate the storyboard's progressive development stages for the primary system.

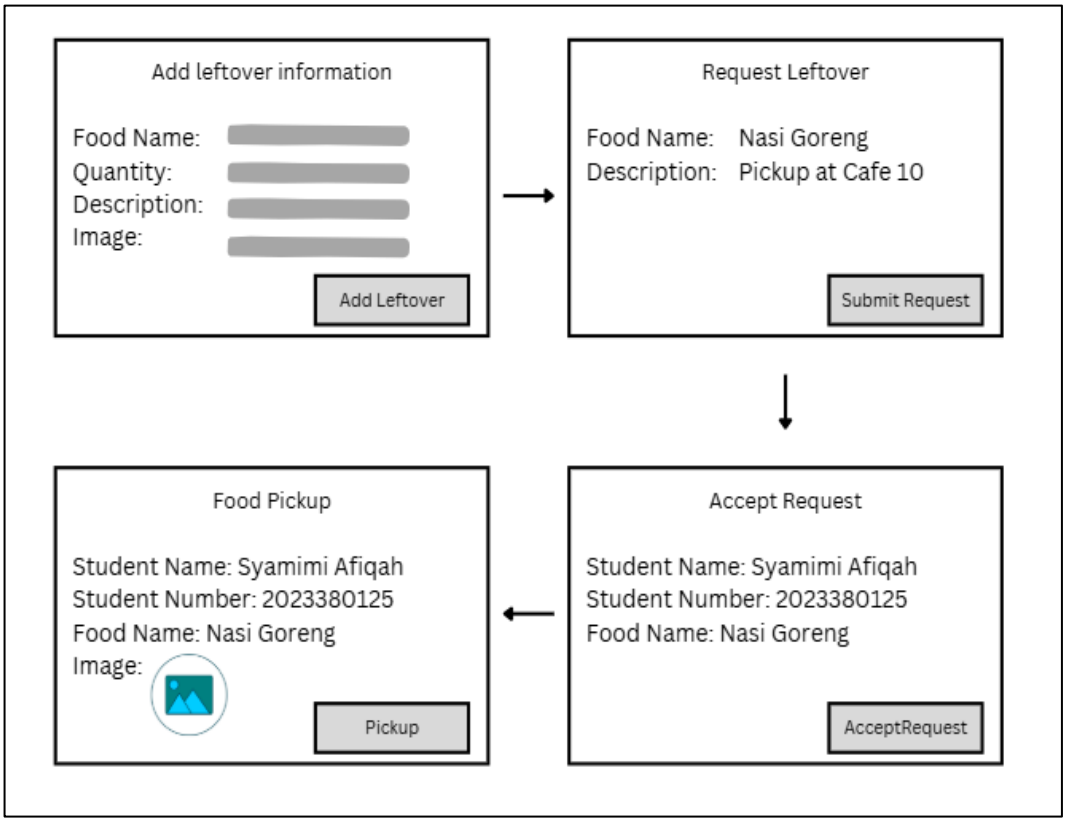


Figure 4.5 Storyboard for the process of adding leftovers until pickup leftovers.

Figure 4.5 shows the storyboard image illustrates the user interface of a FoodShare system designed to manage leftover food between cafeteria owners and students.



Figure 4.6 Storyboard for food booking

Figure 4.6 shows the Food Booking storyboard where users can create a new food order for an event.

4.2.2 Design System (Design Class Diagram)

The diagram presents the FoodShare: Connecting Surplus To Need, Reducing Food Waste class structure. The systems design presents the essential components of the system alongside their relationship patterns.

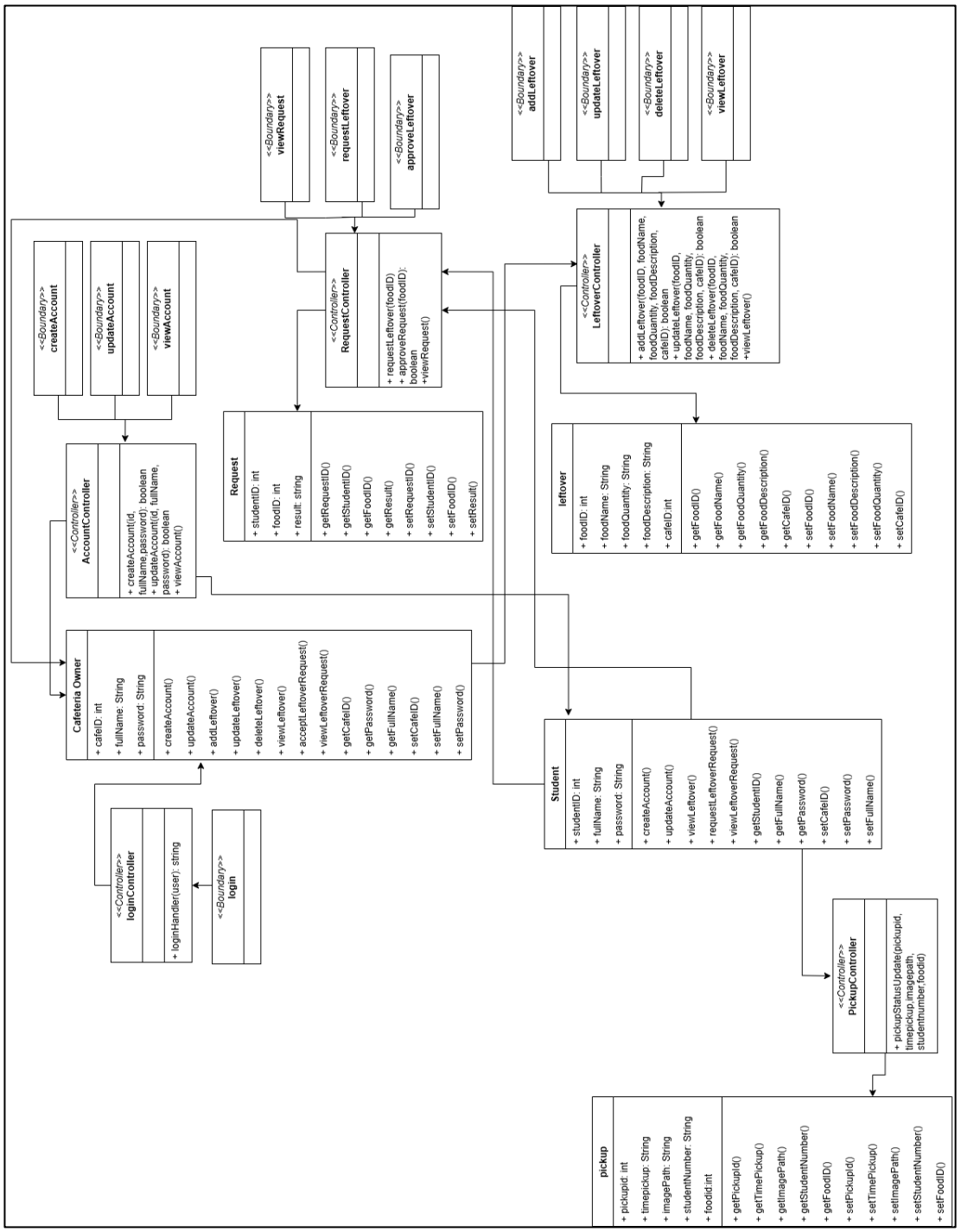


Figure 4.6 Design Class Diagram for FoodShare

4.2.3 Database Design (Entity Relationship Diagram)

This Entity Relationship Diagram (ERD) illustrates the relationships and cardinality among the main entities in the FoodShare system: Cafeteria Owner, Student, Leftover, Request, Booking, and Pickup. Each entity is represented as a table with attributes, including primary keys (PK) and foreign keys (FK), which facilitate the connection between tables and ensure data integrity.

This Entity Relationship Diagram (ERD) illustrates the relationships and cardinality among the main entities in the FoodShare system: Cafeteria Owner, Student, Leftover, Request, Booking, and Pickup. Each entity is represented as a table with attributes, including primary keys (PK) and foreign keys (FK), which facilitate the connection between tables and ensure data integrity. Figure 4.7 demonstrates the links between different tables, highlighting the use of primary keys and foreign keys to define relationships between them.

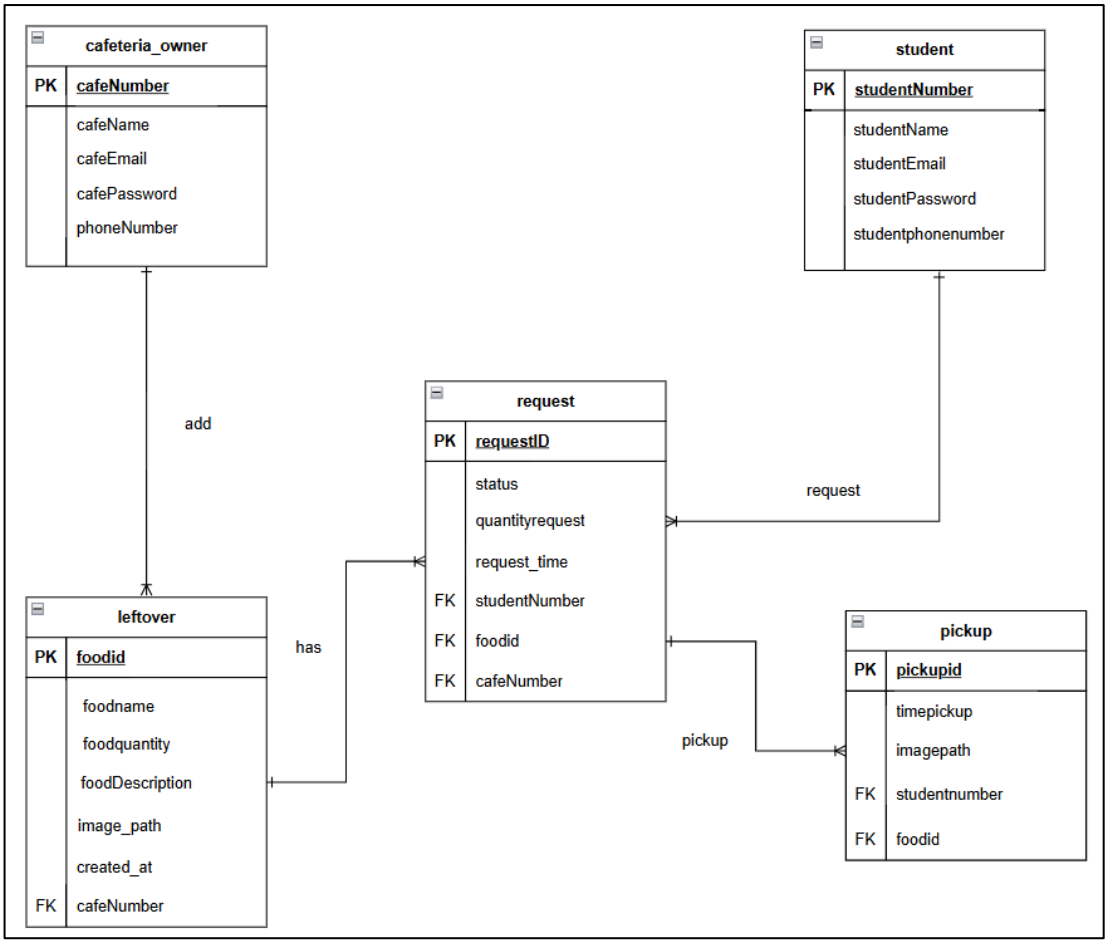


Figure 4.7 Entity Relationship Diagram

In summary, the ERD clearly defines the cardinality and relationships among the entities, ensuring an organized and structured connection within the FoodShare system. This design supports efficient management of leftover food, student requests, and pickups, thereby promoting effective food distribution and waste reduction.

4.2.4 Multilayer Sequence Diagram

Multilayer sequence diagrams are powerful tools for analyzing, designing, and communicating about complex systems with multiple interacting parts. These diagrams allow for a layered view of data and control flows within the system, making it easier to understand how different components connect and work together. By visualizing interactions across layers, multilayer sequence diagrams help translate design ideas into functional parts.

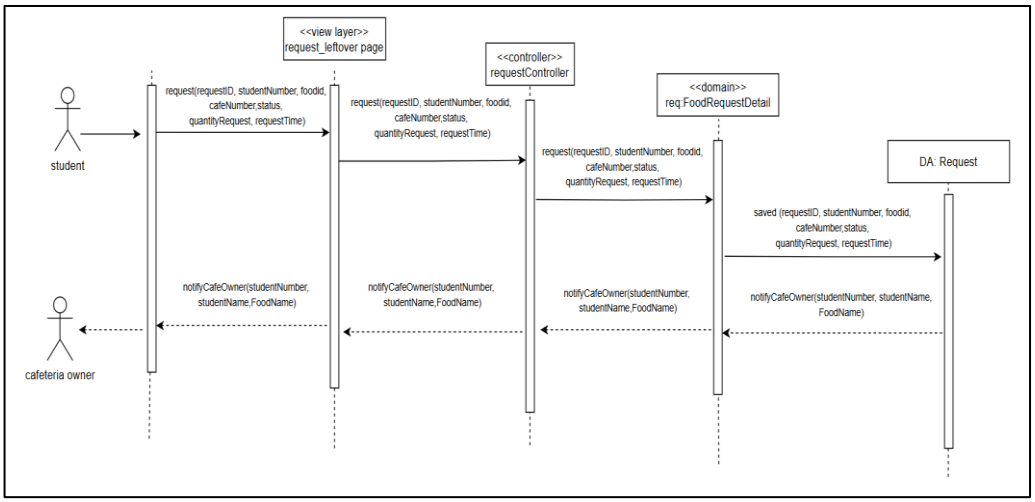


Figure 4.8 Multilayer sequence diagram for add leftover use case

Figure 4.8 presents the multilayer sequence diagram for adding leftover food by the cafeteria owner. This sequence diagram illustrates the process of a student requesting leftover food from the cafeteria owner through a multi-layered system architecture. The interaction begins when the student initiates a request via the request_leftover page in the view layer, providing details such as the requestID, studentNumber, foodID, cafeNumber, status, quantityRequest, and requestTime.

The communication flow is structured and adheres to a clear separation of concerns between the layers: the view layer handles user interaction, the

controller layer coordinates the process, the domain layer manages business logic, and the data access layer ensures data storage and retrieval. This layered design promotes modularity, reusability, and maintainability of the system.

4.2.5 System Design Diagram

The system design diagram for the FoodShare system provides a visual representation of how various components of the system interact with each other. It outlines the key functionalities, such as managing leftover food information, handling user requests, and notifying both cafeteria owners and students. The diagram illustrates how the cafeteria owner can add, update, or delete leftover food details, while students can request food and track the status of their requests. Additionally, the diagram highlights the flow of information between users and the system, ensuring that communication and food distribution are managed efficiently. This design serves as a blueprint for the development and implementation of the FoodShare platform, ensuring that all required features are clearly defined and organized.

4.3 Implementation Phase

The final phase of the Waterfall Methodology is the Implementation stage, where the system is fully developed and continuously refined over time. During this phase, various features and components are constructed as the system is gradually assembled and enhanced. Key activities include writing the actual code and setting up a database to store essential data, ensuring the system is both functional and ready for use.

4.3.1 User Interface of the System

In the initial phase, the system's user interfaces are developed based on the FoodShare system’s use case diagram, with each interface aligned to its specific use cases. The user interface plays a vital role as it bridges the interaction between users and the system, acting as a medium of mutual understanding. It is the main point of access for users to interact with and

utilize the system's features. The design of each interface adheres to the previously created storyboard, ensuring visual and functional consistency throughout.

Additionally, user feedback and usability principles are incorporated into the design, enhancing the overall user experience. This iterative refinement process is focused on meeting user needs and creating a more intuitive system. Subsequent phases will involve usability testing and further adjustments based on feedback, ensuring that any issues are identified and resolved, resulting in a more polished and user-friendly system. Below are the interfaces that will be showcased:

4.3.1.1 Use case: Create Account Cafeteria owner

Figure 4.11 shows the registration interface designed for cafeteria owners. Each interface includes fields for entering essential information such as a unique identifier such as Cafe Number for cafeteria owners, Name, Email, and Password.

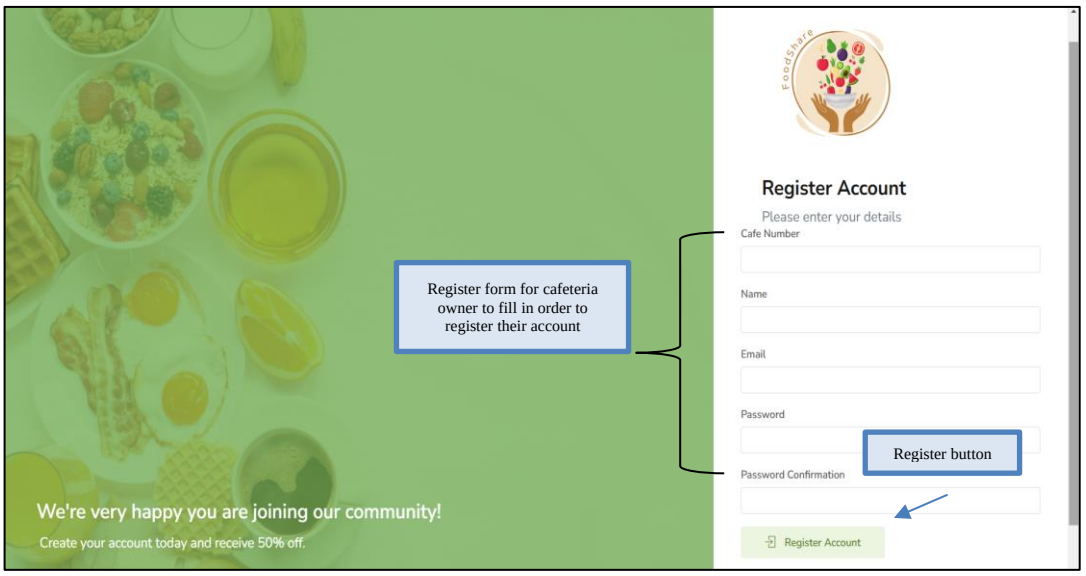


Figure 4.10 Register interface for cafeteria owner

4.3.1.2 Use case: Manage Account Cafeteria owner

Figure 4.12 shows the user interface where the cafeteria owner can view and update their account information. This interface allows the owner to check

and modify important details about their cafeteria. The information displayed includes the Cafe Number, a unique identifier for the cafeteria. Cafe Name, which is pre-filled but can be changed if needed and the Email address, which the owner can update if a different contact email is required. After making any changes, the owner can click the Update button to save the updates. This section of the interface is designed to provide the cafeteria owner with an easy way to manage and maintain their account information accurately.

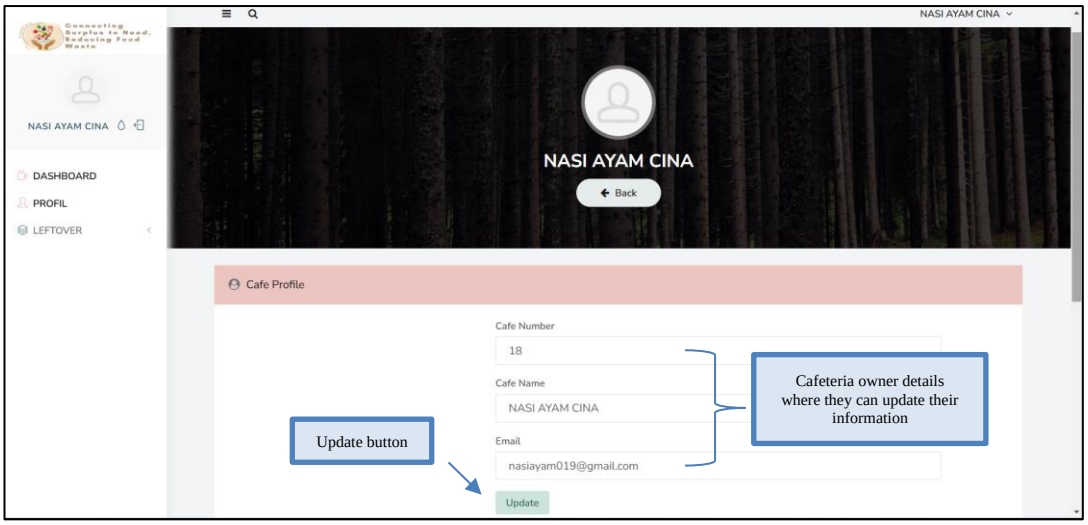


Figure 4.12 Manage Account for cafeteria owner

4.3.1.3 Use case: Create Account Student

Figure 4.13 shows interface shows the registration age for students that has not been registered in Foodshare system.



Figure 4.13 Register interface for student

Through this interface users can start building new accounts by providing their

individual information. Having completed account registration students must enter their Student Number along with Name and Phone Number as well as Email Address. Students must make a Password selection before entering it twice into the Confirmation Password box. By entering their information into all required fields students can move forward with account registration by clicking the Register Account button. Students must complete their data inputs on this page so the system can establish their account.

4.3.1.4 Use case: Manage Account Student

Through Figure 4.14 students can visualize their profile details and edit them for updates. The Student Number along with Name and Email and Phone Number details appear on the current page for viewing. Students can verify the accuracy and currency of their information using this interface. Students can edit required fields on the screen then click Update for their profile details changes to save.

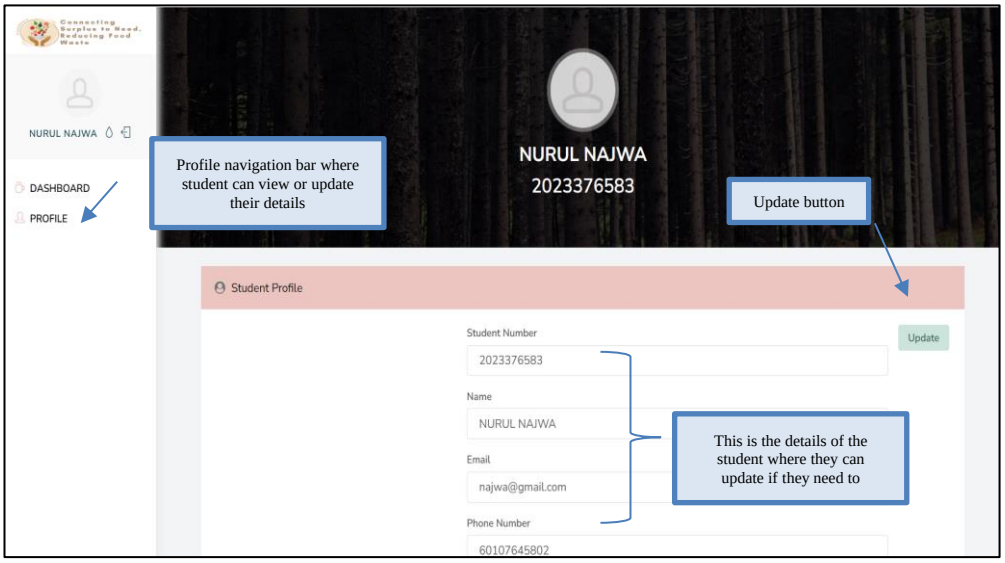


Figure 4.14 Update and view interface for student

4.3.1.5 Use case: Add Leftover information

The Figure 4.15 interface enables cafeteria owners to transmit information regarding leftover food. Using this interface shown in Figure 4.15 Food Name along with Food Quantity Food Description and Food Image upload enable the owner to supply information about available leftovers. After completing all required information input the cafeteria owner uses the Add

Leftover button to send their submission to the system.

The system delivers an automatic notification about added leftovers through the owner's registered phone number shown in Figure 4.15's right section. The system notifies students through Whatsapp notifications providing them with essential details about the remaining food inventory such as names and quantities and descriptions. Students can request the available food through this system before the items disappear.



Figure 4.15 Add leftover

4.3.1.6 Use case: Manage leftover information

Manage leftover information gives the cafeteria owner the ability to modify or remove stored information about leftover foods through Figure 4.16.

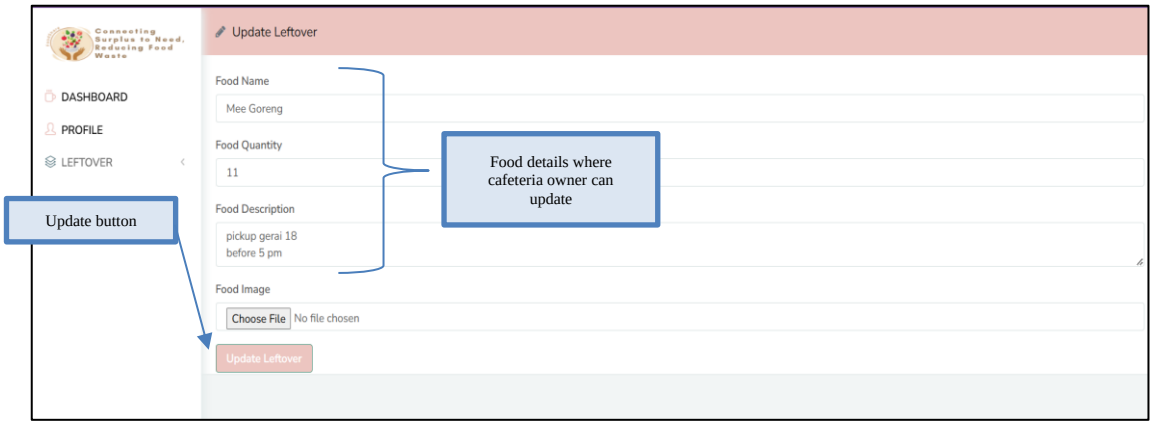


Figure 4.16 Update leftover interface for cafeteria owner

After making necessary changes the owner can press Update Leftover to save the fresh information. Through this feature the cafeteria owner maintains control over dynamic leftover food information management for

real-time accurate system updates as shown in Figure 4.16

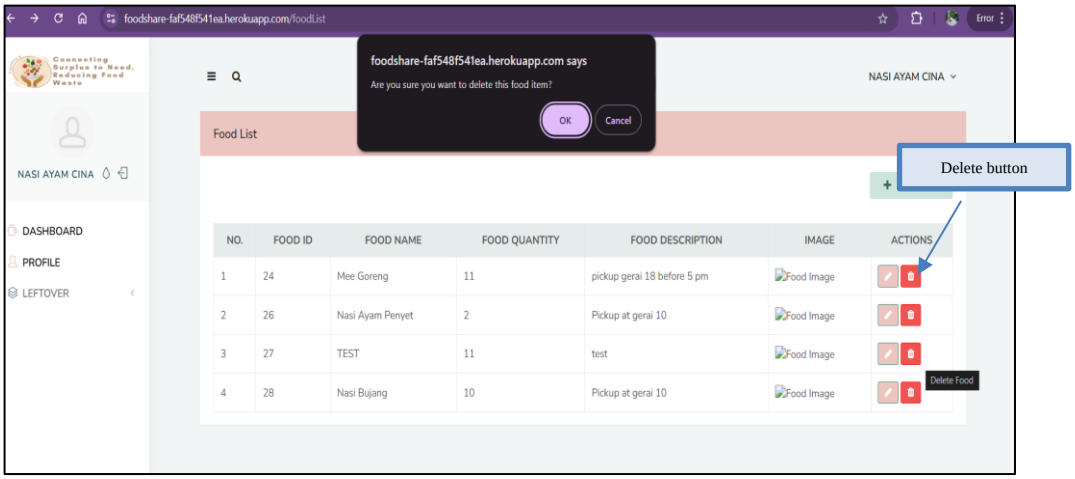


Figure 4.17 Delete leftover interface for cafeteria owner

Figure 4.17 shows that by clicking the delete button, a prompt of confirmation will appear to confirm the deletion of the leftover food.

4.3.1.7 Use case: View leftover information

Cafeteria Owner can access Figure 4.18 to explore a list of leftover food items which were added by the cafeteria owner. The displayed food list contains important entry information including Food ID together with Food Name while presenting Food Quantity information along with Food Description and including an image of the food. Each food entry contains an interface with both Update and Delete action options.

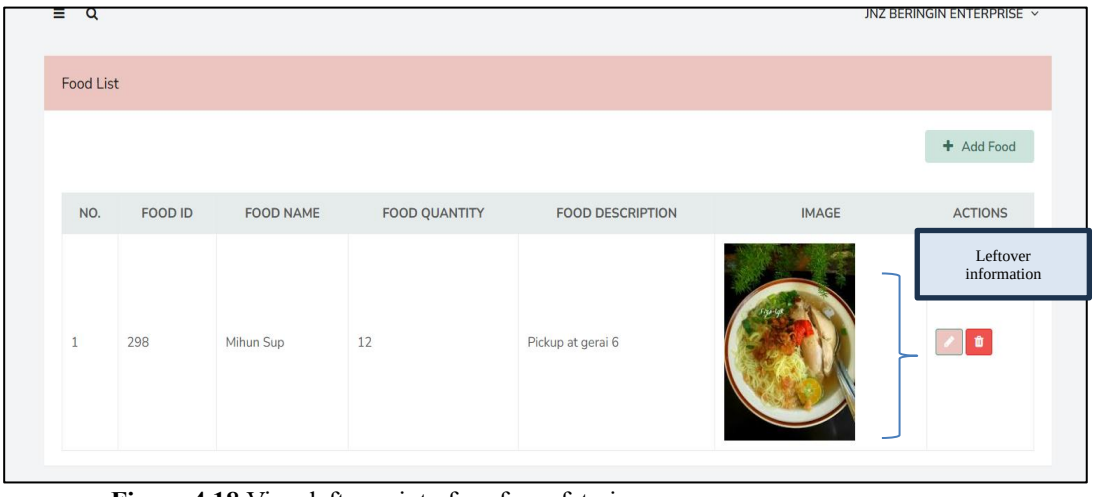


Figure 4.18 View leftover interface for cafeteria owner

The example food entry displayed in Figure 4.17 shows "Mihun Sup" as Food

ID 298. Food leftovers from today amount to 12 portions. A user can collect the food they ordered at "gerai 6," stall number 6 according to the Food Description. Users gain visual insight into the Mihun Sup by viewing an associated image. The cafeteria owner can edit any data through the pencil icon Edit button while the trash can icon Delete button helps remove entries.

4.3.1.8 Use case: Request Leftover, Accept leftover Request, View leftover request

Figure 4.19 shows the interface where students can view available leftover food, including details such as the food name, quantity, and description. When a student decides to request a specific leftover, they click the "Request" button, which prompts them to confirm their selection. Once confirmed, the request is submitted, and the cafeteria owner receives a notification containing the student's name, ID, and details of the requested item. The cafeteria owner then reviews the request through the "Food Request" interface, where they can choose to accept or reject it. If the request is accepted, the student receives a notification confirming the approval, along with pickup instructions. If the request is rejected, the student is notified with a message explaining the denial. This structured process ensures smooth communication and clarity for both parties.

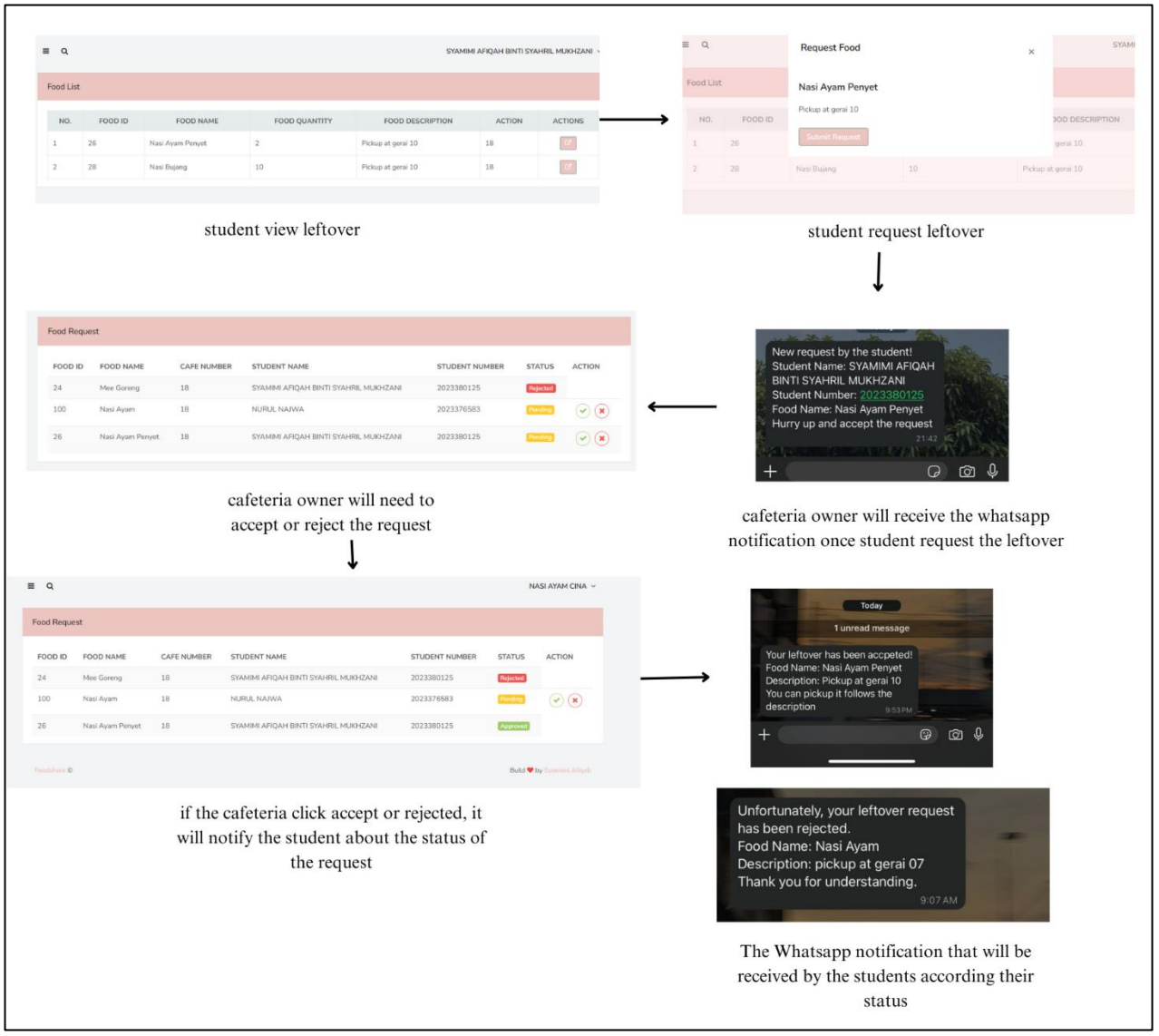


Figure 4.19 View leftover interface for cafeteria owner

4.3.1.9 Use case: Food pickup status update

In this section, students are required to provide evidence to confirm that they have received the leftover food. The pickup page will only appear if the request has been accepted by the cafeteria owner. If a student has not made a request or if their request has not been accepted yet, an alert will be displayed indicating that they are not eligible for pickup. Figure 4.20 illustrates the interface of the alert box that pops up.

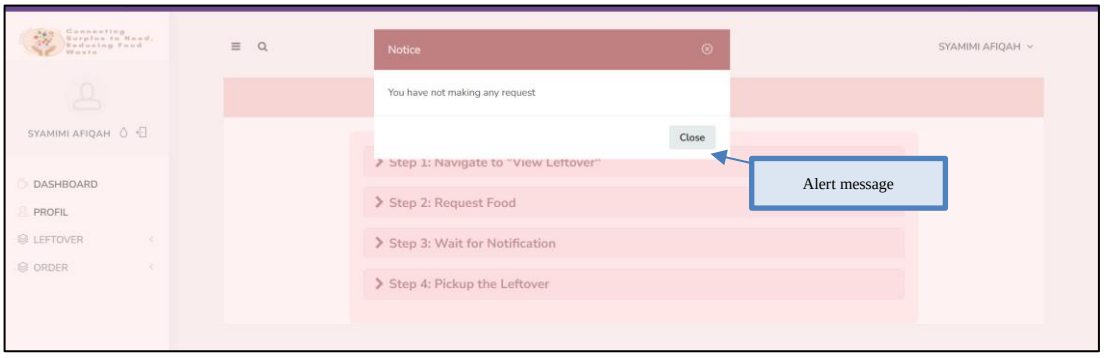


Figure 4.20 Alert box

Figure 4.21 shows the form that students need to fill out to confirm that they have successfully received the food.

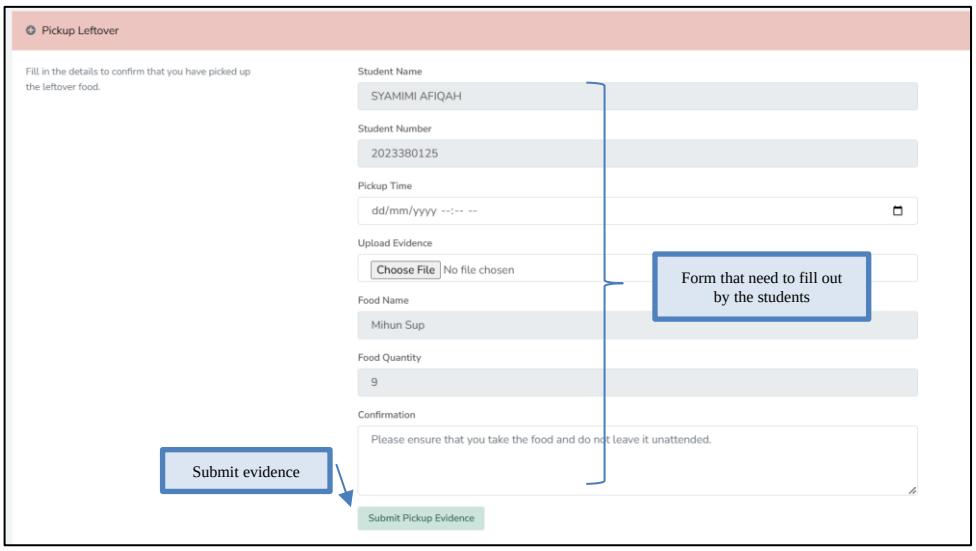


Figure 4.21 Pickup leftover interface

4.4 Summary

The project's outcomes and conclusions, which are covered in detail in this chapter, were made possible by the requirements gathering, requirement analysis, design, and implementation phases in the preceding chapter. The chapter featured screenshots of the user interface along with insights into database design, management system implementation, and integration with the overall system.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

This chapter provides conclusion for the project with an overview of the FoodShare: Connecting Surplus to Need, Reducing Food Waste System. The project's limitations and recommendations for future enhancement have been thoroughly addressed and elaborated upon.

5.1 Conclusion

The main objective of this project was to develop the FoodShare: Connecting Surplus to Need, Reducing Food Waste System, and it has been proven to be a success as all three project objectives were successfully accomplished. These objectives included gathering data and analyzing the requirements of the FoodShare system, designing the system, and ultimately developing the system. Each of these objectives was achieved through a structured and methodical approach.

In conclusion, this study, alongside the implementation of the FoodShare: Connecting Surplus to Need, Reducing Food Waste System, has effectively fulfilled its primary aims and goals. By collecting an impressive depth of knowledge, identifying an extensive list of requirements, and meticulously designing the system down to the smallest detail, it has been possible to accomplish the project’s intended objectives successfully. Furthermore, the project serves as a testament to its significant impact on improving the distribution process of leftover food while also contributing to the efficient operation of the business in the future.

5.1.1 Objective 1: To gather data and analyze requirements of FoodShare: Connecting Surplus to Need, Reducing Food Waste System.

This objective was achieved during the requirement gathering and analysis phase, in which a total of four activities were completed, resulting in the successful achievement of this objective. Before interviewing the stakeholders, questions for the interviews were developed very carefully. In this way, the business activities and problems were explored with the help of the face-to-face and WhatsApp interviews with the stakeholders. Taking record of the interview data, the synchronized texts for the purpose of future reference in the form of transcripts were prepared. All of these efforts resulted into the positive conclusion of all the activities in the first phase and presentation of the Software Requirements Specification (SRS) enclosed in the appendix B.

5.1.2 Objective 2: To design the system of FoodShare: Connecting Surplus to Need, Reducing Food Waste System.

During this phase, the primary goal is to assess the obtained data and requirements before developing the proper system architecture. Data from interviews were collected before being evaluated and translated into applicable requirement diagrams such as the Use Case Diagram, Domain Class Diagram, and Activity Diagram. The deliverables were then organized and documented in SRS. Finally, case-based reasoning knowledge has been obtained. Diagrams such as the Multilayer Sequence Diagram (MLSD) and Design Class Diagram (DCD) are created throughout the design process to demonstrate the requirements that have been received. The storyboard was created to help designers and engineers understand how the system operates. After the system's designs are completed, a Software Design Document (SDD) is prepared. The second aim was met by producing and completing all of the deliverables during this period. The deliverable for this objective, Software Design Specifications (SDD), is included in Appendix C.

5.1.3 Objective 3: To develop a system of FoodShare: Connecting Surplus to Need, Reducing Food Waste System.

Objective 3 aims to create a FoodShare: Connecting Surplus to Need, Reducing Food Waste System, a centralized management platform specifically targeted at UiTM Jasin students and cafeteria owners. This system is designed to address the challenges of food waste and food insecurity by streamlining the redistribution of surplus food. It will simplify the process for cafeteria owners to share leftover food with students who need it and reduce reliance on inefficient manual communication methods like Telegram groups. Additionally, the platform includes a booking feature for event organizers, enabling them to coordinate large food orders with cafeteria owners more effectively. The system will optimize the food-sharing process by providing real-time notifications about food availability, ensuring equal access among students, and minimizing waste.

5.2 Project Strength

The FoodShare: Connecting Surplus to Need, Reducing Food Waste System demonstrates several key strengths that enhance its effectiveness and impact within UiTM Jasin's community. Firstly, the system improves accessibility and transparency by providing a centralized platform for managing surplus food distribution. This ensures that both the owners of the cafeterias and the student gets real time information on foods that are likely to be left over. This makes the process transparent and thus holds FoodShare as well as the stakeholders responsible for ensuring that food redistribution is done efficiently across the UiTM Jasin.

Additionally, FoodShare reduces the inefficiencies and risks associated with the current manual processes of sharing surplus food. Notification and distribution are done within the platform thereby avoiding communication breakdown that leads to delays, wastage, or unfair share. The reliability of food surplus makes it possible to have a reliable system to support cafeteria owners in minimizing food waste and at the same time,

ensuring that students experiencing hunger have sufficient food support in UiTM Jasin.

Finally, FoodShare encourages proper storage and categorization of food surplus instead of using unregulated informal means such as Telegram groups. Thus, the system consolidates information and normalizes processes to minimize overlapping activities and achieve a more orderly division of food. This approach not only reduces time and costs but also promotes sustainability throughout the campus. Consequently, UiTM Jasin can respond more adequately to food issues, increase utilization, and encourage consumers and cafeteria owners to contribute more toward sustainability.

5.3 Limitations

The FoodShare: Connecting Surplus to Need, Reducing Food Waste System, while effective in addressing food surplus management, has some limitations that can be further improved to enhance its functionality and usability:

i) Lack of Mobile Application Support

The current system is web-based, which may limit its accessibility and convenience for users who prefer to interact with the system on-the-go. A mobile application would allow both cafeteria owners and students to receive notifications, request food, and manage leftover information more easily from their smartphones. This would enhance user engagement and ensure that the system is accessible anytime, anywhere, providing greater flexibility and improving real-time responses.

ii) Limited Scalability

The current system may struggle to handle a large number of users or food items, particularly in larger institutions or across multiple locations. As the number of cafeteria owners, students, and food transactions grows, the system may experience slower performance or even downtime. To address this, the system should be optimized for

scalability, allowing it to efficiently manage increased demand without compromising speed or reliability. Implementing cloud-based infrastructure or a more robust database could help improve the system's ability to scale and accommodate future growth.

iii) No Integration with External Donation Platforms

This system is implemented in UiTM Jasin only but there is no option of connecting it with the other food donation platforms. This restricts its capability to link to other bigger food donation networks that should make the delivery of food items get to even more of the targeted recipients.

5.4 Recommendation for Future Enhancement

To enhance the project, the following improvements and additional features are recommended:

i) **Mobile Application Support**

Expanding the system to include a mobile application would make the platform more accessible to users on-the-go. Both cafeteria owners and students could easily interact with the system, receive notifications, manage food requests, and track food availability directly from their smartphones. A mobile app would enhance user engagement and offer greater convenience, ensuring that the system remains effective in various situations and contexts.

ii) Scalability Improvements

To ensure the system can handle increased demand and grow with the needs of different institutions, implementing scalability enhancements is essential. Transitioning to cloud-based infrastructure or upgrading the database management system would ensure that the platform can accommodate a growing number of users and transactions without compromising performance. This enhancement would allow the system to support large-scale operations, ensuring sustainability as the

platform expands.

iii) Integration with Other Donations Platform

Further expand the ability of the system by connecting it to public food donation systems. This would allow food waste not only within the UiTM Jasin area only but other communities too and help save more food from wastage.

5.5 Summary

This chapter highlighted the assessment of the realization of three of the objectives of the project. Outsourcing also had its restrictions explained with recommendations on how the project could be done in the future to enhance the results. In the future, the system may be developed and the efficiency of the system may be used maximizing the use of constraints related to the project. At the same time, it is possible to consider that the recommendations could act as guidelines for the further development of the system.

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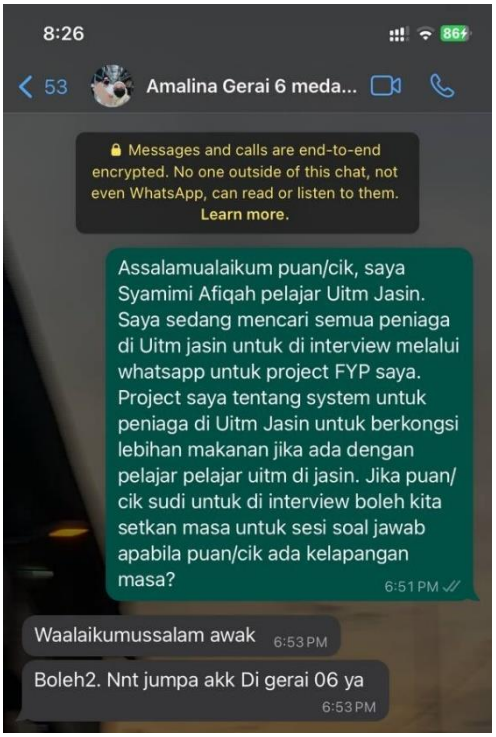
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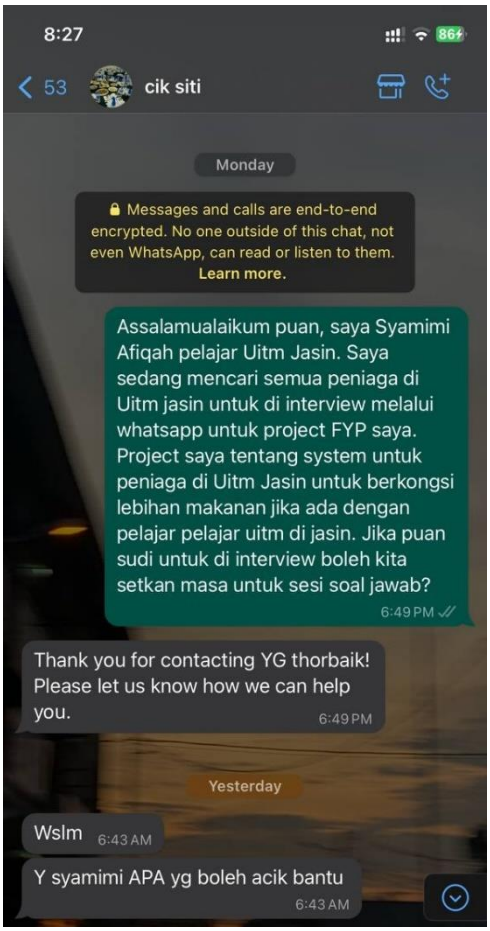
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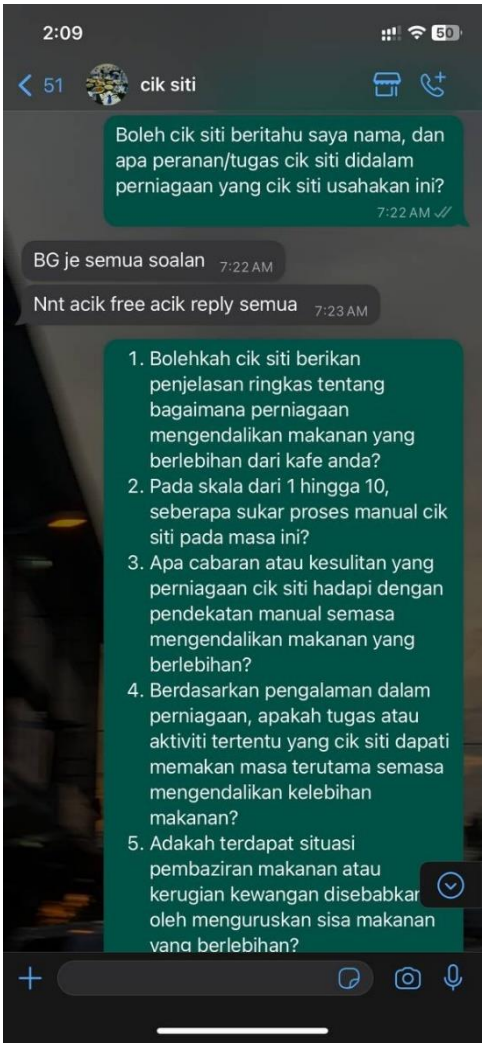
APPENDIX A

No	Question	Answer
1.	May I have a brief explanation regarding how you handle if there is leftover food from your cafe?	If there are some leftovers that we cannot save, we usually ask random students from the cafeteria to take the food if they want it. If they do not want it, usually we will ask for help from them to bring it to the college to distribute it to the students. There are times when there are no students left at the cafeteria, so we bring it home to give out to the mosque or eat by ourselves.
2.	On a scale of 1 to 10, how difficult is your manual process currently?	7
3.	What challenges or difficulties do you encounter with the current manual approach to handling leftover food?	It is hard to find students to distribute the food. Sometimes, we felt bad for asking for help since they needed to share in the telegram group. We also did not know the status of the food once the students brought it to the college.
4.	Based on your experience, what specific tasks or activities do you find particularly time-consuming while distributing the surplus?	Communicate with students to take the food.
5.	Have there been any instances of food wastage or financial losses due to the current process of distributing leftovers?	No, since we only give out what cannot be kept for the next day, there will be no financial losses since we think it's about charity.
6.	In your opinion, what improvements or enhancements could be made to streamline the process of handling leftover food?	If students can quickly know there is leftover food from the cafeteria, they can just come and take it.
7.	Do you see any benefits of having this system that could improve the process of handling surplus?	Yes, it will help us distribute our leftover food efficiently if we have any.
8.	If there is a system, what specific features or functionalities would you expect from a computerized system designed to handle leftover food in the cafe?	• we can share the food information that we have

		• update if there is someone who has already taken it
9.	What measurable outcomes or benefits would you hope to achieve by implementing a computerized system for leftover food management, such as cost savings, improved inventory management, or reduced environmental impact?	We hope that we can help the students who are in need. At least the food will not just go wasted.
10	If there's an opportunity to solve the problem of handling leftovers with students by proposing the FoodSharing system, will you accept it as a stakeholder?	Yes absolutely.


8:26
53
Amalina Gerai 6 meda...
Messages and calls are end-to-end encrypted. No one outside of this chat, not even WhatsApp, can read or listen to them. Learn more.
Assalamualaikum puan/cik, saya Syamimi Afiqah pelajar Uitm Jasin. Saya sedang mencari semua peniaga di Uitm jasin untuk di interview melalui whatsapp untuk project FYP saya. Project saya tentang system untuk peniaga di Uitm Jasin untuk berkongsi lebihan makanan jika ada dengan pelajar pelajar uitm di jasin. Jika puan/cik sudi untuk di interview boleh kita setkan masa untuk sesi soal jawab apabila puan/cik ada kelapangan masa?
6:51 PM
Waalaikumussalam awak 6:53 PM
Boleh2. Nnt jumpa akk Di gerai 06 ya
6:53 PM


8:27
53
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Monday
Messages and calls are end-to-end encrypted. No one outside of this chat, not even WhatsApp, can read or listen to them. Learn more.
Assalamualaikum puan, saya Syamimi Afiqah pelajar Uitm Jasin. Saya sedang mencari semua peniaga di Uitm jasin untuk di interview melalui whatsapp untuk project FYP saya. Project saya tentang system untuk peniaga di Uitm Jasin untuk berkongsi lebihan makanan jika ada dengan pelajar pelajar uitm di jasin. Jika puan sudi untuk di interview boleh kita setkan masa untuk sesi soal jawab?
6:49 PM
Thank you for contacting YG thorbaik! Please let us know how we can help you.
6:49 PM
Yesterday
Wslm 6:43 AM
Y syamimi APA yg boleh acik bantu
6:43 AM


2:09
51
cik siti
Boleh cik siti beritahu saya nama, dan apa peranan/tugas cik siti didalam perniagaan yang cik siti usahakan ini?
7:22 AM
BG je semua soalan 7:22 AM
Nnt acik free acik reply semua 7:23 AM
1. Bolehkah cik siti berikan penjelasan ringkas tentang bagaimana perniagaan mengendalikan makanan yang berlebihan dari kafe anda?
2. Pada skala dari 1 hingga 10, seberapa sukar proses manual cik siti pada masa ini?
3. Apa cabaran atau kesulitan yang perniagaan cik siti hadapi dengan pendekatan manual semasa mengendalikan makanan yang berlebihan?
4. Berdasarkan pengalaman dalam perniagaan, apakah tugas atau aktiviti tertentu yang cik siti dapati memakan masa terutama semasa mengendalikan kelebihan makanan?
5. Adakah terdapat situasi pembaziran makanan atau kerugian kewangan disebabkan oleh menguruskan sisa makanan yang berlebihan?

APPENDIX B



SOFTWARE REQUIREMENTS SPECIFICATION FOR FOODSHARE: CONNECTING SURPLUS TO NEED, REDUCING FOOD WASTE SYSTEM

Prepared by:
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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Initial SRS-Version 1.0	Syamimi Afiqah	Added SRS Template	09/08/24
Added Introduction-Version 1.0	Syamimi Afiqah	Added and checked all the introductions well explained.	09/08/24
Added Use case diagram-Version 1.0	Syamimi Afiqah	Added Use Case Diagram and checked it.	10/08/24
Added Use case description-Version 1.0	Syamimi Afiqah	Added Use Case Description.	10/08/24
Added other Non-Functional Requirements-Version 1.0	Syamimi Afiqah	Added and checked non-functional requirements.	10/08/24
Added Project Description-Version 1.0	Syamimi Afiqah	Added and checked project description.	10/08/24
Added Class Diagram-Version 1.0	Syamimi Afiqah	Added and checked Class Diagram.	13/08/24
Added Business Activity Diagram-Version 1.0	Syamimi Afiqah	Added and checked Business Activity Diagram.	13/08/24
Added System Sequence Diagram-Version 1.0	Syamimi Afiqah	Added and checked System Sequence Diagram.	13/08/24

1. Introduction

1.0 Document Purpose

This document specifies the software requirements for the item, including its revision or release number. To determine the product and its specific version or release number, further context or information is required. Regarding the scope of the item encompassed by this Software Requirements Specification (SRS). The information presented does not clearly specify which particular components of the system or subsystem are addressed by the SRS, unless more context is given. The document will explain the objectives and characteristics of the software, the interfaces it includes, its intended functionalities, the limitations it must adhere to, and its response to external stimuli. This document is designed for both the software's end users and developers.

1.1 Product Scope

FoodShare: Connecting Surplus to Need, Reducing Food Waste refers to a system that allows cafeteria owners at UITM Jasin to share their leftovers with students. This system simplifies the process for cafeteria owners by providing an alternative to discarding or distributing leftovers elsewhere. The software offers a user-friendly graphical interface, enabling users to perform various functions included in the FoodShare system where cafeteria owners can input details about available surplus, set pick-up times, and notify students about the availability of food. This streamlined process saves time and effort, making it an attractive option for those looking to manage their resources more effectively. The purpose is to guide developers in selecting a design that can accommodate a full-scale application. In summary, the FoodShare system not only assists cafeteria owners in effectively distributing leftovers but also plays a vital role in helping students who are in need. By reducing food waste and promoting resource-sharing, the system contributes to a more sustainable and inclusive campus environment.

1.2 Intended Audience and Document Overview

All stakeholders use the Software Requirements Specification (SRS) document for their respective tasks. Developers use the SRS to understand what needs to be created and how it should function. Project managers rely on it to plan and manage the project, including scheduling and resource allocation. Testers reference the SRS to determine what aspects need to be tested, evaluated, and verified. Documentation writers use the SRS to understand what must be documented and how it should be presented. This section provides an overview of the software and its functions, defining what the software must do and how it should work. It also discusses the constraints and requirements that the software must meet in terms of performance, security, and other factors. Additionally, this section covers the software's appearance, user interface, and its interactions with other systems. The document is intended for requirements engineers, programmers, developers, UI designers, system analysts, and project managers.

1.3 Definitions, Acronyms and Abbreviations

Table 1.1 Acronyms and Definition

ACRONYMS/ABBREVIATIONS	DEFINITION
UI	User Interface
DB	Database
CRUD	Create, Read, Update, Delete
SRS	Software Requirement Specification

1.4 Document Conventions

This document was created based on IEEE formatting requirements. It uses the followingconvention:

Table 1.2 Document Conventions

Convention	Description
Font Size	Font Arial with size of 14 is used for the headings and the samefont with size of 11 is used for other writings.
Bold	It is used for header highlighting and to emphasize some points.
Cross Reference	It is an online document with blue underlined text that indicates hyperlink that can be redirected to the attached material when it is clicked.
Screen Text	It identifies the text as read on the screen.
Spacing	Text should be single spaced and have 1-inch margins

1.5 References and Acknowledgments

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2 Overall Description

2.0 Product Perspective

At UITM Jasin, there are 13 distinct cafeterias, each of which operates independently and provides a distinctive selection of food and beverages. These cafeterias specialize in a wide variety of food options, including well-known dishes like nasi ayam and nasi lemak. Each cafeteria is committed to delivering exceptional meals that are tailored to the preferences and tastes of students, staff, and visitors. The cafeterias at UITM Jasin are instrumental in improving the campus food experience by dedicating themselves to quality and variety.

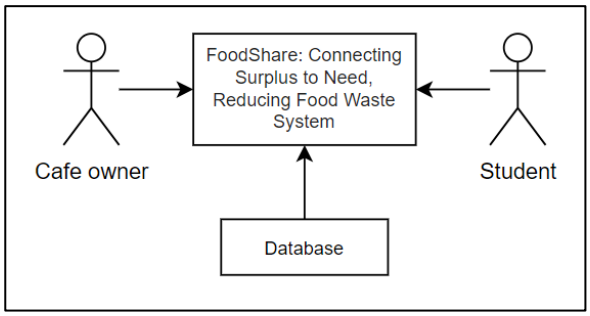


Figure 2.0: Block diagram of FoodShare@UITM

2.1 Product Functionality

The following are the major functions that the FoodShare SRS (Software RequirementsSpecification) document should include:

a) Leftover Food Distribution: allow the cafeteria owner to distribute the leftover food to students including the push-up notifications to notify the students.
b) Online Access: provide private website access for cafeteria owners and students including the ability to upload the leftovers, view the availability of leftovers and access the food information.

2.2 Users and Characteristics

The system will be used by both cafeteria owners and students at UITM Jasin. Cafeteria owners will utilize the system to log into their accounts and manage leftover food information by adding, updating, and deleting surplus details. Meanwhile, students will benefit from receiving notifications about available surplus food and can request items directly through the system. This efficient interaction ensures that surplus food is distributed effectively, helping to reduce waste and support students in need.

2.3 Operating Environment

An operating environment, also referred to as an integrated applications environment, is the setting in which cafeteria owners and students run software on a computer. This environment typically includes a user interface for application administrators and an application programming interface (API) for application managers. The software requirements are minimal, requiring only an online server and an Internet connection, as the software is hosted and operated through the online server.

2.4 Design and Implementation Constraints

The developers face specific challenges, including browser compatibility, hardware limitations, network limitations, system accessibility, security, and GUI customization. Regarding browser compatibility, cafeteria owners and students might use older or less capable web browsers. Additionally, users may access the system on older or less powerful devices, which could impact performance and limit optimization options. For security considerations, both cafeteria owners and students must have encryption authentication. Users will also manage certain aspects of GUI customization. Finally, network limitations may affect the database server, as users could experience limited network connectivity.

2.5 User Documentation

User documentation is designed to assist end-users to use the system. It consists of several components such as user manuals, online help and tutorials.

a) User manual acts as a guide for new users by providing step by step on how to use the systems.
b) Online help will help users when they want to ask questions regarding the systems.
c) Tutorial videos will show a walk-through illustration on how to use the app one by one.

2.6 Assumptions and Dependencies

1. **Availability and Compatibility of Third-Party Libraries or APIs**
   The FoodShare system relies on certain third-party libraries and APIs for its functionality. If a required library or API becomes unavailable or incompatible with the project's technology stack, it may cause significant delays or necessitate the search for alternative solutions. It is crucial to ensure that these third-party components are well-maintained and will continue to be supported in the future to avoid disruptions.

2. **Technical Constraints Such as Browser Compatibility, Hardware Requirements, Network Infrastructure, and Security Policies**
   The FoodShare system must consider technical constraints like browser compatibility, hardware requirements, network infrastructure, and security policies. These constraints must be accurately reflected in the system requirements. Changes in these technical constraints could necessitate modifications to the project's requirements or design. Additionally, security policies will dictate specific requirements for data protection and privacy, which must be incorporated to ensure compliance.

3. **User Behavior and Preferences**
   Understanding how cafeteria owners and students are likely to interact with the FoodShare system, along with their specific preferences, is vital for developing requirements that meet their needs. However, user behavior and preferences can change over time, requiring periodic updates to the requirements to reflect these changes. Failing to consider user behavior and preferences could result in a system that is difficult to use or does not meet users' needs, potentially impacting user adoption and satisfaction negatively.

4. **Adoption and Integration with Existing Systems and Processes**
   If the FoodShare project is intended to integrate with existing systems or processes at UITM Jasin, it is essential to understand and document these dependencies within the requirements. Proper documentation and design will ensure compatibility with existing systems and processes, minimizing disruptions and ensuring a smooth transition. Conversely, failing to consider the need for integration could lead to difficulties during implementation and a system that does not meet user needs.

3 System Features

3.1USE CASE DIAGRAM

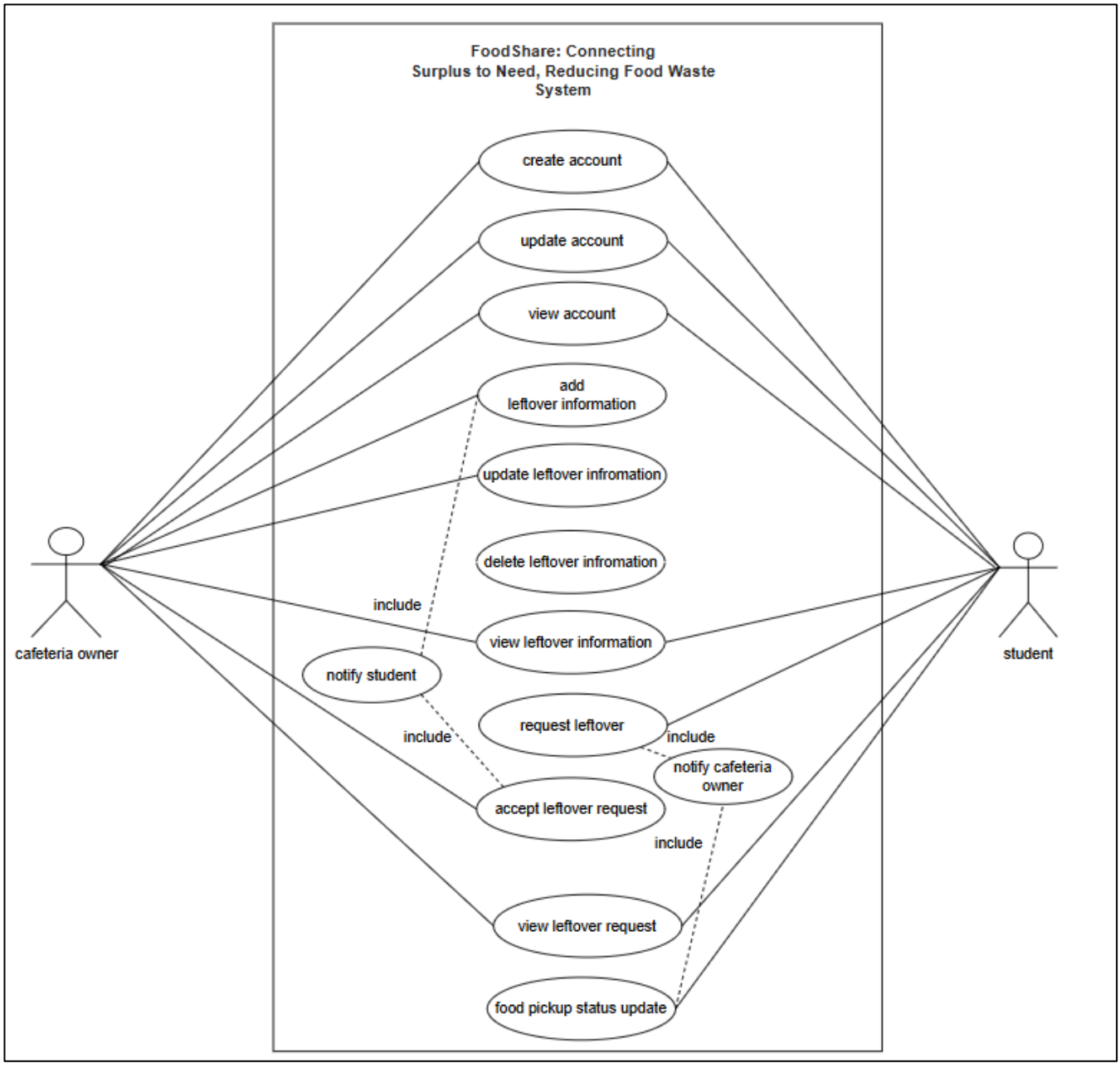


Figure 3.1 Use Case Diagram

3.2 USE CASE DESCRIPTION

3.2.1 UCD_001 Create Account

Use Case ID:	UCD_001		
Use CaseName:	Create Account		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	10/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	Cafeteria owner and student	
Description:	The cafeteria owner and student enter new information and credentials.	
Preconditions:	Cafeteria owner and students accounts must be available.	
Postconditions:	The Cafeteria owner and student information must be created and saved. Account must be created and saved.	
Priority:	High.	
Frequency of Use:	Medium.	
Normal Course of Events:	**Actor**	**System**
	1. The cafeteria owner or student creates an account and enters basic information.	1.1. System creates a cafeteria owner or studentsx account. 1.2 System verifies authorization for username and password. 1.3 System displays cafeteria owner or student account details.
Alternative Courses:	N/A	
Exceptions:	1.1 Data entry is invalid, data entry is undefined, click the cancel button	
Includes:	N/A.	
Notes and Issues:	Basic information (full name, username, password)	

3.2.2 UCD_002 Update Account

Use Case ID:	UCD_002		
Use CaseName:	Update Account		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	10/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	Cafeteria owner, student	
Description:	This enables cafeteria owner or student to view and update their account information similarly.	
Preconditions:	Cafeteria owner or student accounts must be available.	
Postconditions:	Account details can be updated and saved.	
Priority:	High.	
Frequency of Use:	Medium.	
Normal Course of Events:	**Actor**	**System**
	1. The actors view their account details. 2. The actor updates an existing account information.	1.1 System finds the exact actor's account in the databases. 1.2 System updates and view the credential and information.
Alternative Courses:	N/A	
Exceptions:	1.1 Data entry is invalid, data entry is undefined, click the cancel button	
Includes:	Create account use case.	
Notes and Issues:	Account information (full name, username, password).	

3.2.3 UCD_003 View Account

Use Case ID:	UCD_003		
Use CaseName:	View Account		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	10/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	Student, cafeteria owner	
Description:	The student or cafeteria owner view their account information.	
Preconditions:	Student and cafeteria owner accounts must be available.	
Postconditions:	The student or cafeteria information must be created and saved. Account must be created and saved.	
Priority:	High.	
Frequency of Use:	Medium.	
Normal Course of Events:	**Actor**	**System**
	1. The student or cafeteria owner clicks on view account	1.1. System displays student or cafeteria owner account details.
Alternative Courses:	N/A	
Exceptions:	1.1 Data entry is invalid, data entry is undefined, click the cancel button	
Includes:	N/A.	
Notes and Issues:	Basic information (full name, username, password)	

3.2.4 UCD_004 Notify Cafeteria Owner

Use Case ID:	UCD_004		
Use Case Name:	Notify Cafeteria Owner		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	10/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	System (Automatic Process)	
Description:	Wants to receive notifications about student requests, food availability, and any changes to the system to facilitate timely actions.	
Preconditions:	The cafeteria owner is logged into the system. The system has a pending student request or updated food information.	
Postconditions:	The cafeteria owner is successfully notified about relevant updates, such as student requests, new leftover food, or status updates.	
Priority:	High.	
Frequency of Use:	Medium.	
Normal Course of Events:	**Actor**	**System**
	1. The system detects a new student request and food pickup status update.	1.1 System identifies the need to notify the cafeteria owner (e.g., new request or food pickup status update).
Exceptions:	1.1 If the system fails to send the notification, an error message is displayed.	
Includes:	N/A	
Notes and Issues:	Cafeteria Phone number must be valid	

Alternative Courses:	N/A

3.2.5 UCD_005 Notify Student

Use Case ID:	UCD_005		
Use Case Name:	Notify Student		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	10/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	System (Automatic Process)	
Description:	Wants to be informed about available leftover food and the status of their requests.	
Preconditions:	The student is logged into their account. The system has new leftover food available or status updates for a student's request.	
Postconditions:	he student is successfully notified of food availability or the status of their request.	
Priority:	High.	
Frequency of Use:	Medium.	
Normal Course of Events:	**Actor**	**System**
	1. The system detects new leftover food availability or updates to an existing food request.	1.1 The system automatically sends a notification to the student, informing them of the new or updated information.
Exceptions:	1.1 If the system fails to send the notification, an error message is displayed.	
Includes:	N/A	

Notes and Issues:	Students number must be valid
Alternative Courses:	N/A

3.2.5 UCD_006 Add leftover information

Use Case ID:	UCD_006		
Use CaseName:	Add leftover information		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	10/08/2024	**Date LastUpdated:**	17/10/2024
Actor:	Cafeteria owner		
Description:	The cafeteria owner adds new leftover information to the system for students to view and request.		
Preconditions:	The cafeteria owner must be logged into the system.		
Postconditions:	The leftover information is successfully added and available for students to view.		
Priority:	High.		
Frequency of Use:	Medium.		
Normal Course of Events:	**Actor**	**System**	
	1. The cafeteria owner clicks on "Add Leftovers. 2. The cafeteria owner fills out the leftover details. 3. The cafeteria owner submits the form.	1.1 The system displays a form to enter leftover details. 2.1 The system validates the input and adds the information to the database. 3.1 The system confirms that the leftover information has been successfully added.	
Alternative Courses:	N/A		
Exceptions:	1.1 Data entry is invalid. 1.2 Data entry is undefined. 1.3 Click the cancel button.		

Includes:	Notification to the student
Notes and Issues:	Ensure that all relevant details are entered to provide accurate information to students.

3.2.6 UCD_007 Update Leftover Information

Use Case ID:	UCD_007		
Use Case Name:	Update leftover information		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024
Actor:	Cafeteria owner		
Description:	This use case allows the cafeteria owner to manage surplus food information by updating. The system verifies changes and updates the available surplus list accordingly.		
Preconditions:	The leftover food information must already exist in the system.		
Postconditions:	Updated information is reflected and available for students to view.		
Priority:	Medium.		
Frequency of Use:	Medium.		
Normal Course of Events:	**Actor**	**System**	

	1 The cafeteria owner accesses the leftover food details 2 The cafeteria owner updates the relevant details	1.1 The system displays the current information. 2.1 The system verifies and saves the updates, confirming that the update was successful.
Alternative Courses:	N/A	
Exceptions:	1.1 Data entry is invalid, data entry is undefined, click the cancel button	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.7 UCD_008 Delete Leftover Information

Use Case ID:	UCD_008		
Use Case Name:	Delete leftover information		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024
Actor:	Cafeteria owner		
Description:	This use case allows the cafeteria owner to remove outdated or no longer available leftover food from the system.		
Preconditions:	The leftover food information must already exist in the system.		
Postconditions:	Updated information is reflected and available for students to view.		

Priority:	Medium.	
Frequency of Use:	Medium.	
Normal Course of Events:	**Actor**	**System**
	1 The cafeteria owner accesses the leftover food details 2 The cafeteria owner deletes the food from the system.	1.2 The system displays the current information. 2.2 The system confirms the removal
Alternative Courses:	N/A	
Exceptions:	1.1 If the food has been requested by student, it will not be able to be deleted	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.8 UCD_009 View Leftover Information

Use Case ID:	UCD_009		
Use Case Name:	View leftover information		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024
Actor:	Cafeteria owner, student		

Description:	Users view available leftover food information.	
Preconditions:	The system must have leftover information available.	
Postconditions:	The user views the information.	
Priority:	High.	
Frequency of Use:	High.	
Normal Course of Events:	**Actor**	**System**
	1. The user accesses the leftover food section. 2. The user selects an item to view details.	1.1 The system displays the list of available leftovers. 2.1 The system displays detailed information about the selected item.
Alternative Course :	N/A	
Exceptions:	No available leftovers to display.	
Includes:	N/A	
Notes and Issues:	The display should be optimized for ease of use.	

3.2.9 UCD_010 Request Leftover

Use Case ID:	UCD_010		
Use Case Name:	Request Leftover		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024
Actor:	Student		
Description:	A student requests a portion of available surplus food.		
Preconditions:	The student must be logged into the system.		
Postconditions:	The request is logged, and the cafeteria owner is notified.		

Priority:	High.	
Frequency of Use:	High.	
Normal Course of Events:	**Actor**	**System**
	1. The student selects the surplus food item to request 2. The student submits the request.	1.1 The system confirms the availability of the item. 2.1 The system logs the request and notifies the cafeteria owner.
Alternative Course :	N/A	
Exceptions:	If the student is not logged in, the system prompts them to log in before proceeding.	
Includes:	Notify cafeteria owner	
Notes and Issues:	N/A	

3.2.10 UCD_011 Accept leftover Request

Use Case ID:	UCD_011		
Use Case Name:	Accept leftover request		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	Cafeteria owner	
Description:	The cafeteria owner reviews and accepts a student's request for surplus food.	
Preconditions:	The cafeteria owner must be logged into the system. The leftover must be available.	
Postconditions:	The request is accepted and student will be notified.	
Priority:	High.	
Frequency of Use:	High.	
Normal Course of Events:	**Actor**	**System**
	1. The cafeteria owner clicks the accept button.	1.1 The system logs the request and notifies the cafeteria owner.
Alternative Course :	N/A	
Exceptions:	If no requests are available to accept, the system displays a message indicating no pending requests.	
Includes:	Notify students	
Notes and Issues:	N/A	

3.2.11 UCD_012 View leftover Request

Use Case ID:	UCD_012		
Use Case Name:	View leftover request		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	Cafeteria owner,student	
Description:	The cafeteria owner and student can view the status request	
Preconditions:	The cafeteria owner and student must be logged into the system.	
Postconditions:	N/A	
Priority:	High.	
Frequency of Use:	High.	
Normal Course of Events:	**Actor**	**System**
	1. The cafeteria owner and student clicks on the view request button	1.1 The system logs display the request leftover food
Alternative Course :	N/A	
Exceptions:	If there are no pending requests, the system displays a message saying there are no requests.	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.11 UCD_013 Food Status Pickup Update

Use Case ID:	UCD_013		
Use Case Name:	Food status pickup update		
Created By:	Syamimi Afiqah	**Last Updated By:**	Syamimi Afiqah
Date Created:	13/08/2024	**Date LastUpdated:**	17/10/2024

Actor:	Student	
Description:	Students need to click pickup to notify the cafeteria owner about the status	
Preconditions:	Students need to pick up the food	
Postconditions:	When food is already pickup the cafeteria owner will be notified.	
Priority:	High.	
Frequency of Use:	High.	
Normal Course of Events:	**Actor**	**System**
	1. The student clicks the pick up button.	1.1 The system logs the status of pickup and notifies the cafeteria owner.
Alternative Course :	N/A	
Exceptions:	Item no longer available.	
Includes:	Notify cafeteria owner	
Notes and Issues:	N/A	

3 Other Non-functional Requirements

4.1 Performance Requirements

The FoodShare system must centralize leftover food data to eliminate the need for cafeteria owners to manage surplus manually. The application will allow easy access to information about available leftovers, enabling cafeteria owners to efficiently add, update, and delete leftover information in the system. Students can view and request available surplus through the system, streamlining the distribution process. Cafeteria owners must have the capability to manage leftover information via any device with internet access, allowing them to perform actions such as viewing, accepting, and updating surplus requests digitally. Similarly, students must be able to access notifications about available surplus food and request items easily. The system should support real-time updates and notifications to ensure timely information dissemination and efficient surplus distribution.

3.2 Safety and Security Requirements

Users of the FoodShare system, including cafeteria owners and students, must authenticate themselves according to their user category before accessing the system. The system will implement industry-standard security practices for user authentication, including data encryption and password protection, to prevent unauthorized access and protect user data, such as usernames and passwords, from malicious entities. The system must not contain any harmful components that could compromise the current hardware and software of the users' devices. Additionally, the system will include functionalities to protect against data leakage and ensure that user information is only accessible to authorized parties. Users must agree to the terms and conditions and follow the guidelines for using the system to ensure compliance with security protocols.

3.3 Software Quality Attributes

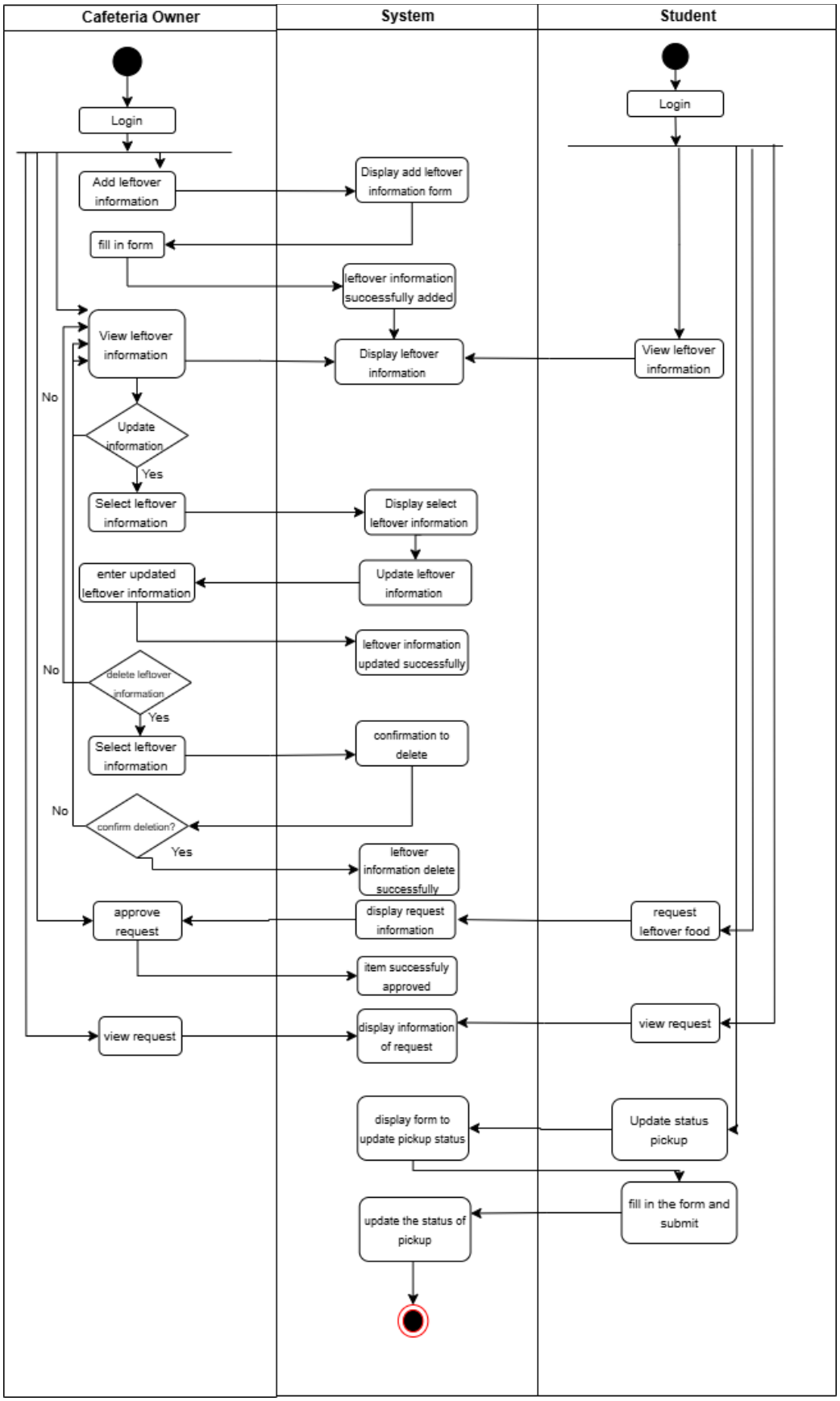
The FoodShare system must adhere to the following software quality attributes to ensure a high level of performance and user satisfaction:

1. **Availability**: The system must be available 24/7, allowing cafeteria owners and students to access it at any time, provided there is an internet connection.
2. **Adaptability**: The system is designed to integrate seamlessly with existing processes at UITM Jasin, allowing cafeteria owners and students to use it alongside other university systems.
3. **Correctness**: The system must accurately display and manage surplus food information, ensuring that users receive correct and up-to-date data about available leftovers and requests.
4. **Reusability**: The system should allow cafeteria owners to reuse leftover information efficiently, facilitating quick updates and modifications to surplus details as needed.
5. **Maintainability**: The system must be easy to maintain and support updates and changes cost-effectively, ensuring long-term usability and adaptability to evolving requirements.

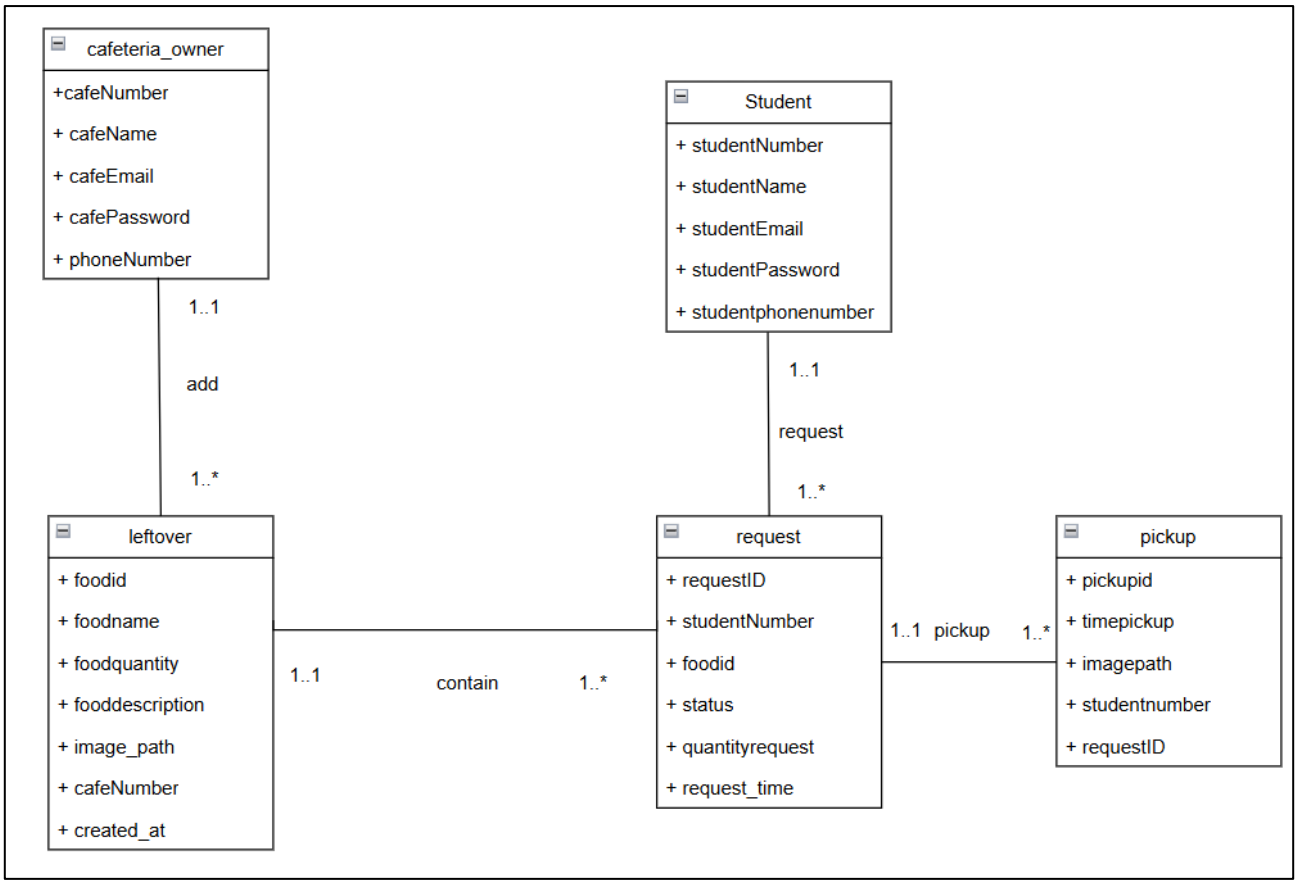
By ensuring these quality attributes, the FoodShare system will effectively meet the needs and expectations of its users, facilitating the reduction of food waste and supporting students at UITM Jasin.

Appendix A – REQUIREMENT MODELS

1. BUSINESS ACTIVITY DIAGRAM

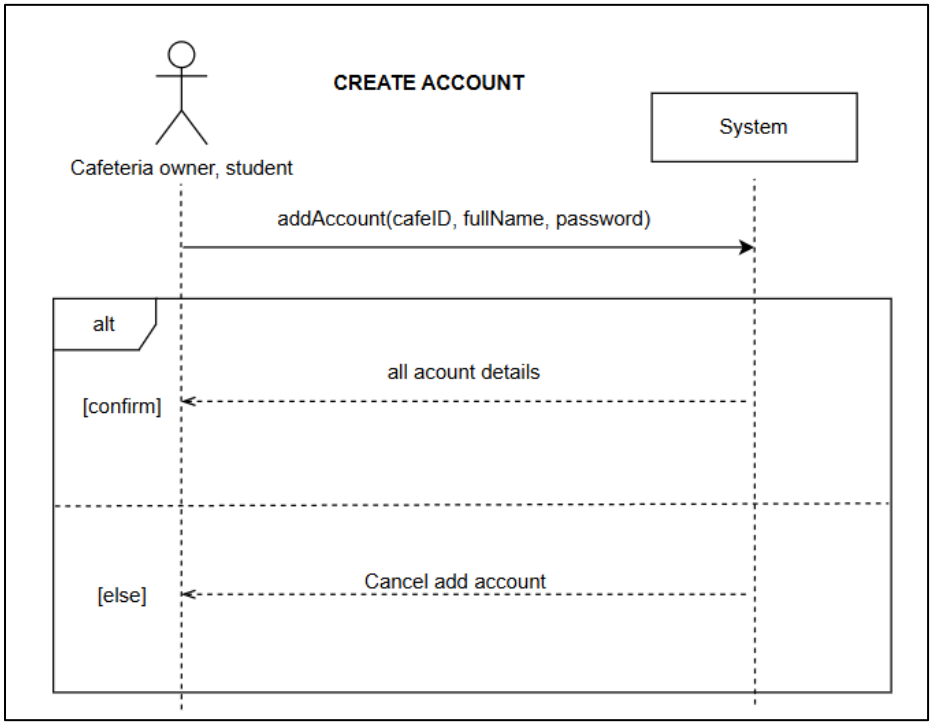

Cafeteria Owner
System
Student
Login
Login
Add leftover information
Display add leftover information form
fill in form
leftover information successfully added
View leftover information
Display leftover information
View leftover information
No
Update information
Yes
Select leftover information
Display select leftover information
enter updated leftover information
Update leftover information
leftover information updated successfully
No
delete leftover information
Yes
Select leftover information
confirmation to delete
No
confirm deletion?
Yes
leftover information delete successfully
approve request
display request information
request leftover food
item successfuly approved
view request
display information of request
view request
display form to update pickup status
Update status pickup
fill in the form and submit
update the status of pickup

2. CLASS DIAGRAM

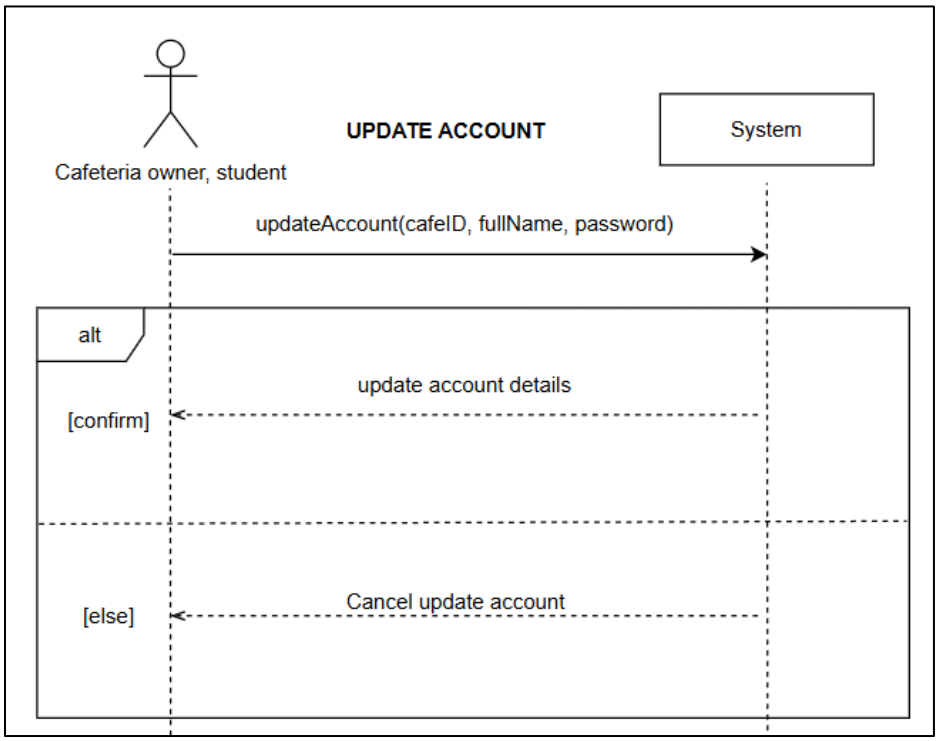


3. SYSTEM SEQUENCE DIAGRAM

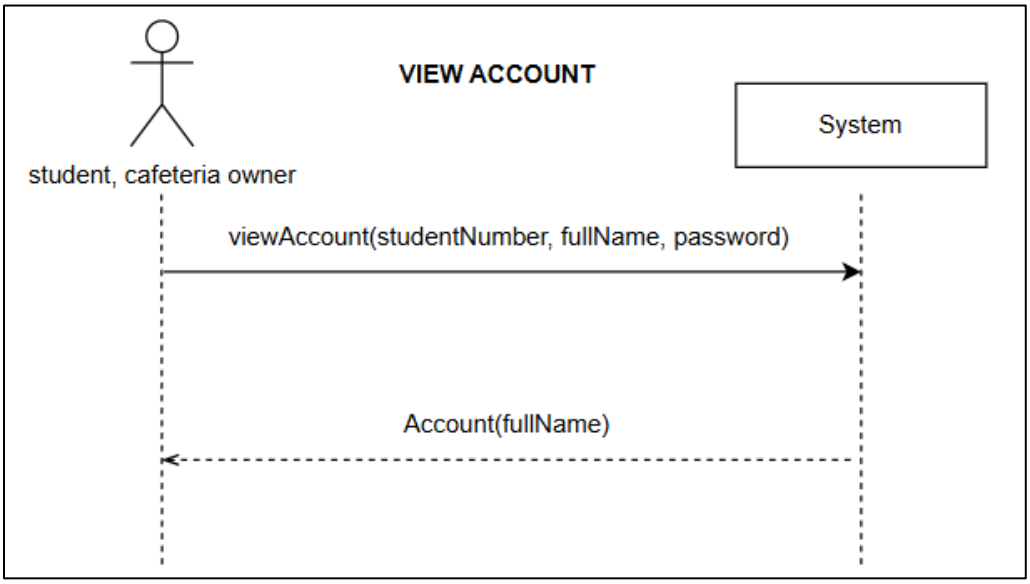
3.1 Create Account



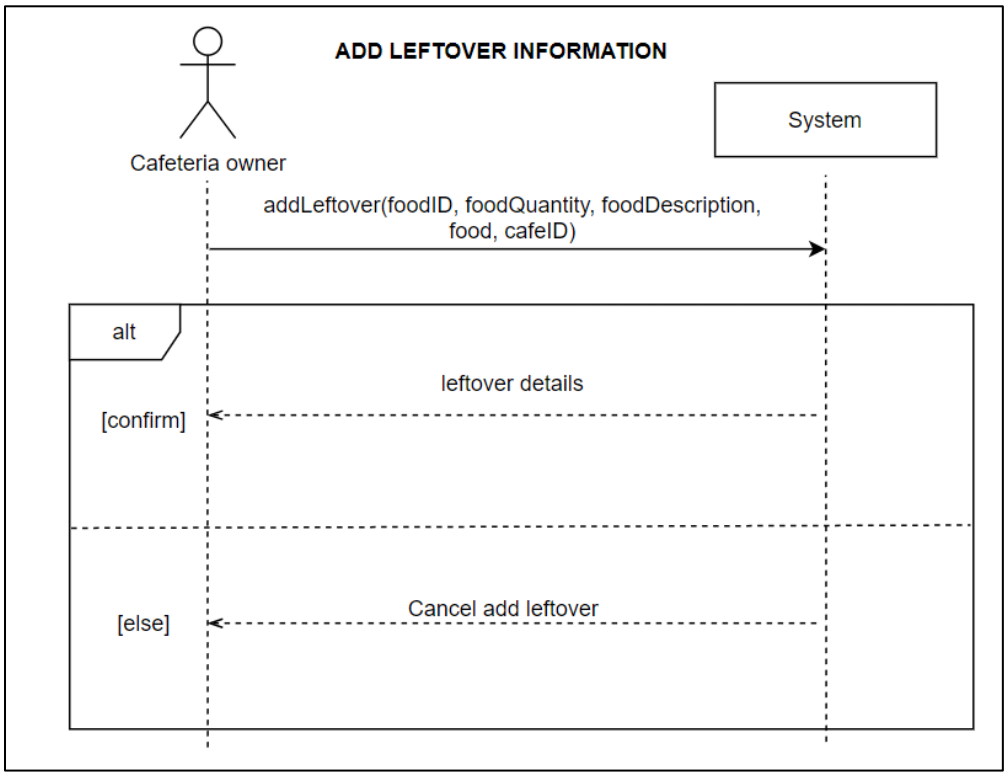
3.2 Update Account



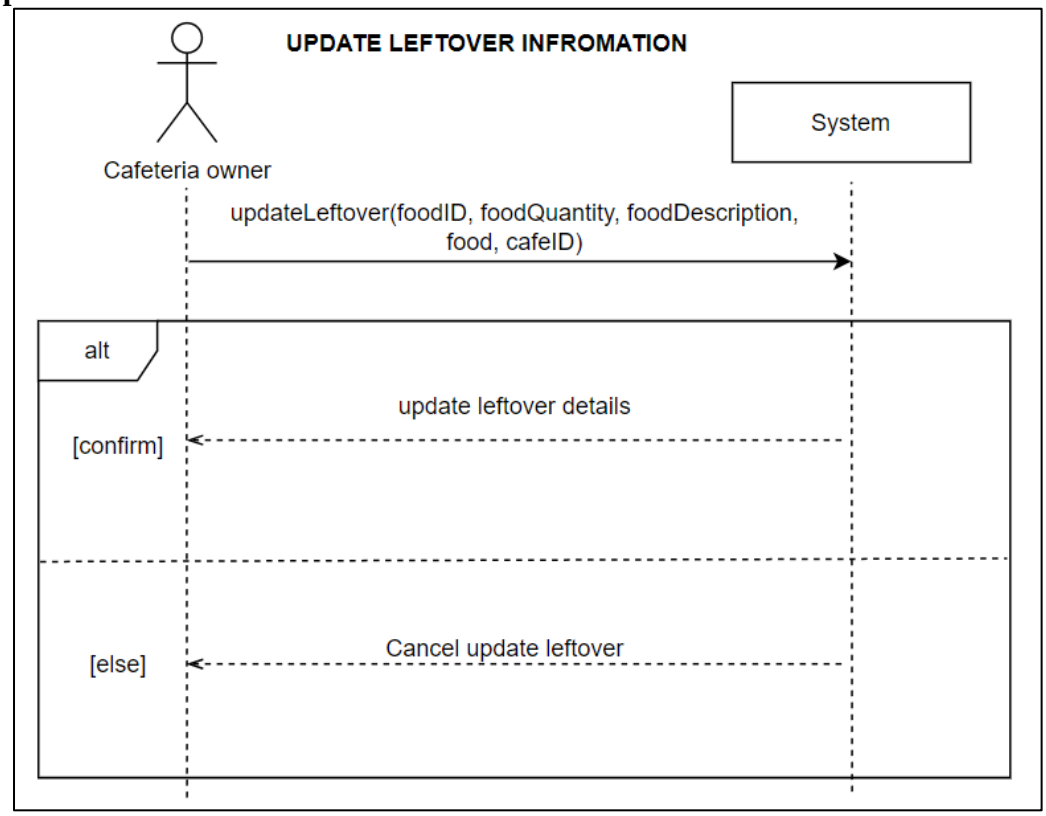
3.3 View Account



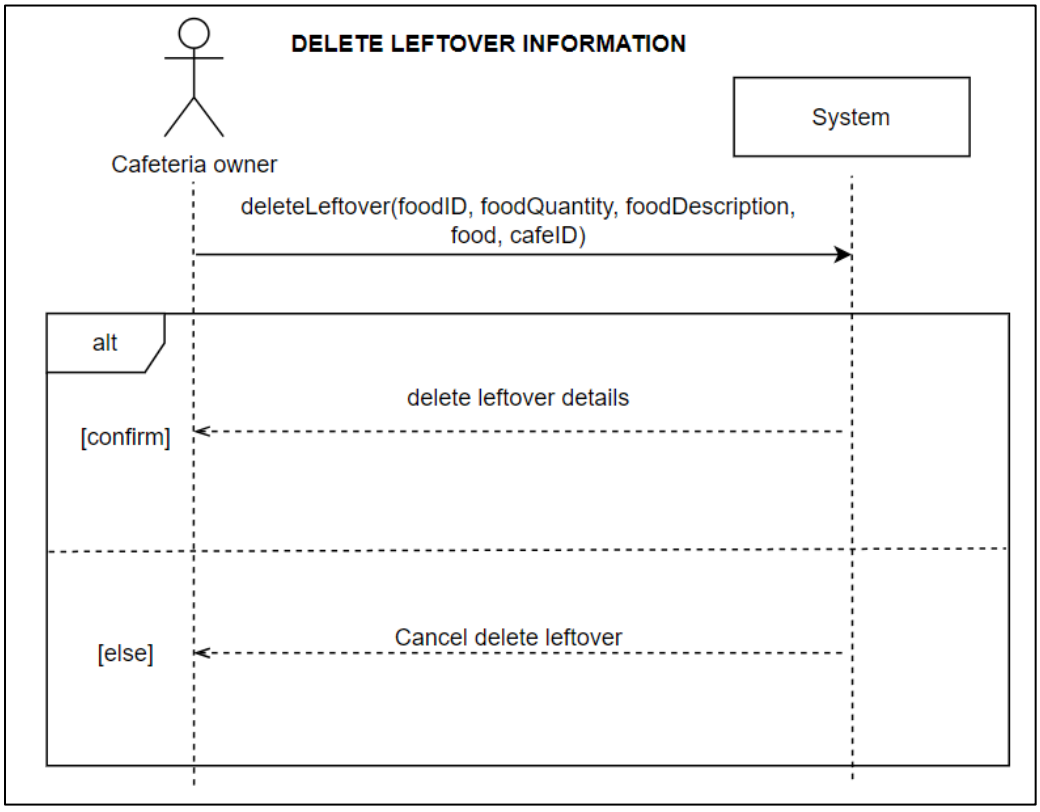
3.4 Add Leftover information



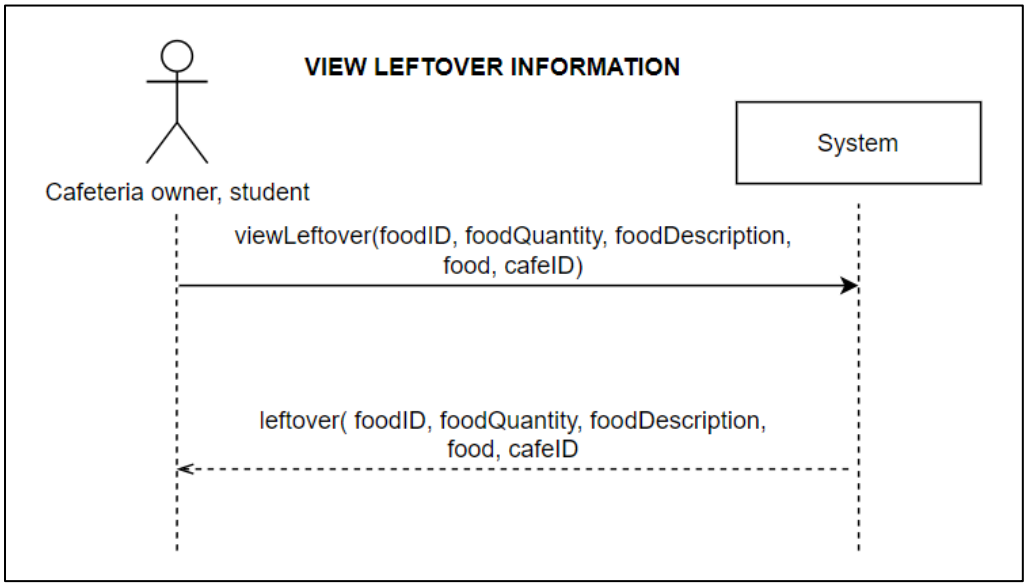
3.5 Update Leftover information



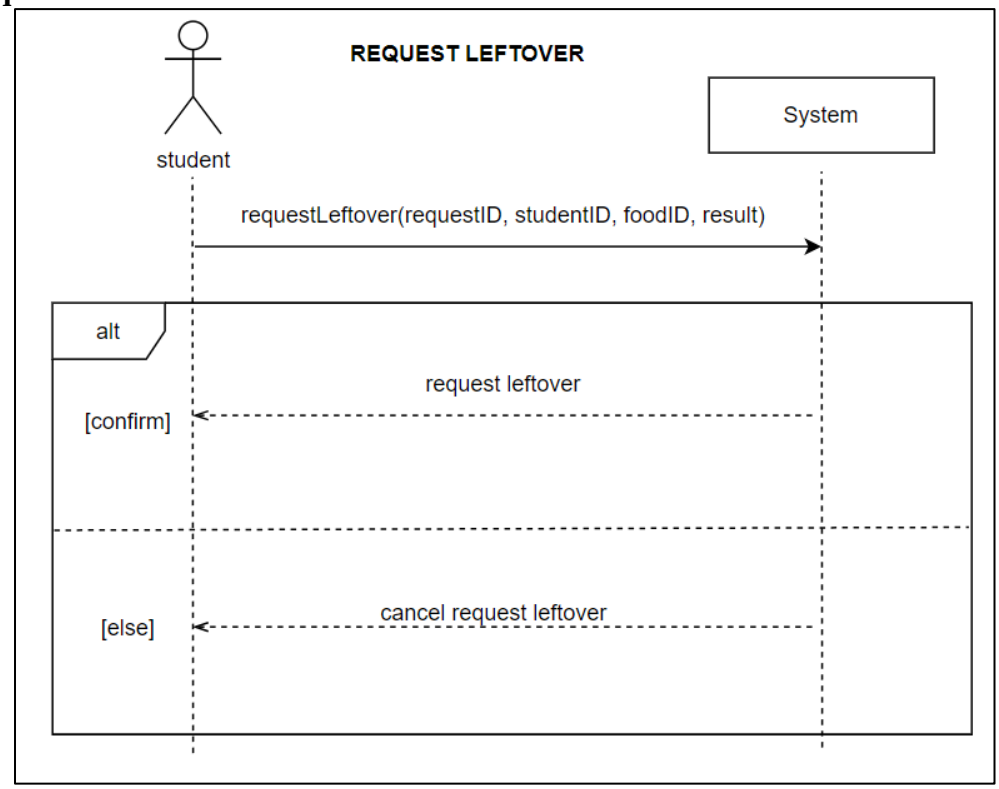
3.6 Delete Leftover information



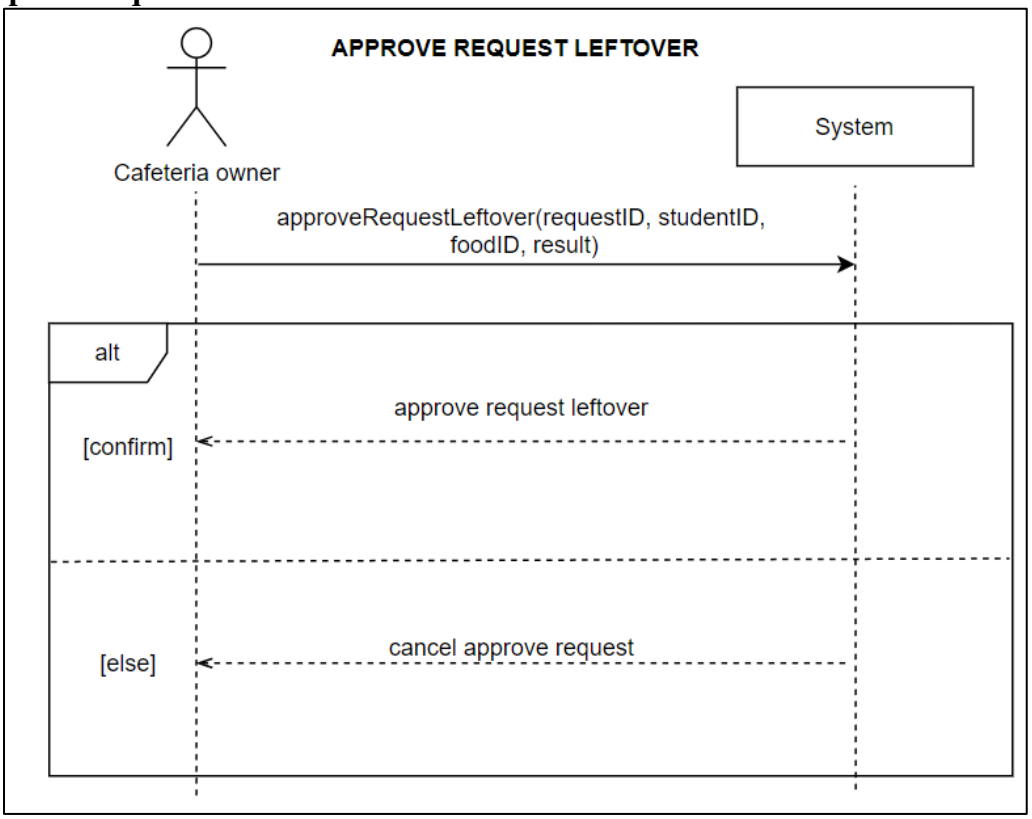
3.7 View Leftover information



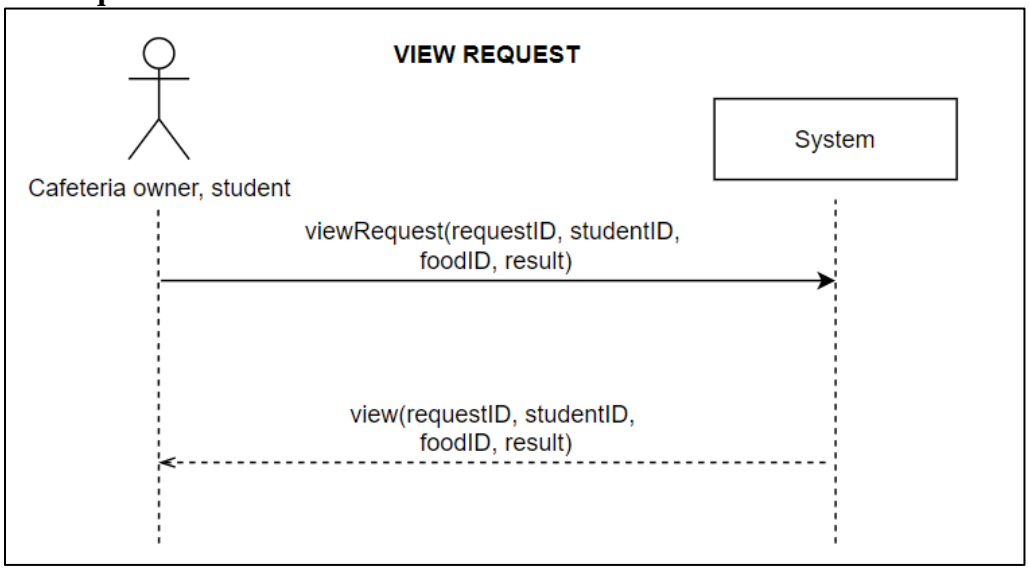
3.8 Request Leftover



3.9 Approve request leftover



3.10 View request leftover



APPENDIX C



SOFTWARE DESIGN DOCUMENT FOR FOODSHARE: CONNECTING SURPLUS TO NEED, REDUCING FOOD WASTE SYSTEM

Prepared by:
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1. Introduction

1.1 Purpose

This Software Design Document (SDD) outlines the architecture and system design of FoodShare: Connecting Surplus to Need, Reducing Food Waste. The system will provide a web-based platform enabling cafeteria owners to share leftover food with students. The purpose of this SDD is to present a comprehensive overview of the software's design, including its components, modules, interfaces, and data flow. This document serves as a reference for developers, designers, and stakeholders involved in the development and deployment of FoodShare, ensuring a shared understanding of its design principles, features, and technical specifications.

The document will describe the software's purpose and key functionalities, the user interfaces, and how the system operates under various constraints. Additionally, it will detail the software's responses to external factors. This SDD is intended for both the end users and developers.

1.2 Scope

a) Allows cafeteria owners to input details about surplus food, set pick-up times, and notify students.
b) Offers a user-friendly graphical interface for easy management of food distribution.
c) Reduces the need to discard or distribute leftovers elsewhere.
d) Saves time and effort for cafeteria owners, making resource management more efficient.
e) Helps students in need by providing access to surplus food.
f) Reduces food waste and promotes resource-sharing on campus.
g) Helps students to create orders for event.

1.3 References

- *System Sequence Diagram: A Complete Tutorial | EdrawMax*. (2013). Edrawsoft.

  https://www.edrawsoft.com/article/uml-system-sequence-diagram.html.

1.4 Document Structure

All stakeholders use the SDD for their own work. Developers will be using the SDD to learn and follow what has to be created and how it works. Project managers are required to plan and administer the project, including schedule and available tools. Testers have to understand what needs to be tested, evaluated and checked. For documentation writers using SDD, they need to learn what must be documented and how it should be done. An overview of the software and its function is given in this section. This section defines what the software must do and how it should work. The restraints and requirements that the software must adhere to in terms of performance, security, and other factors are discussed in this section. The software's appearance, feel, and user interface are all covered in this section. The software's relationships with other systems and how it contacts them are covered in this section. Additionally, the document is intended for requirements engineer, programmer, developer, ui designer, system analyst and project manager.

1.5 Definitions, Acronyms and Abbreviations

a) UI : User Interface
b) DB : Database
c) CRUD : Create, Read, Update, Delete
d) SDD : Software Design Description

1.6 System Overview

The FoodShare: Connecting Surplus to Need, Reducing Food Waste system is designed to help cafeteria owners at UITM Jasin efficiently share surplus food with students, reducing waste and promoting resource-sharing. The system allows cafeteria owners to input details about leftover food, set pick-up times, and notify students about available surplus. The goal of the system is to streamline the process of food distribution, save time, and reduce the effort required for managing surplus food. This, in turn, helps meet the needs of students facing food insecurity, while contributing to a more sustainable campus environment.

The system is designed to be user-friendly, with a simple graphical interface, ensuring easy navigation for cafeteria owners. It will be scalable to accommodate the needs of the campus, adaptable for different food distribution patterns, and efficient in managing resources.

The following are key functionalities of the system:

- Food Management: Cafeteria owners can input food availability, pick-up times, and quantity, and students can view, reserve food and create order for booking.
- Notification System: Students will be notified when food becomes available, ensuring timely pick-up.
- Online Access: Users will have access through a web-based platform, enabling easy interaction between cafeteria owners and students.

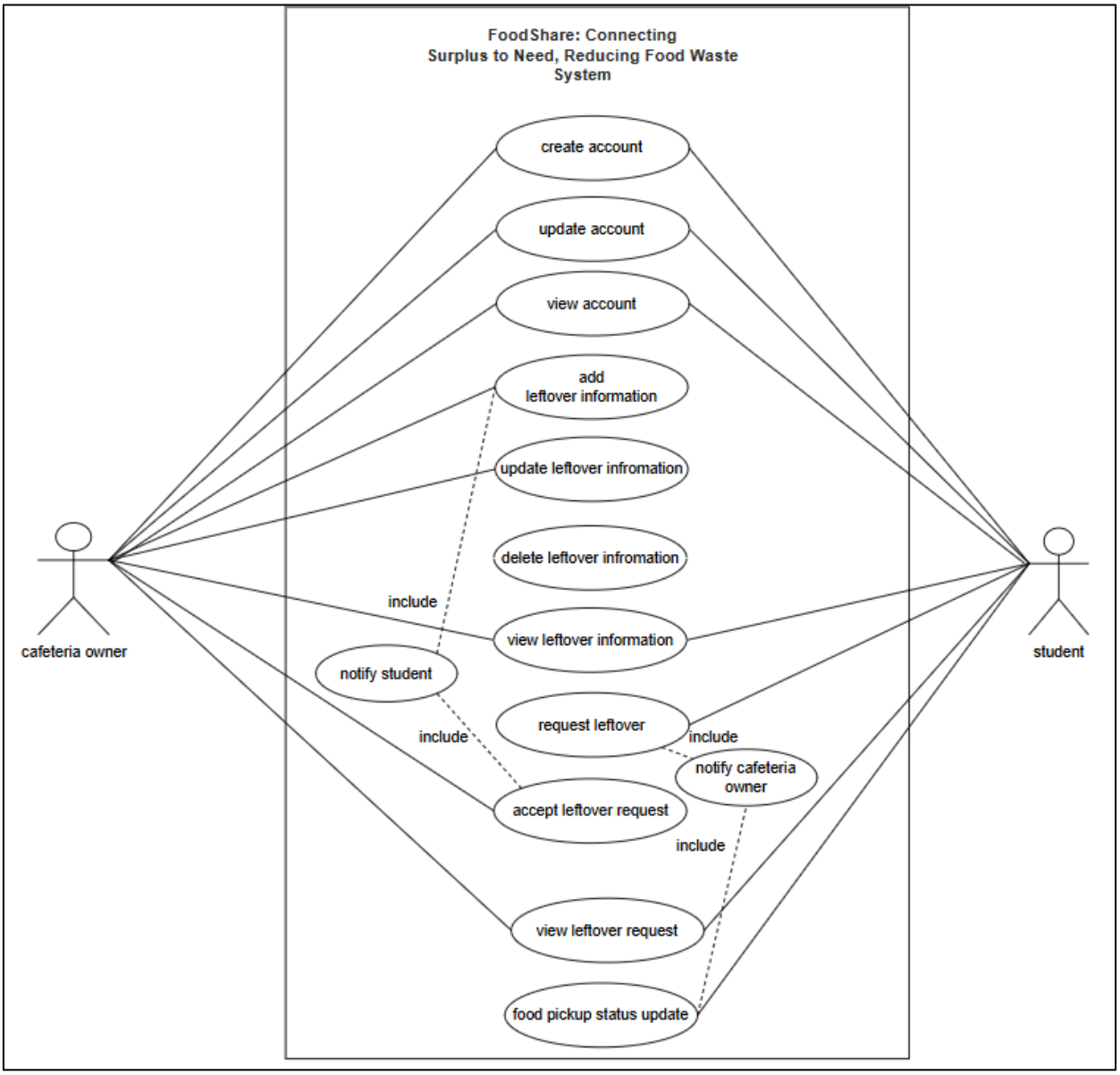
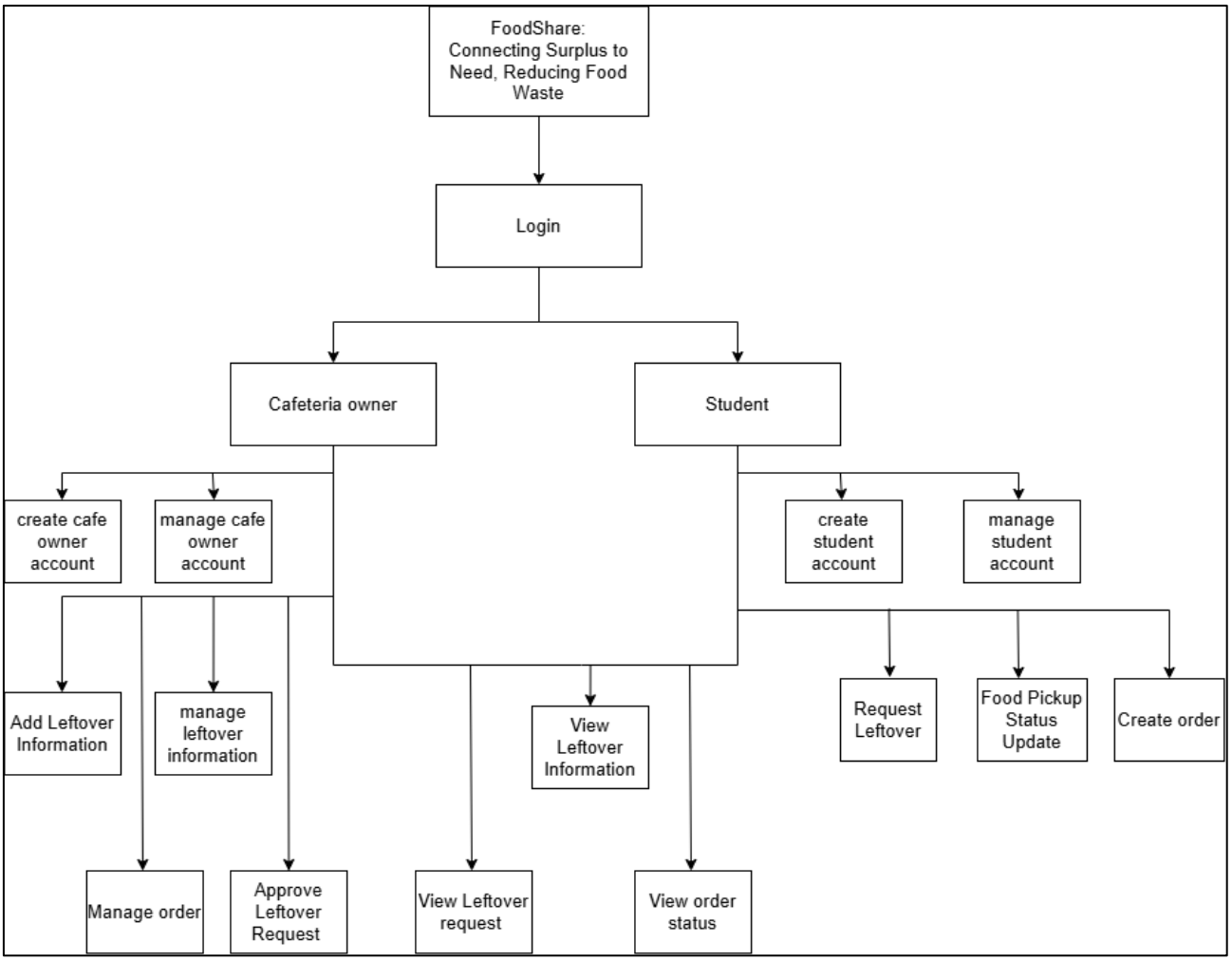


Figure 1.1 Use Case Diagram

2. System Architecture

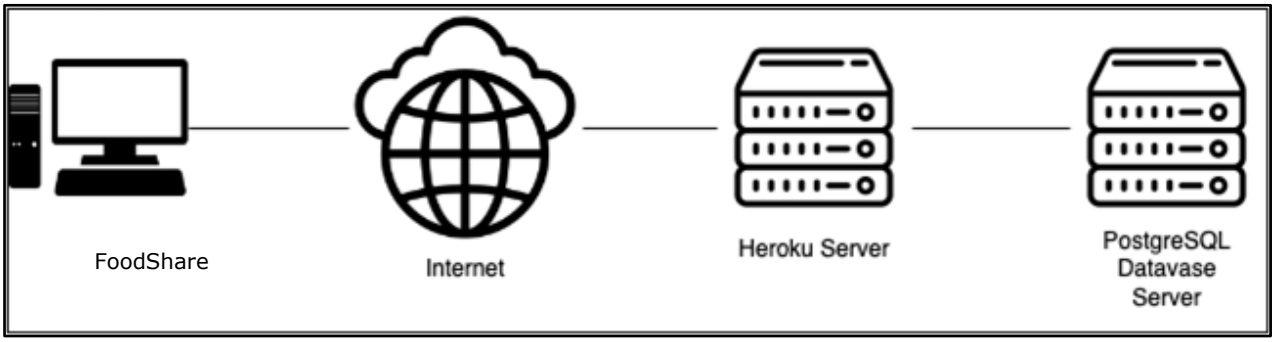
2.1 Architectural Description



2.1 Block Diagram of the major subsystem

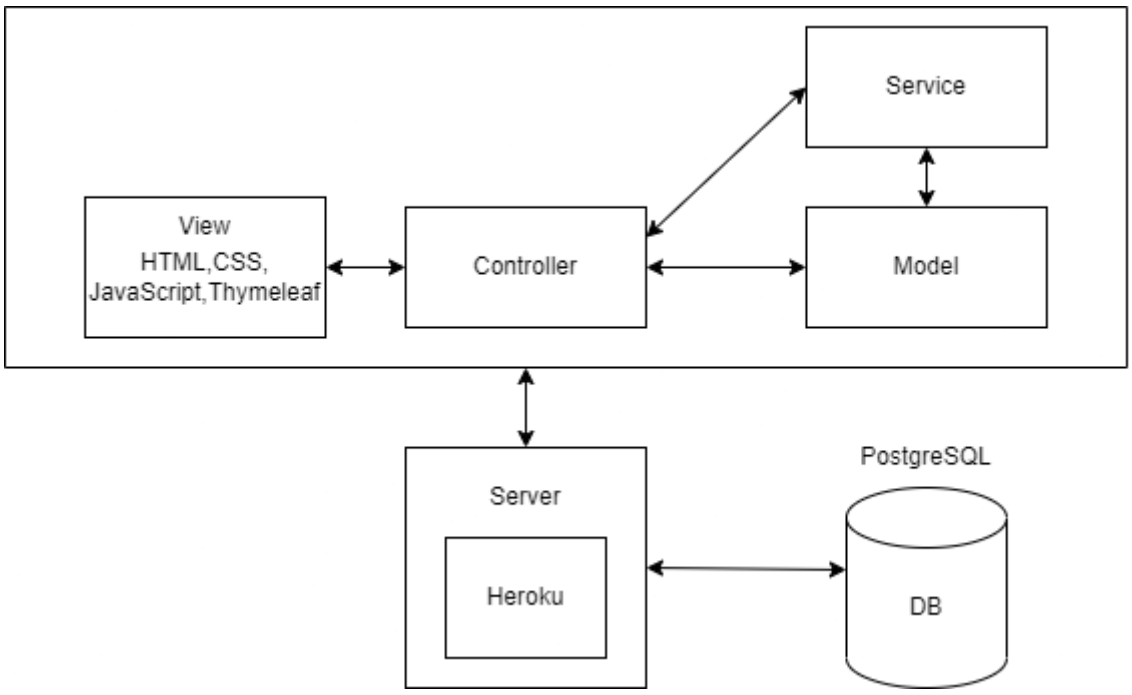
This is a high-level overview of how responsibilities of the systems were partitioned and then assigned to subsystems.

The system will be using Heroku services which includes PostgreSQL that allows the system to be connected to a database. Heroku supports Java, HTML, CSS, JavaScript and SQL Query which are some of the main languages used for this system. Heroku server will execute all the necessary files that are used for the system and display it over to the internet for client usage.



2.2 Network Diagram

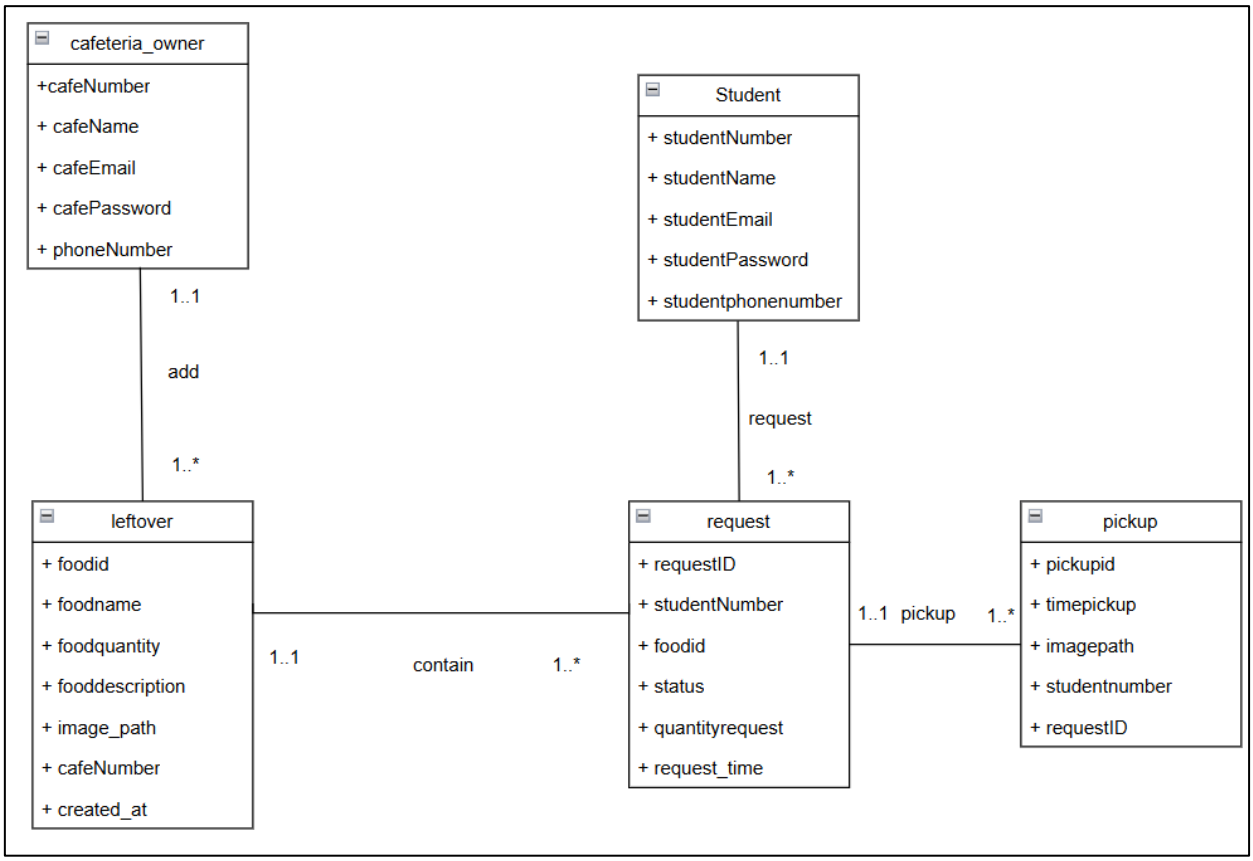
2.2 Design Rationale



2.3 Design Rationale Diagram.

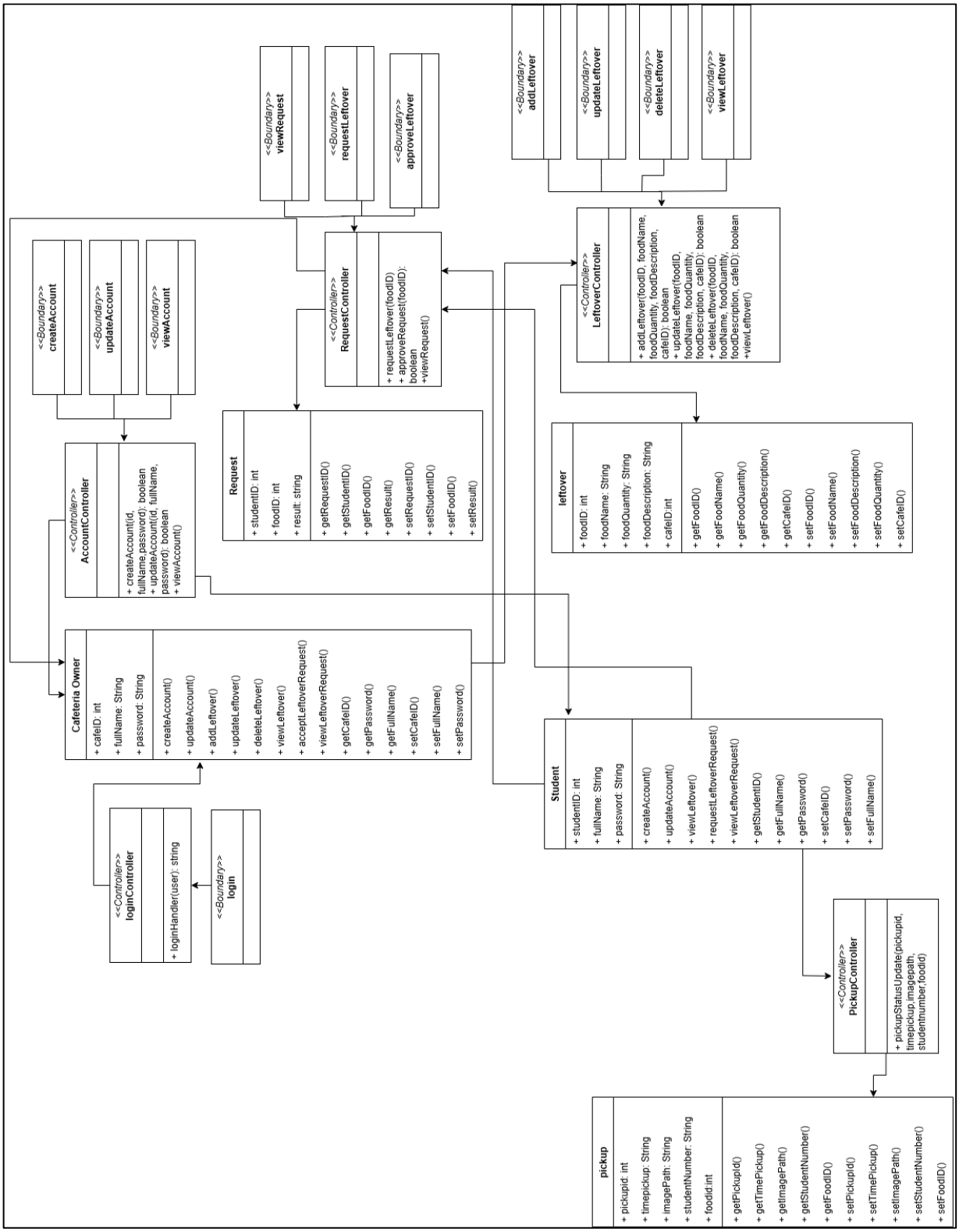
The process of documenting the logic, trade-offs, and judgements behind the design of a system or product is referred to as design rationale. It captures the thought processes, restrictions, and considerations that influenced design decisions, allowing for a clear understanding of why particular decisions were taken. Design reasoning promotes successful communication, cooperation, and knowledge transfer among designers and stakeholders, assisting in the evaluation and refinement of the design, tracking decisions through time, and ensuring that the design stays intelligible and maintainable over the lifecycle of the system.

2.3 Domain Class Diagram



2.4 Domain Class Diagram.

2.4 Design Class Diagram



2.5 Design class diagram.

3. Data Design

3.1 Database Description

The database that this system used to store the data is relational database. The database is located on cloud and is secured using a combination of physical and logical security measures. Moreover, the size of the database is a medium-sized database with 10,000 rows of data. Hence, the database is able to handle the expected load and is affordable.

3.2 Data Dictionary

Cafeteria Owner						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
cafeNumber	Id for cafe owner	text	Max=2	-	Y	Y
cafeName	cafeName	text	max=50	-	Y	N
cafeEmail	Email of cafe owner	text	max=60	-	Y	N
cefePassword	Cafe password	text	Max =	-	Y	N
phoneNumber	Cafe owner phone number	text	Max =12	-	y	N

Student						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
studentNumber	Id for student	text	max=10	-	Y	Y
studentName	Full name of student	text	max=50	-	Y	N
studentEmail	Email of student	text	max=60	-	Y	N
studentphonenumber	Phone number student	text	Max =12	-	Y	N
studentPassword	Password of student	text	Max=30	-	y	N

Leftover						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
foodID	Id of leftover	int	Auto increment	-	Y	Y
foodName	Name of items	Character varying(255)	max=255	-	Y	N
foodQuantity	Quantity of items	int	-	-	Y	N
Image_path	Image storing	Character varying(255)	max=255	-	Y	N
cafeNumber	Id for cafeteria owner	text	reference cafeteria owner(cafeNumber)	Available	Y	Y

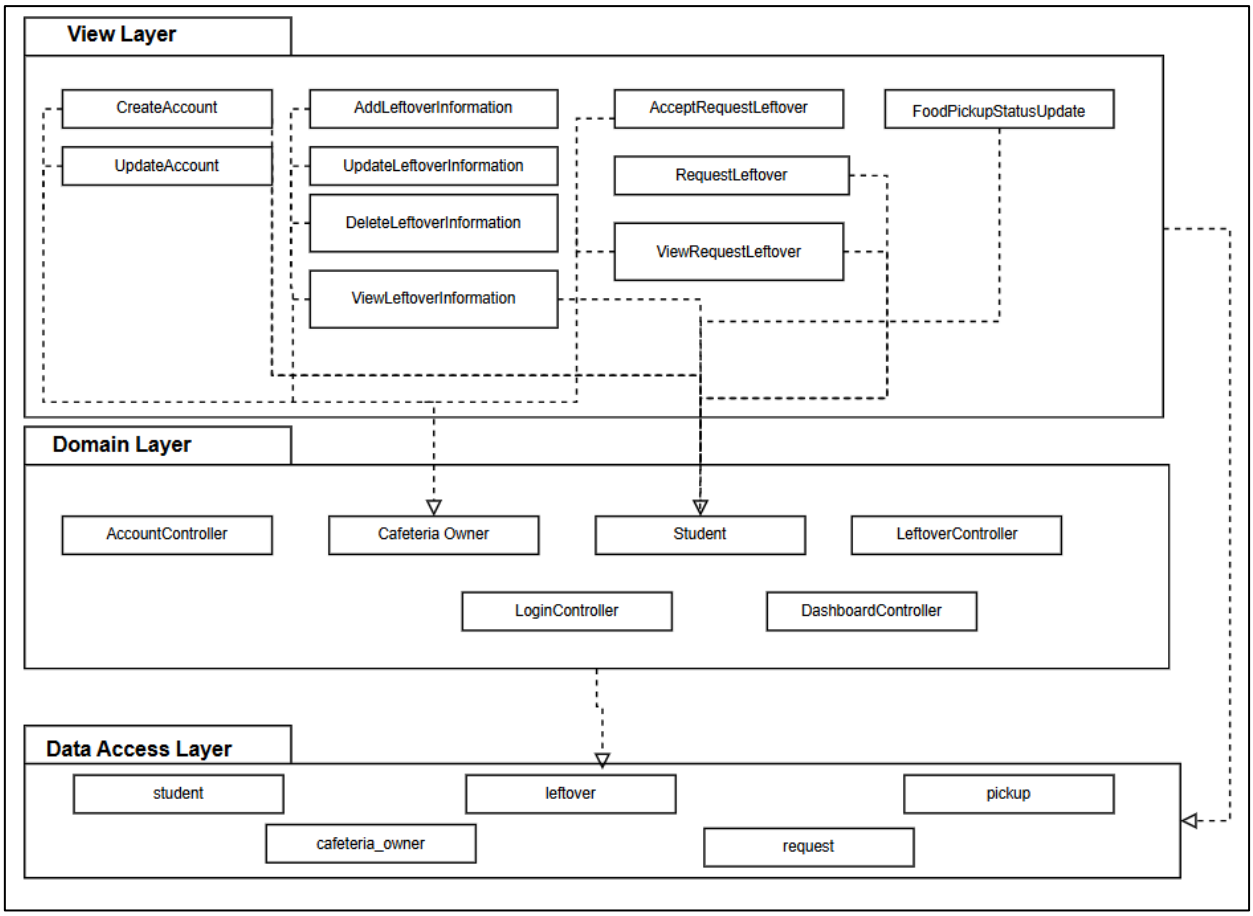
Created_at	Time the food was add	timestamp	-	-	Y	N
Place_to_pickup	Place to pickup the food	Character varying(255)	max=255	-	Y	N
Pickup_time	Time to pickup	Character varying(255)	max=255	-	Y	N

Request						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
requestID	Id of request	text	Auto increment	-	Y	Y
quantityRequest	Amount request	int	-		Y	N
Request_time	Time that student request	timestamp	-	-	Y	N
studentNumber	Id of student	text	reference student(studentNumber)	-	Y	Y
cafeNumber	Id of cafe	text	reference cafeteria owner(cafeNumber)	-	Y	Y
foodID	Id of food	int	reference leftover(foodid)		Y	Y
status	status of request	Character varying(20)	Pending/Accepted/Rejected	-	Y	N

Pickup						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
pickupID	Id of pickup	int	Auto increment	-	Y	Y
timepickup	Time students pickup	timestamp	-	-	Y	Y
Image_path	Image storing	Character varying(255)	max=255	-	Y	N
studentNumber	Id of student	text	reference student(studentNumber)	-	Y	Y
foodID	Id of food	int	reference leftover(foodid)		Y	Y

4. Package Diagram

4.1 Package Diagram



4.1 Package diagram.

4.2 Package Functions

Table 4.2 Package functions

Package	Package Functions
View Layer (SDD_PKG_100)	● To display login account form, create account form, update account form, delete account form, addleftover information, update leftover information, delete leftover information, view leftover information, request leftover information, accept request leftover and view leftover request. ● Display Data and accept input data from users. ● Edit and validate input data
Domain Layer (SDD_PKG_200)	Process business rules.
Data Access Layer (SDD_PKG_300)	Establish and maintain connection of database for each table.

4.3 Package Content

Table 4.3 Package Content

Package	Class ID	Class Name
SDD_PKG_100	SDD_PKG_101	CreateAccount
	SDD_PKG_102	UpdateAccount
	SDD_PKG_103	ViewAccount
	SDD_PKG_104	AddLeftoverInformation
	SDD_PKG_105	UpdateLeftoverInformation
	SDD_PKG_106	DeleteLeftoverInformation
	SDD_PKG_107	ViewLeftoverInformation
	SDD_PKG_108	RequestLeftover
	SDD_PKG_109	AcceptRequestLeftover
	SDD_PKG_110	ViewLeftoverRequest
	SDD_PKG_111	FoodPickupStatusUpdate
SDD_PKG_200	SDD_PKG_201	CafeteriaOwner
	SDD_PKG_202	Student
	SDD_PKG_203	LoginController
	SDD_PKG_204	DashboardController
	SDD_PKG_205	AccountController
	SDD_PKG_206	LeftoverController
	SDD_PKG_207	RequestController
SDD_PKG_300	SDD_PKG_301	Cafe_Owner
	SDD_PKG_302	student

	SDD_PKG_303	Leftover
	SDD_PKG_304	Request
	SDD_PKG_305	pickup

5. Class Description

5.1 Class ID & NAME

(SDD_PKG_203 - LoginController)

The purpose of the LoginController is a component that handles user login functions. It is most likely made up of procedures and logic for validating user credentials, authenticating users, and managing the login process.

(SDD_PKG_206 - LeftoverController)

The purpose of the ItemController is a system component that handles operations on leftover. It most likely provides methods and logic for item management, such as adding new leftover, retrieving leftover information, modifying leftover details, or deleting leftover information.

(SDD_PKG_207 - RequestController)

The purpose of requestController is to by optimizing the request process of the leftover within the students and the cafeteria which can enhance efficiency, and streamline procurement processes.

5.1.2 Class Attributes & Methods

(SDD_PKG_203 - LoginController)

- Login(id, password) - To validate the provided id and password for user authentication.

(SDD_PKG_206 - LeftoverController)

- ViewLeftover (foodName) - To retrieve and display information about a specific leftover based on its name.
- UpdateItem(foodID) - To modify a leftover's information or properties based on its unique identifier.

(SDD_PKG_207 - RequestController)

- RequestItem(foodID) - To initiate a request for a specific leftover based on its id.

5.1.3 Method Description

5.1.3.1 LoginAccount()

Parameter: id, password
Pseudocode:
Start
Go to login page
Enter userID
Enter password
Click "Login"
IF credentials are correct
  Display "Login Successful"
  Redirect to dashboard
ELSE
  Display "Invalid Credentials"
End

5.1.3.2 CreateAccount()

Parameter: fullname, password.
Pseudocode:
Start
Go to the login page
Click 'Create Account'
IF “cafeteria owner”
  Click “Cafeteria Owner”
Else
  Click “Student”
Enter full name, and password
IF the username and password are valid THEN
  Display account details
ELSE
  Show an error message for invalid or undefined data
IF you want to cancel THEN
  Click 'Cancel'
End

5.1.3.3 UpdateAccount()

Parameter: fullname, password.
Pseudocode:
Start
Log in to the system
Go to 'View Account' page
View existing account details
Edit account information (full name, password)
IF data is valid THEN
  Update the account information
  Display updated account details
ELSE
  Show an error message for invalid or undefined data
IF you want to cancel THEN
  Click 'Cancel'
End

5.1.3.4 ViewAccount()

Parameter:id, password.
Pseudocode:
Start
Log in to the system
Go to 'View Account' page
Retrieve and display account information (full name, password)
End

5.1.3.5 AddLeftoverInformation()

Parameter: food name, quantity, description
Pseudocode:
Start
Go to log in page
Log in as the cafeteria owner
Click on 'Add Leftovers' option
Fill out the form with leftover details (food name, quantity, etc.)
IF the input is valid THEN
  Add the leftover information to the system
  Confirm that the leftover information was successfully added
ELSE
  Show an error message for invalid or undefined data
IF you want to cancel THEN
  Click 'Cancel'
End

5.1.3.6 UpdateLeftoverInformation()

Parameter: food name, quantity, description
Pseudocode:
Start
Log in as the cafeteria owner
Access the leftover details to be updated
Edit the leftover details
IF the updated data is valid THEN
  Save the updated information
  Confirm that the update was successful
ELSE
  Show an error message for invalid or undefined data
IF you want to cancel THEN
  Click 'Cancel'
End

5.1.3.7 Delete LeftoverInformation()

Parameter: foodID
Pseudocode:
Start
Log in as the cafeteria owner
Access the leftover details to be deleted
Click on 'Delete'
Confirm the deletion
IF you confirm THEN
  Delete the information
ELSE
  Click 'Cancel'

End

5.1.3.8 ViewLeftover Information ()

Parameter: food name, quantity, description, cafeID.
Pseudocode:
Start
Log in as the cafeteria owner or student
Go to the 'View Leftovers' section
Select an item to view details
Display the detailed information of the selected item
End

5.1.3.9 RequestLeftover ()

Parameter: studentID, foodID
Pseudocode:
Start
Log in as the student
Go to the 'View Leftovers' section
Select a surplus food item to request
Click 'Request'
IF the item is available THEN
  Display request confirmation
ELSE
  Show 'Item no longer available'
End

5.1.3.10 AcceptLeftoverRequest()

Parameter: foodID, studentID
Pseudocode:
Start
Start
Log in as the cafeteria owner
Go to the 'Requests' section
View student requests for leftover food
Click 'Accept' on a request
Log the accepted request
Notify the student
Display acceptance confirmation
End

5.1.3.11 ViewRequestLeftover()

Parameter: studentID, foodID, foodDescription, foodStatus
Pseudocode:
Start
IF "Student"
    Log in as the student
    Go to the 'Requests' section
    View the list of surplus food requests you have made
    Select a specific request to view its status
    Display request details and status (Pending, Accepted, or Declined)
Else

Log in as the cafeteria owner
Go to the 'Requests' section
View the list of student requests for surplus food
Select a specific request to view its details
Display the request information (student name, item, quantity, etc.)
Optionally, filter requests by status (Pending, Accepted, Declined)
End

5.1.3.12 PickupStatusUpdate()

Parameter: pickupID, timepickup, imagepath, foodid, studentNumber
Pseudocode:
Start
IF "Student"
Log in as the student
Go to the 'Pickup' section
Fill in the form needed
Clicks submit
End

5.1.3.13 CreateNewOrder()

Parameter: bookingID, bookingQuantity, bookingMenu, bookingDate, studentNumber, cafeNumber
Pseudocode:
Start
IF "Student"
Log in as the student
Go to the 'Create New Order' section
Fill in the form of booking
Click Submit
End

6. Human Interface Design (Screens)

6.1 Overview of the User Interface

The user interfaces of the system are shown based on the use case diagram of FoodShare, which are designed according to its use case diagram. The user interface is crucial as it acts as a bridge connecting the user and the system. It's like a language that helps them understand each other. The design of the system's interface follows the storyboard created earlier, ensuring consistency and clarity throughout the user experience. Moving forward, the next phases involve testing the interfaces for usability and making any necessary adjustments based on feedback.

6.1.2 Screen Images

6.1.2.1 Use case: Create Account Cafeteria owner

Figure 6.1 shows the registration interface designed for cafeteria owners. Each interface includes fields for entering essential information such as a unique

identifier such as Cafe Number for cafeteria owners, Name, Email, and Password.

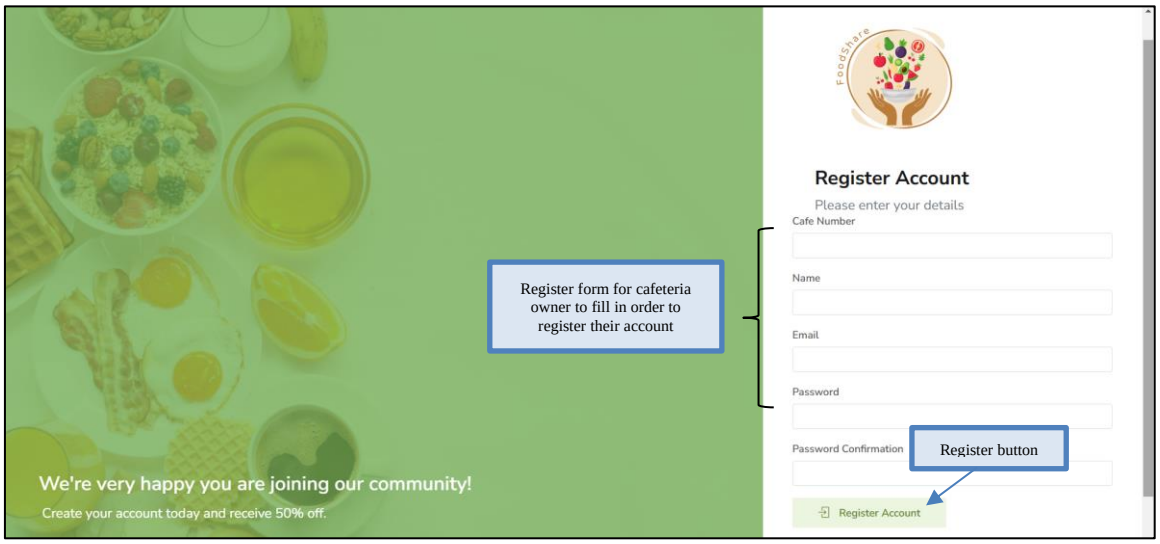


Figure 6.1 Register interface for cafeteria owner

Figure 6.1 shows the registration interface designed for cafeteria owners. Each interface includes fields for entering essential information such as a unique identifier such as Cafe Number for cafeteria owners, Name, Email, and Password.

6.1.2.2 Use case: Manage Account Cafeteria owner

Figure 6.2 hows the interface where the cafeteria owner can view and update their account information.

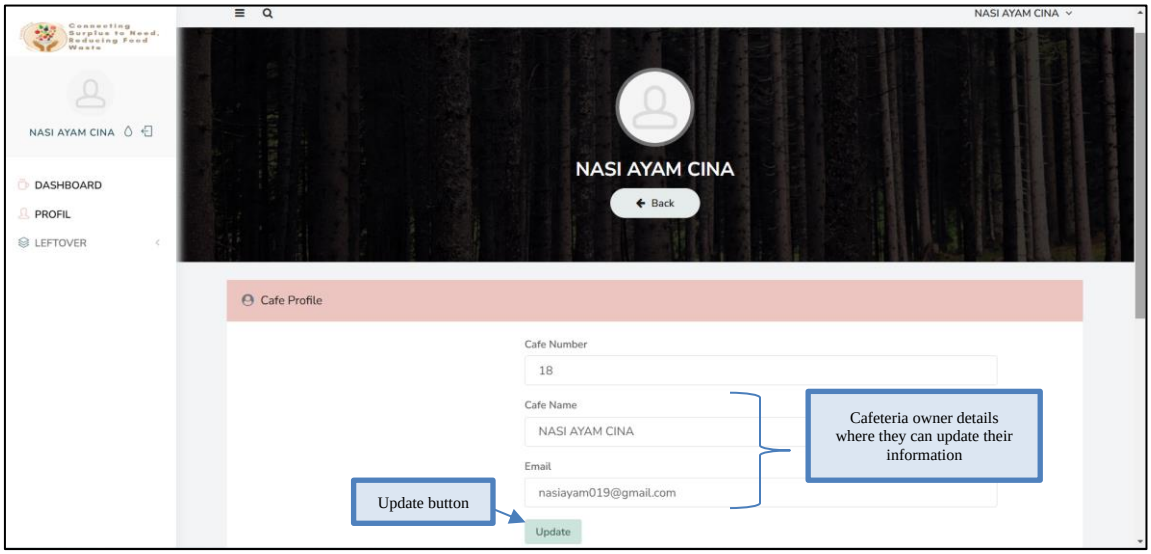


Figure 6.2 Manage Account for cafeteria owner

6.1.2.3 Use case: Create Account Student

Figure 6.3 shows interface shows the registration age for students that has not been registered in Foodshare system.



Figure 6.3 Register interface for student

6.1.2.4 Use case: Manage Account Student

This interface in Figure 6.4 is where the student can view and update their information.

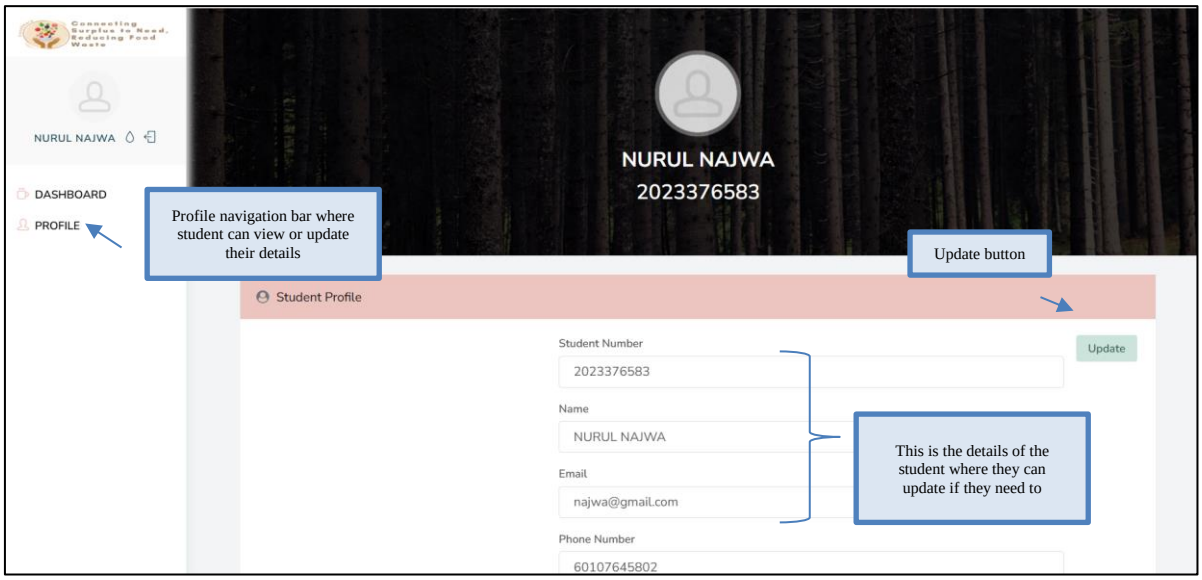


Figure 6.4 Update and view interface for student

6.1.2.5 Use case: Add Leftover information

This is the interface where the cafeteria owner will add the leftover information. Once the cafeteria owner clicks the add button, the student's registered phone number will receive a notification regarding the food availabilities.

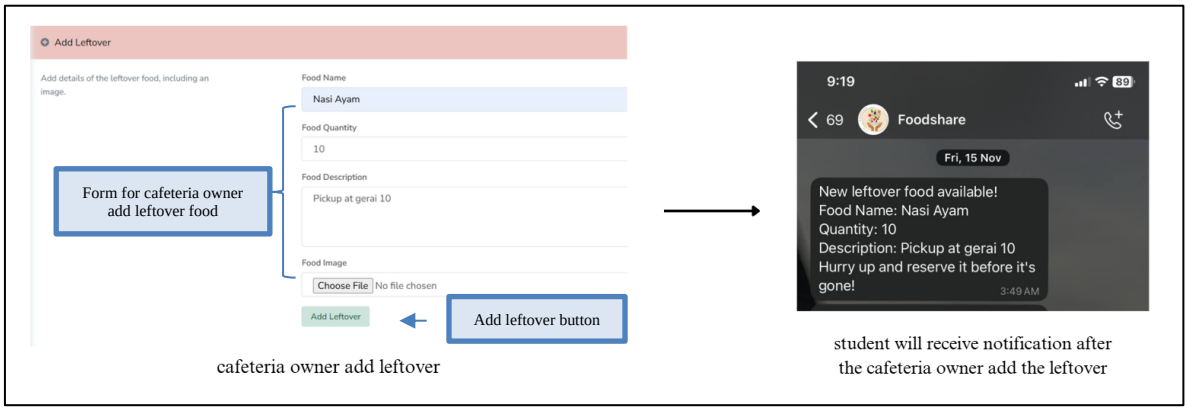


Figure 6.5 Add leftover

6.1.2.6 Use case: Manage leftover information

This use case where the cafeteria owner can update or delete the leftover information that has been added into the system.

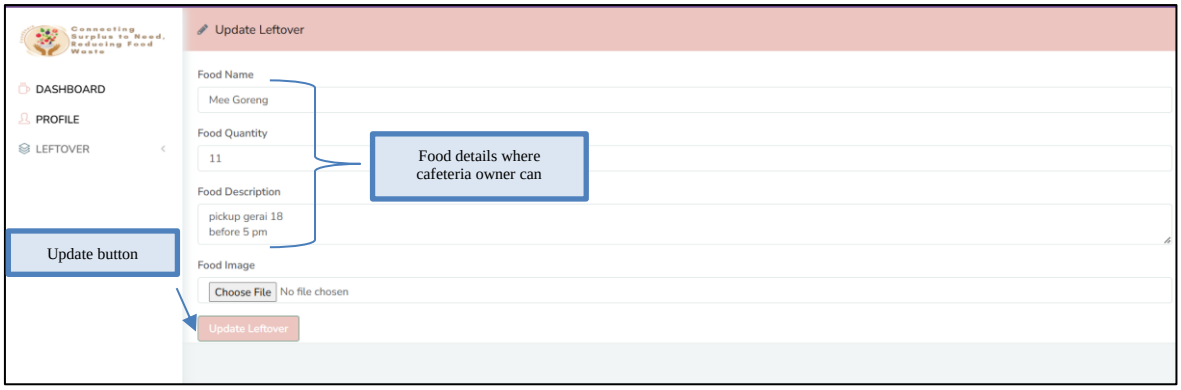


Figure 6.6 Update leftover interface for cafeteria owner

By clicking the update button, cafeteria can update the leftover information when needed.

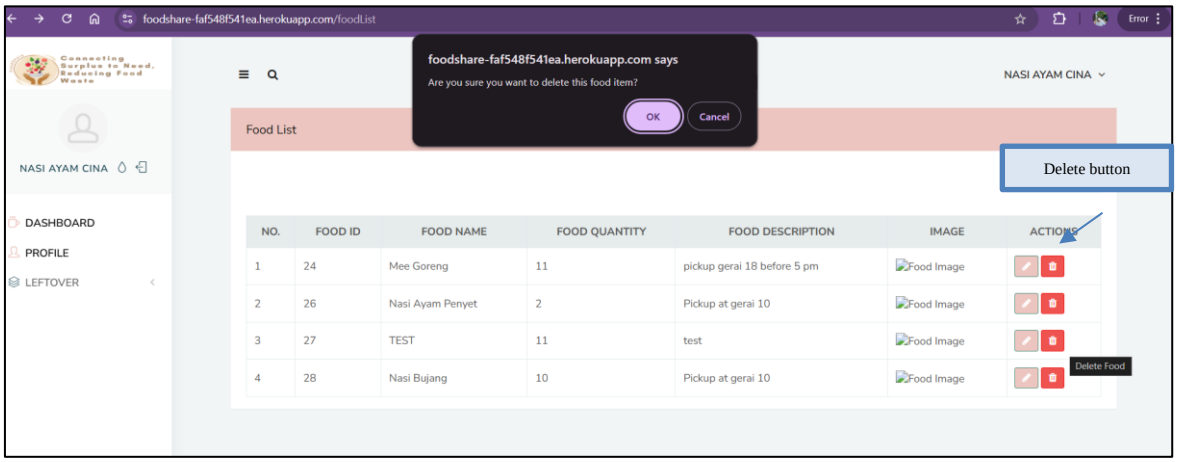


Figure 6.7 Delete leftover interface for cafeteria owner

By clicking the delete button, a prompt of confirmation will appear to confirm the deletion of the leftover food.

6.1.2.7 Use case: View leftover information

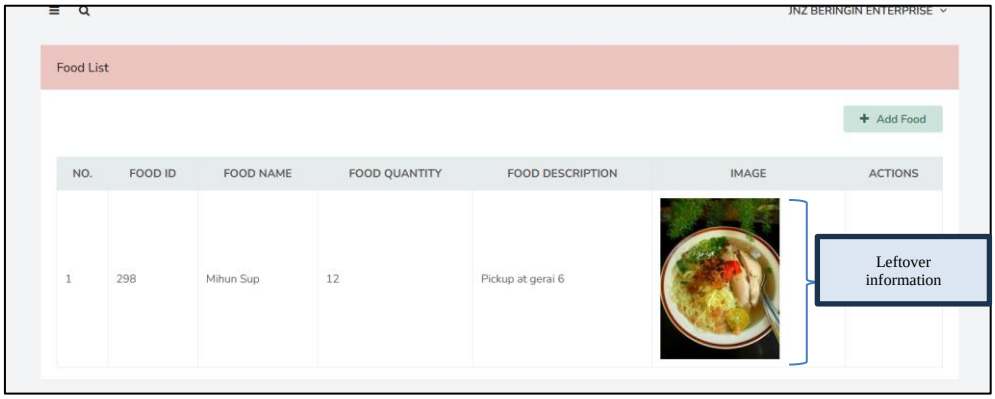


Figure 6.8 View leftover interface for cafeteria owner

This interface will allow the user to view the leftover information that has been added by the cafeteria owner where they can update and delete.

6.1.2.8 Use case: Request Leftover, Accept leftover Request, View leftover request

Figure 6.9 shows the interface where students can view available leftover food, including details such as the food name, quantity, and description.

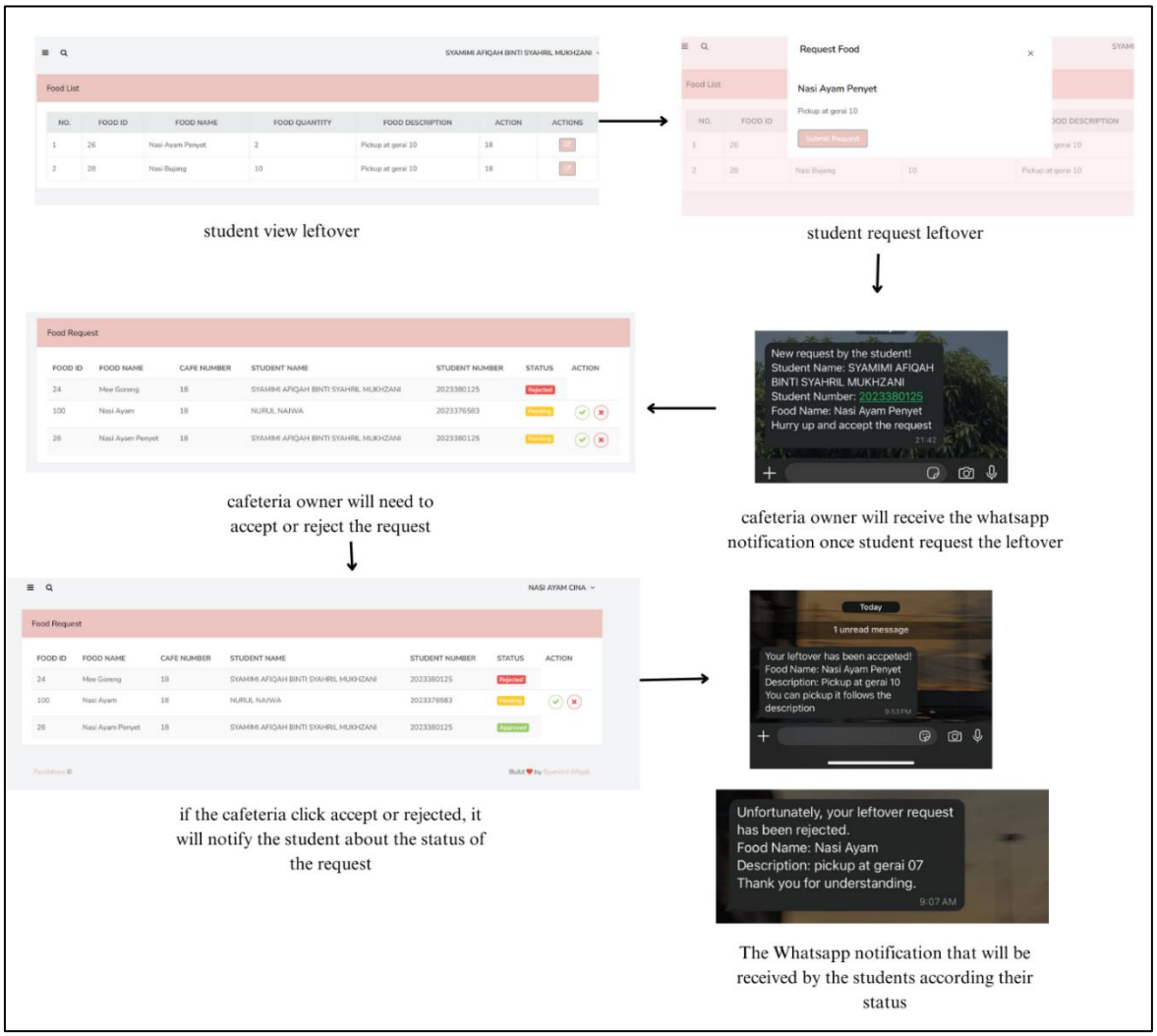


Figure 6.9 View leftover interface for cafeteria owner

When a student decides to request a specific leftover, they click the "Request" button, which prompts them to confirm their selection. Once confirmed, the request is submitted, and the cafeteria owner receives a notification containing the student's name, ID, and details of the requested item. The cafeteria owner then reviews the request through the "Food Request" interface, where they can choose to accept or reject it. If the request is accepted, the student receives a

notification confirming the approval, along with pickup instructions. If the request is rejected, the student is notified with a message explaining the denial. This structured process ensures smooth communication and clarity for both parties.

6.1.2.9 Use case: Food pickup status update

In this section, students are required to provide evidence to confirm that they have received the leftover food. The pickup page will only appear if the request has been accepted by the cafeteria owner. If a student has not made a request or if their request has not been accepted yet, an alert will be displayed indicating that they are not eligible for pickup. Figure 6.10 illustrates the interface of the alert box that pops up.

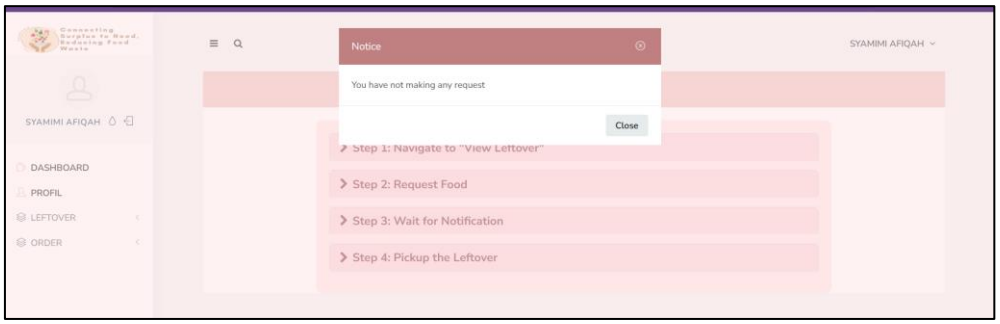


Figure 6.10 Alert box

Figure 6.11 shows the form that students need to fill out to confirm that they have successfully received the food.

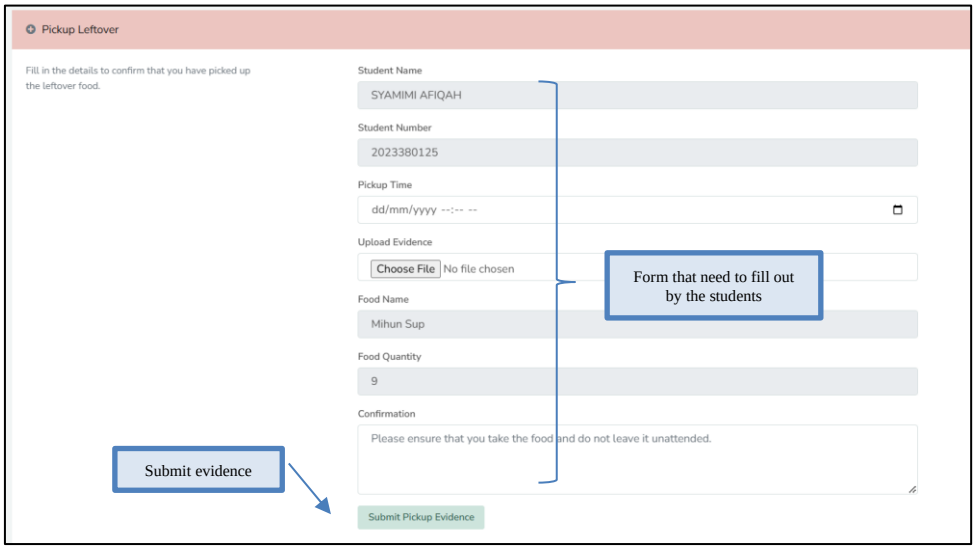


Figure 6.11 Pickup leftover interface

7. Traceability Requirements Matrix

Table 7.1 Package Content

Package	Class ID	Class Name
SDD_PKG_100	SDD_PKG_101	CreateAccount
	SDD_PKG_102	UpdateAccount
	SDD_PKG_103	ViewAccount
	SDD_PKG_104	AddLeftoverInformation
	SDD_PKG_105	UpdateLeftoverInformation
	SDD_PKG_106	DeleteLeftoverInformation
	SDD_PKG_107	ViewLeftoverInformation
	SDD_PKG_108	RequestLeftover
	SDD_PKG_109	AcceptRequestLeftover
	SDD_PKG_110	ViewLeftoverRequest
	SDD_PKG_111	FoodPickupStatusUpdate
SDD_PKG_200	SDD_PKG_201	CafeteriaOwner
	SDD_PKG_202	Student
	SDD_PKG_203	LoginController
	SDD_PKG_204	DashboardController
	SDD_PKG_205	AccountController
	SDD_PKG_206	LeftoverController
	SDD_PKG_207	RequestController
SDD_PKG_300	SDD_PKG_301	Cafe_Owner
	SDD_PKG_302	student
	SDD_PKG_303	Leftover
	SDD_PKG_304	Request
	SDD_PKG_305	pickup

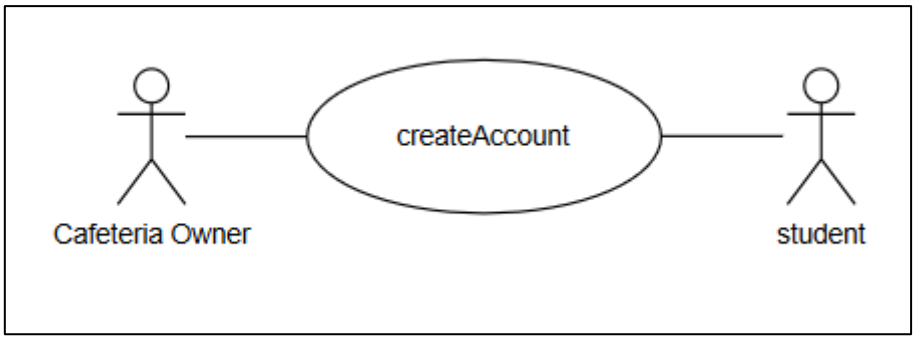
Table 7.2 Use Case Content

Package	Class ID	Class Name
UCD_000	UCD_001	CreateAccount
	UCD_002	UpdateAccount
	UCD_003	ViewAccount
	UCD_004	NotifyCafeteriaOwner
	UCD_005	NotifyStudent
	UCD_006	AddLeftoverInformation
	UCD_007	UpdateLeftoverInformation
	UCD_008	DeleteLeftoverInformation
	UCD_009	ViewLeftoverInformation
	UCD_010	RequestLeftover
	UCD_011	AcceptLeftover
	UCD_012	ViewLeftoverRequest
	UCD_013	FoodPickupStatusUpdate

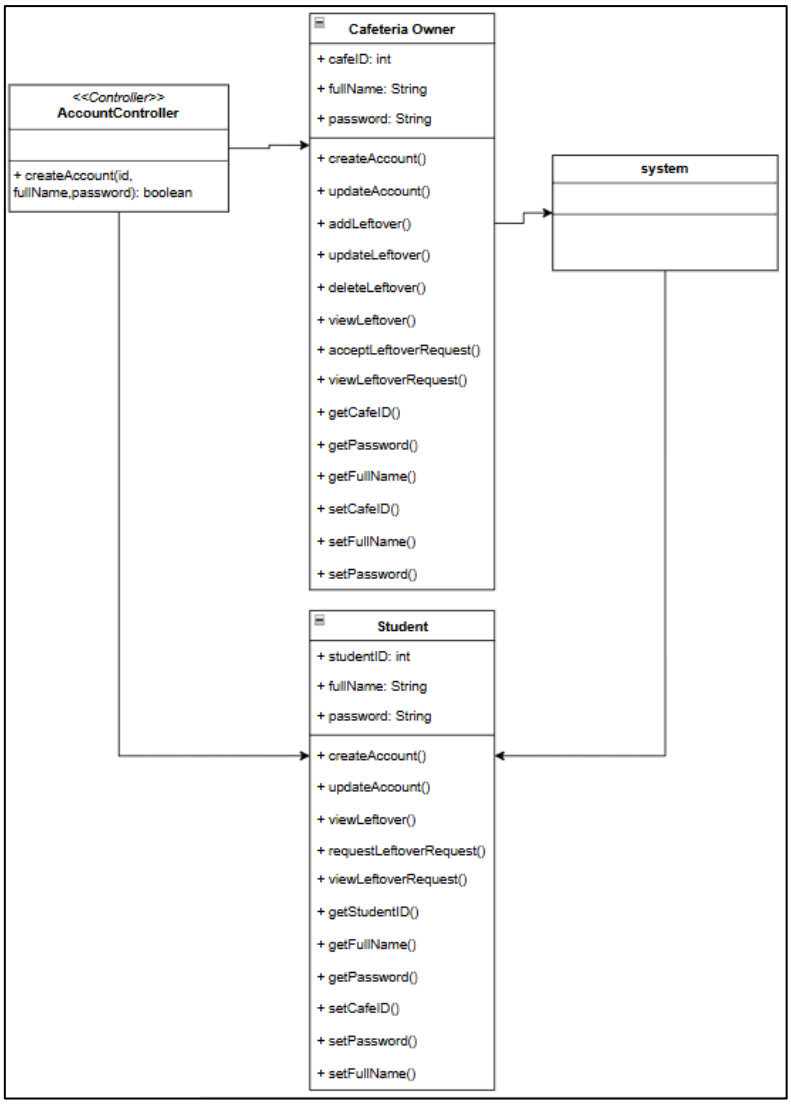
Requirement	Package (SDD)	SDD_PKG_100											SDD_PKG_200							SDD_PKG_100				
	Class SRS / Class SDD	SDD_PKG_101	SDD_PKG_102	SDD_PKG_103	SDD_PKG_104	SDD_PKG_105	SDD_PKG_106	SDD_PKG_107	SDD_PKG_108	SDD_PKG_109	SDD_PKG_110	SDD_PKG_111	SDD_PKG_201	SDD_PKG_202	SDD_PKG_203	SDD_PKG_204	SDD_PKG_205	SDD_PKG_206	SDD_PKG_207	SDD_PKG_301	SDD_PKG_302	SDD_PKG_303	SDD_PKG_304	SDD_PKG_305
UCD_000	UCD_001	√											√	√	√		√			√	√			
	UCD_002		√										√	√	√		√			√	√			
	UCD_003			√									√	√	√		√			√	√			
	UCD_004									√		√												
	UCD_005				√						√													
	UCD_006				√													√				√		
	UCD_007					√												√				√		
	UCD_008						√											√				√		
	UCD_009							√										√				√		
	UCD_010								√										√				√	
	UCD_011									√									√				√	
	UCD_012										√								√				√	
	UCD_013											√							√					√

8. Appendices

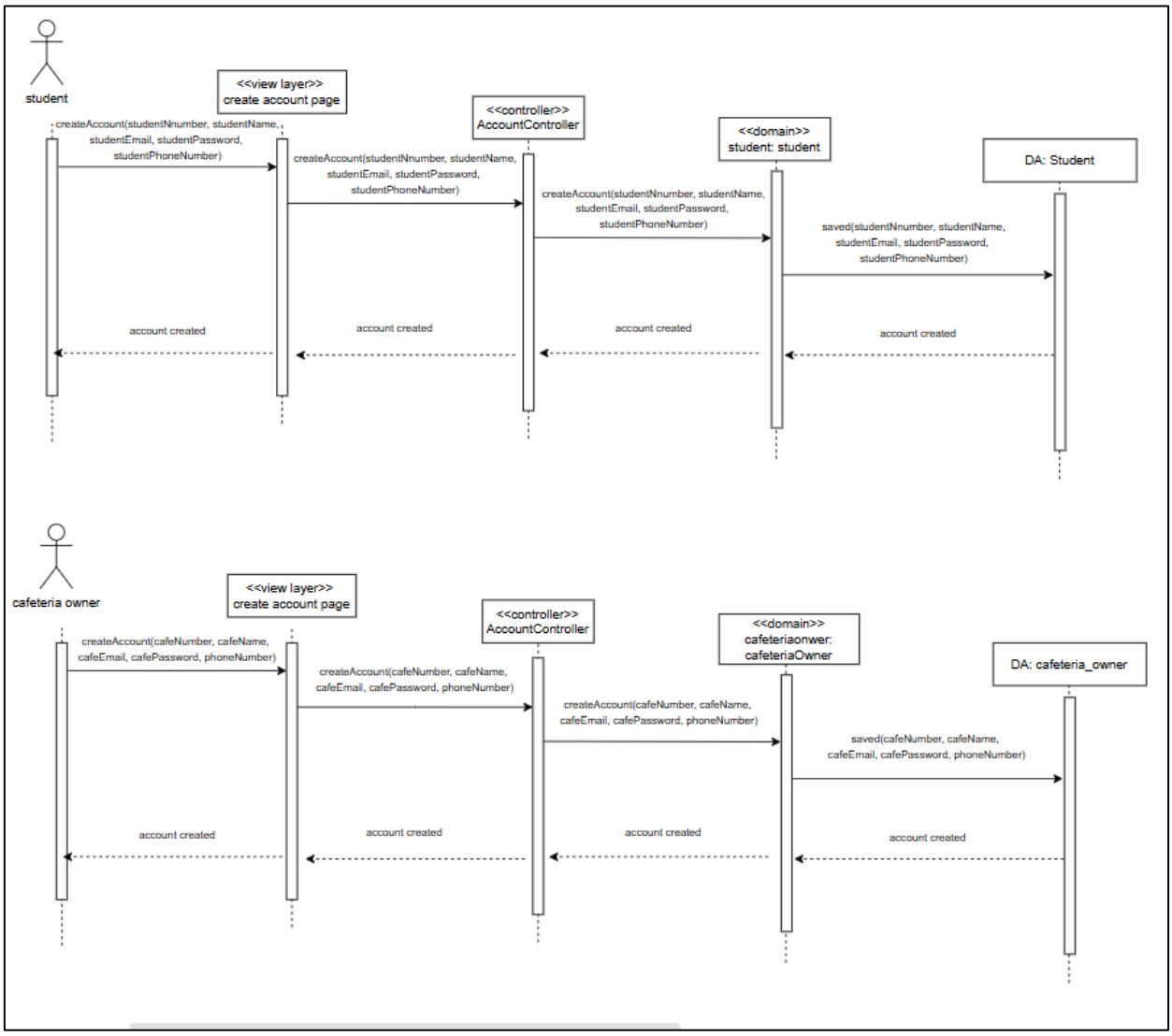
8.1 Create Account



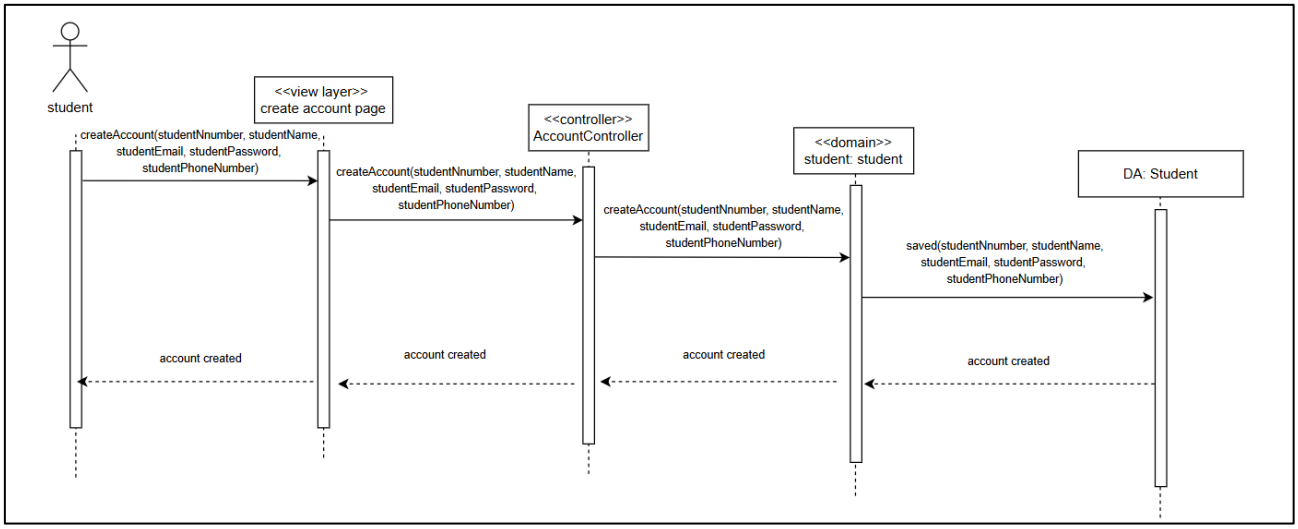
8.1.1 Detail Class Diagram



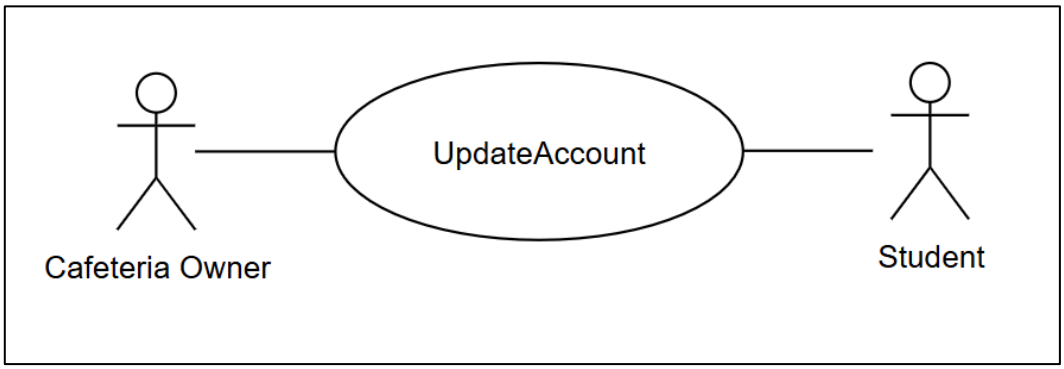
8.1.2 Multilayer Sequence Diagram



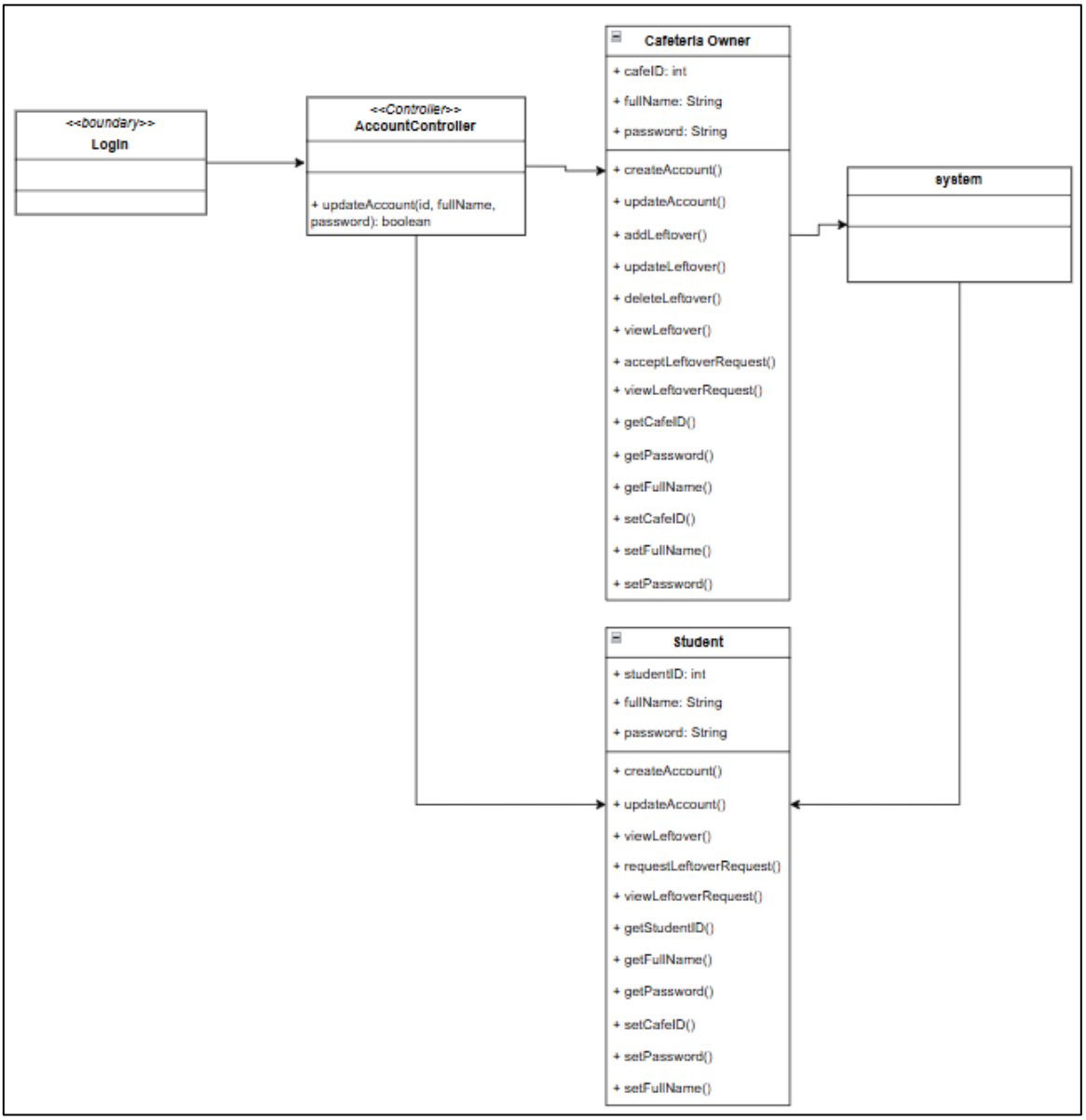
8.1.3 Multilayer Sequence Diagram



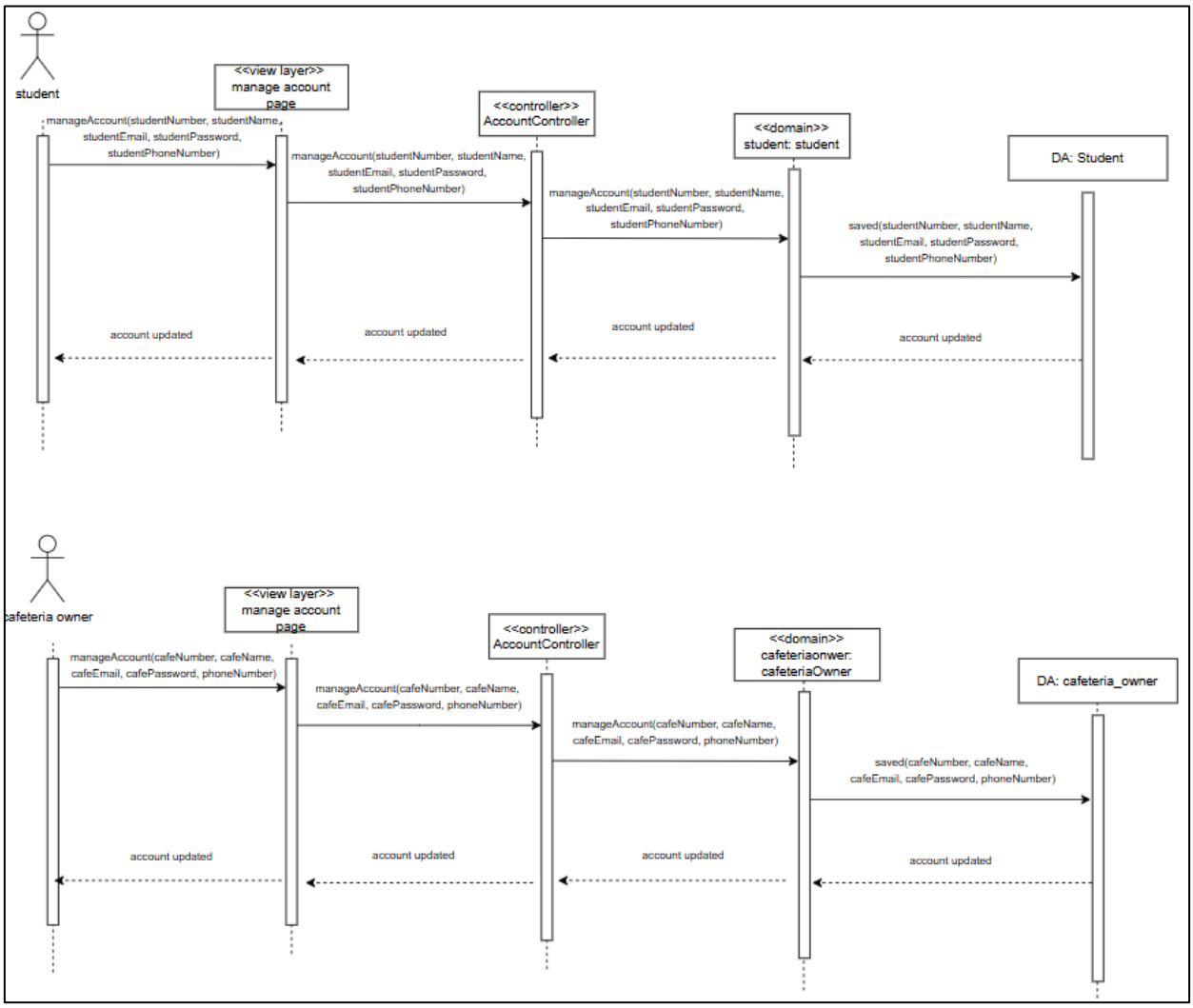
8.2 Update Account



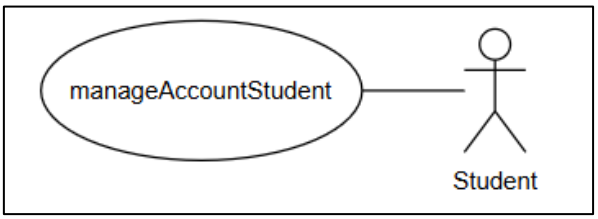
8.2.1 Detail Class Diagram



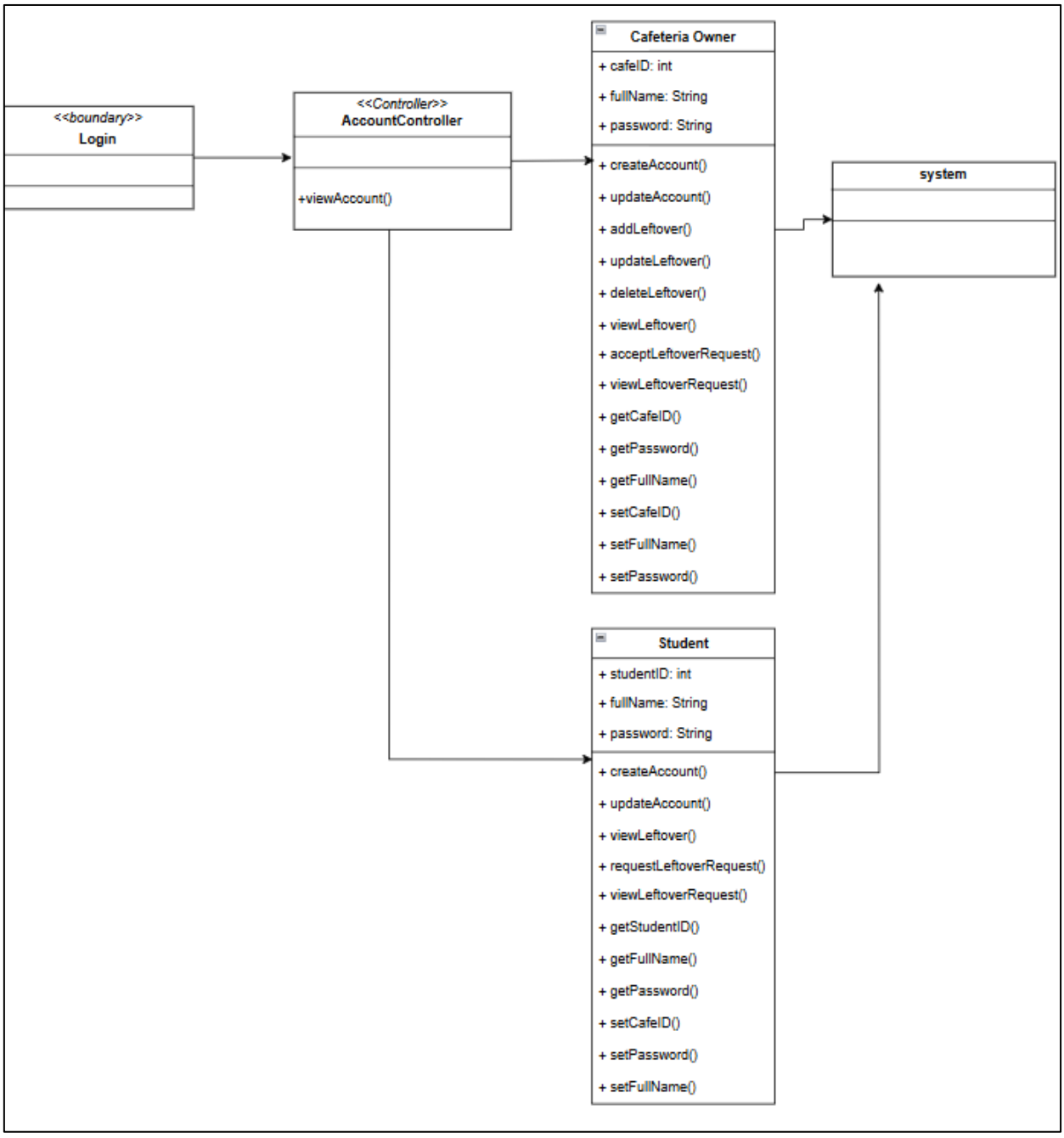
8.2.2 Multilayer Sequence Diagram



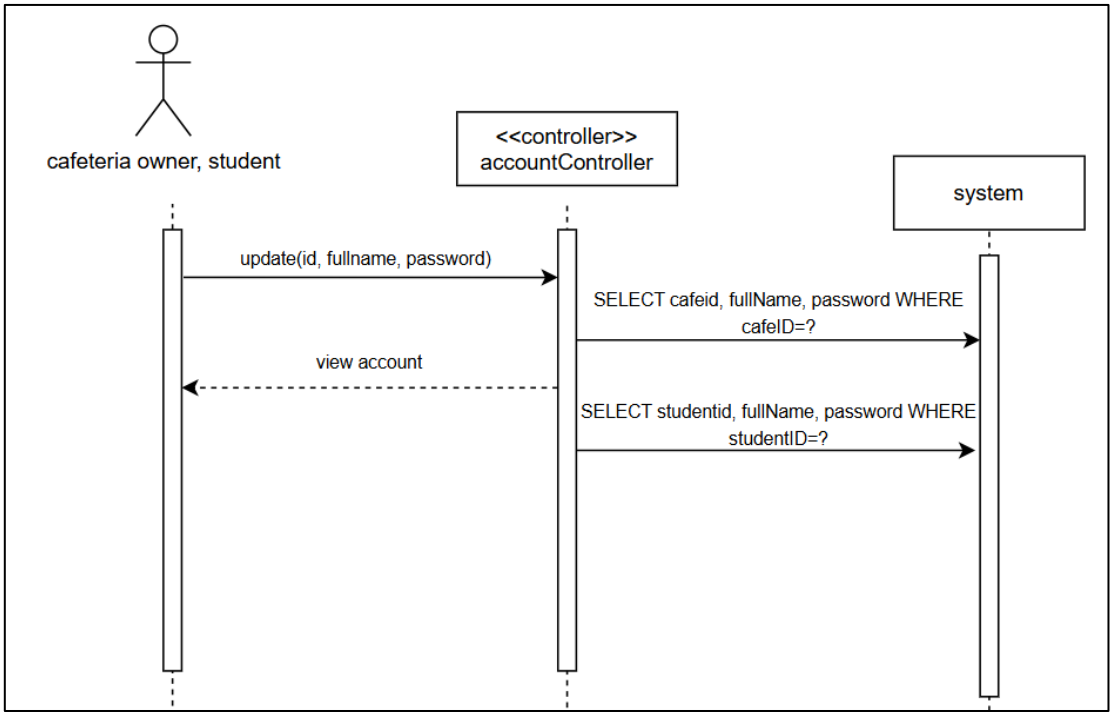
8.3 View Account



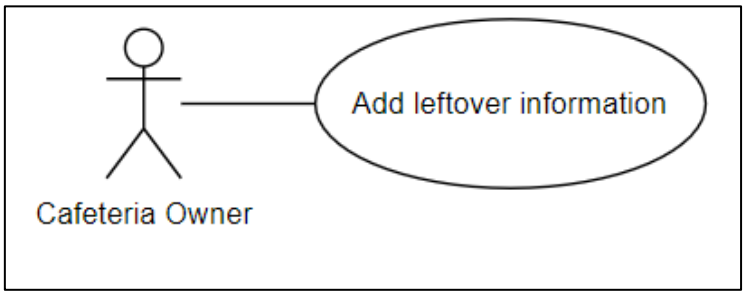
8.3.1 Detail Class Diagram



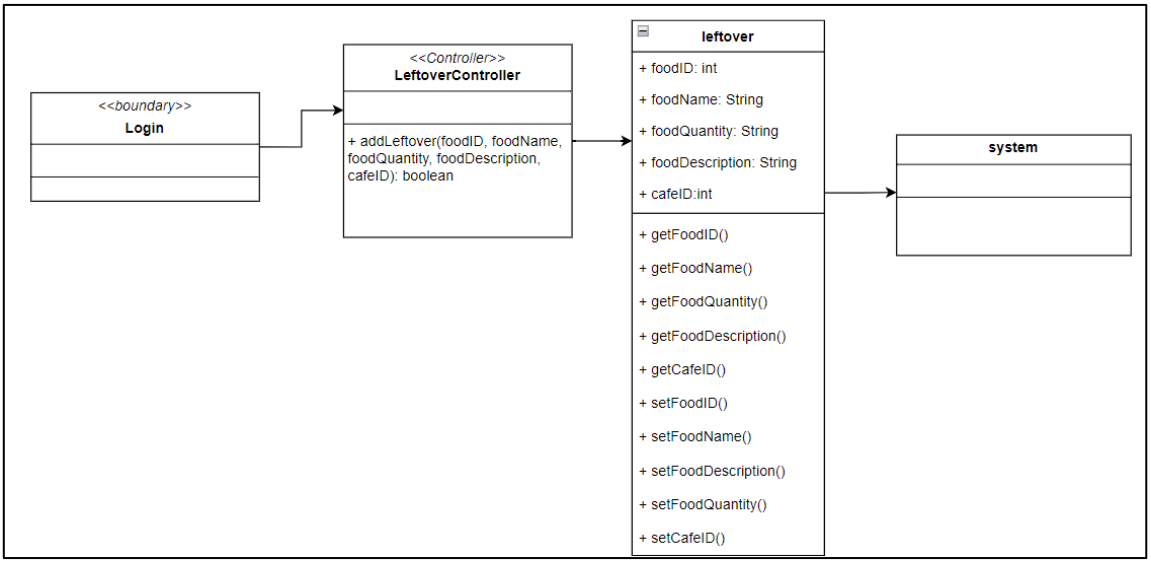
8.3.2 Multilayer Sequence Diagram



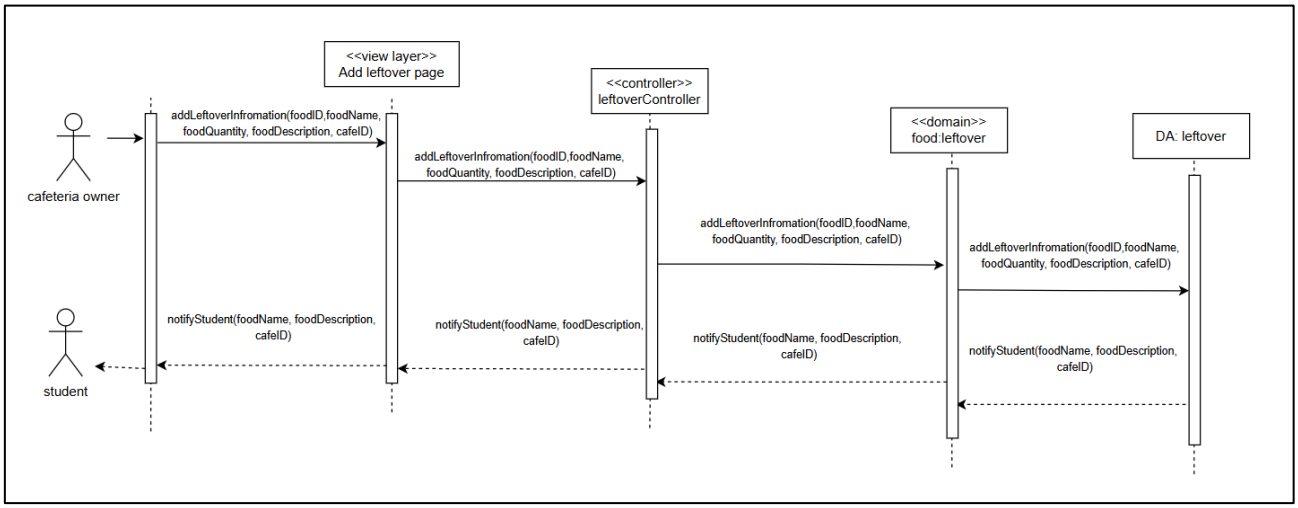
8.4 Add Leftover Information



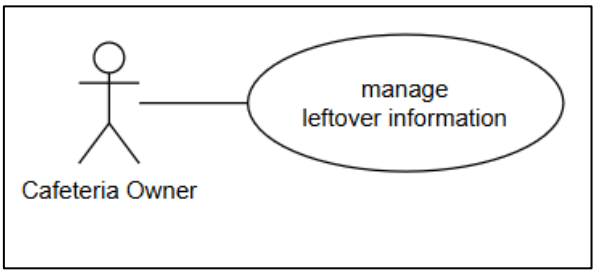
8.4.1 Detail Class Diagram



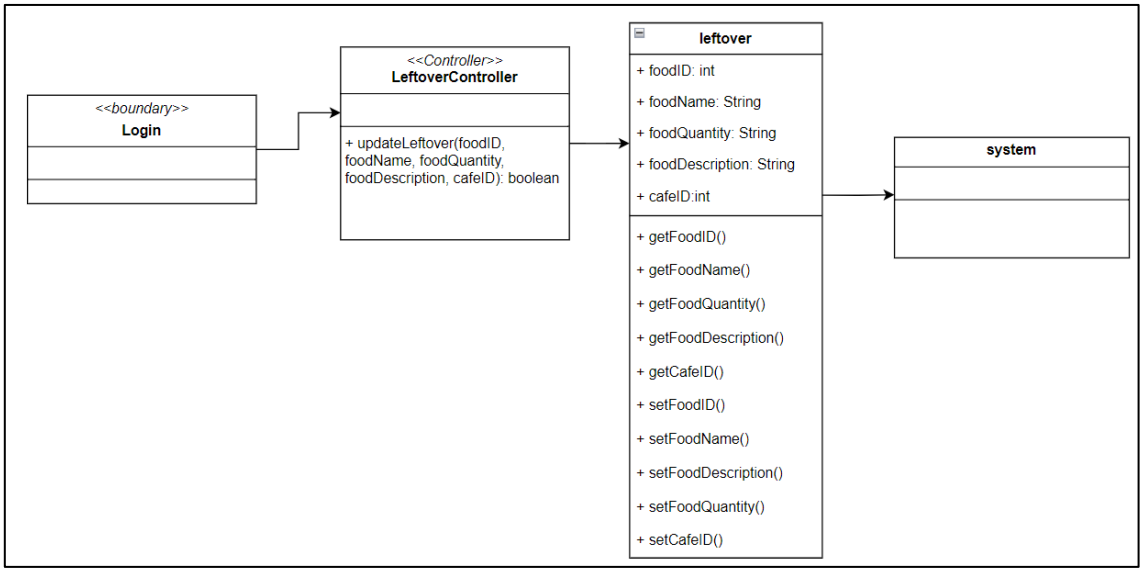
8.4.2 Multilayer Sequence Diagram



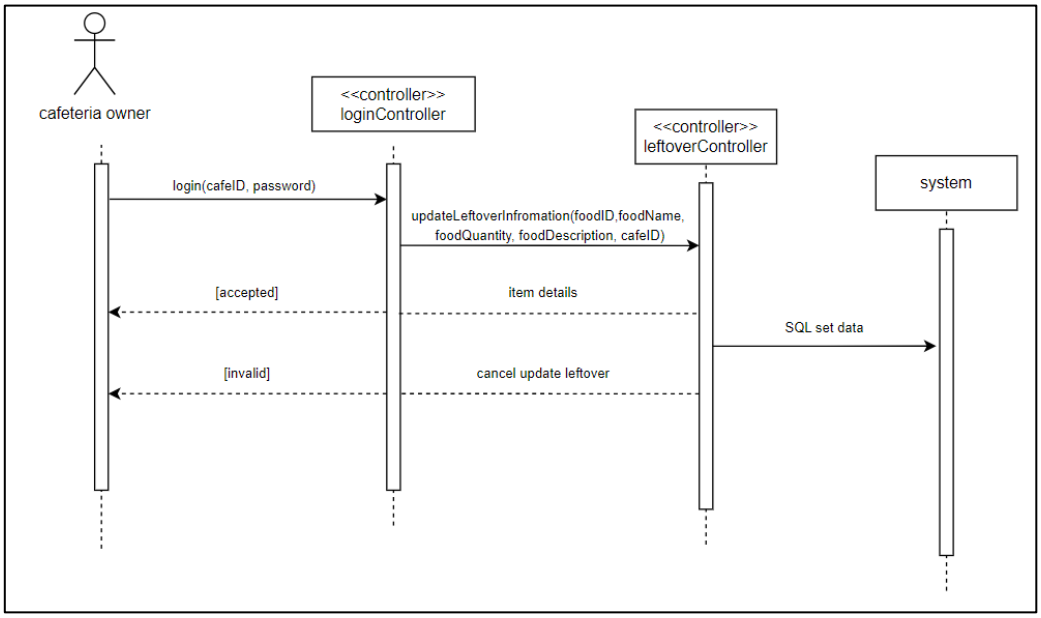
8.5 Manage Leftover Information



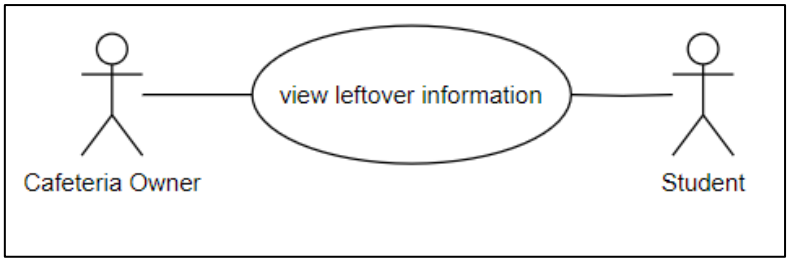
8.5.1 Detail Class Diagram



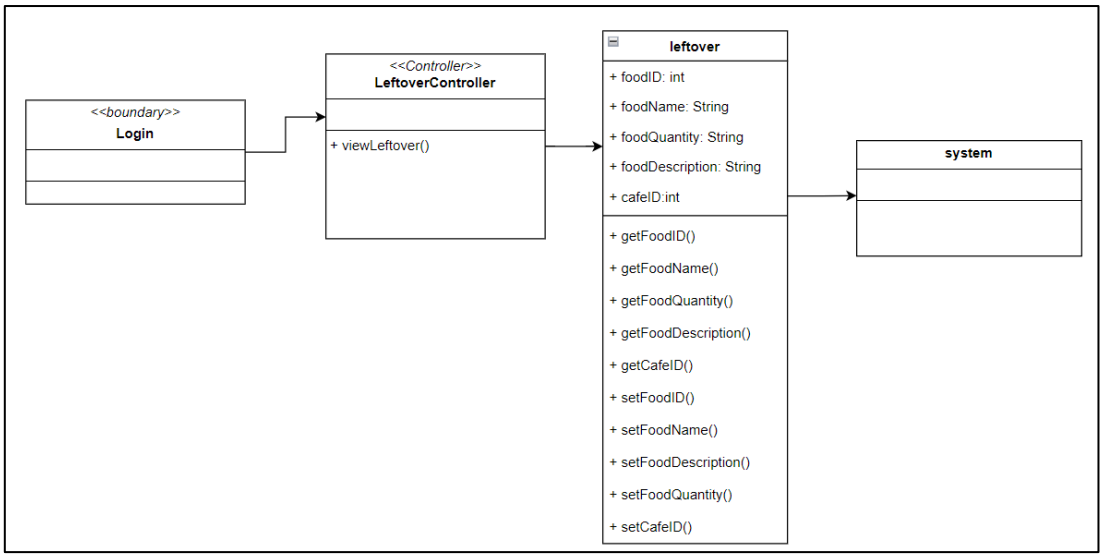
8.5.2 Multilayer Sequence Diagram



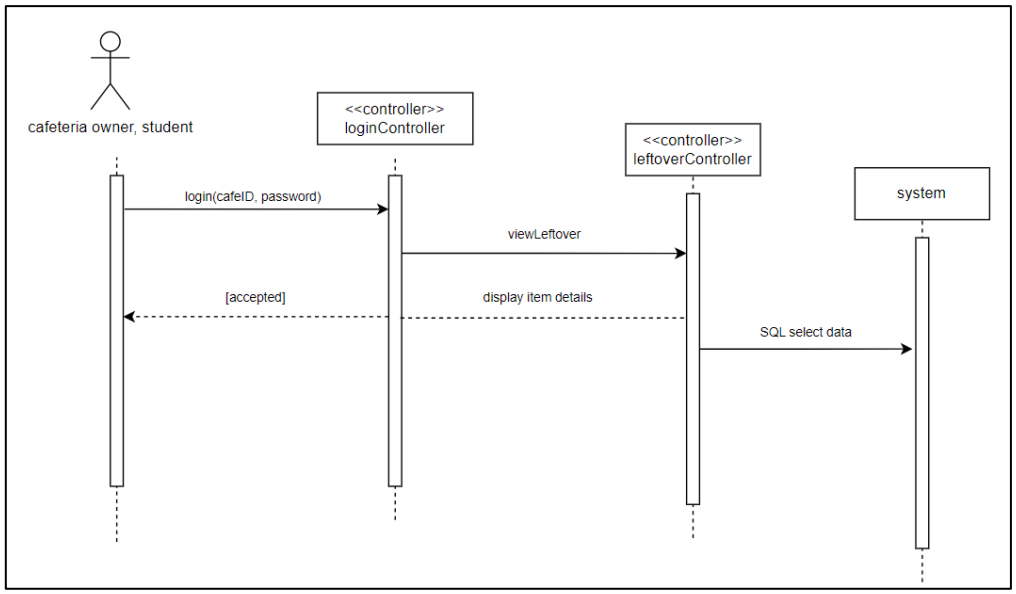
8.6 View Leftover Information



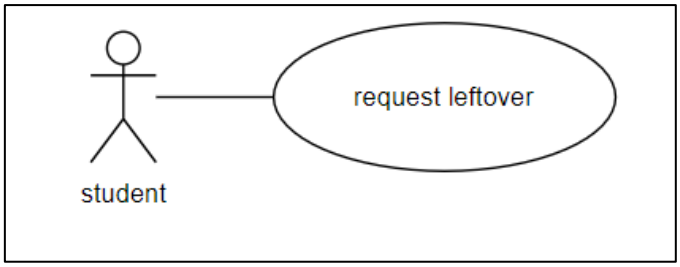
8.6.1 Detail Class Diagram



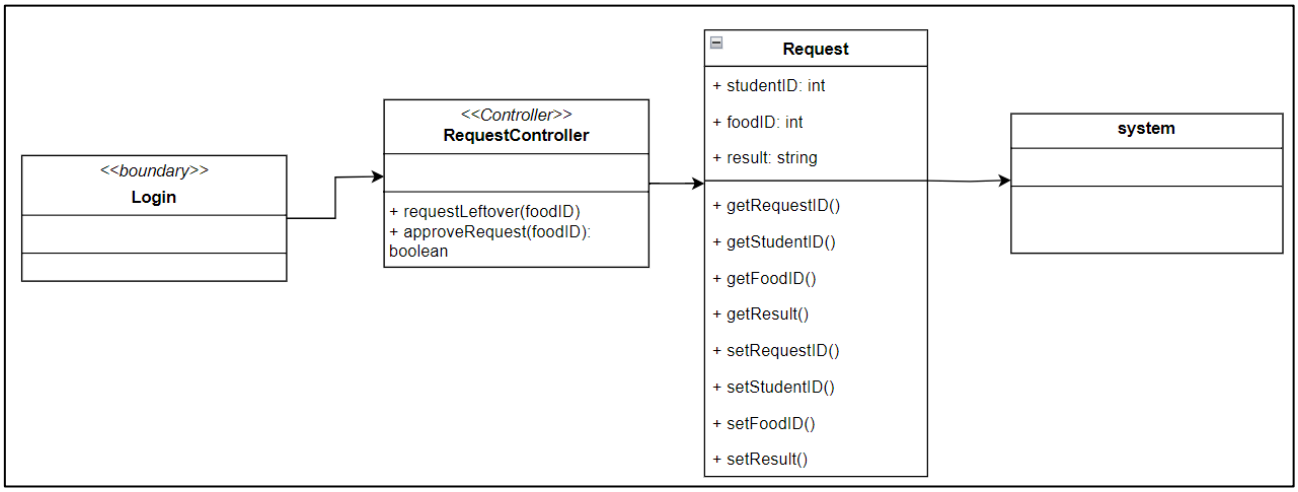
8.6.2 Multilayer Sequence Diagram



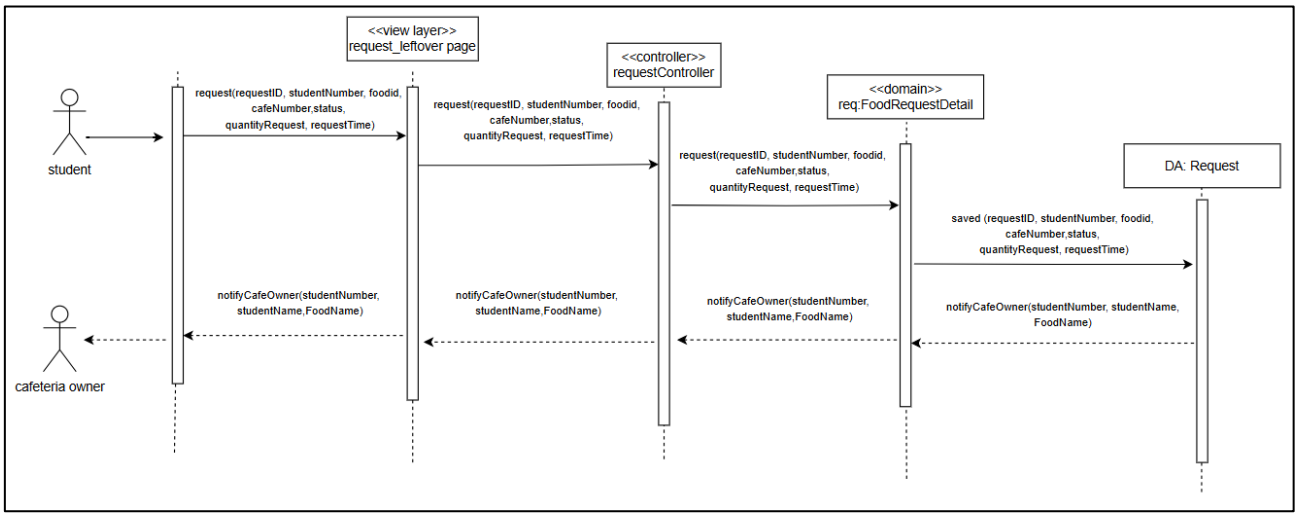
8.7 Request Leftover



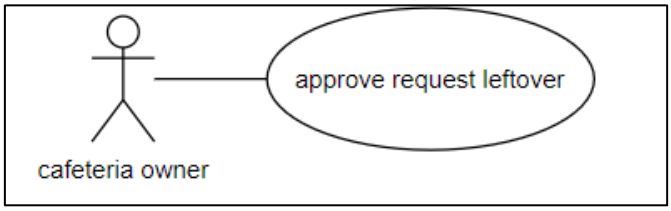
8.7.1 Detail Class Diagram



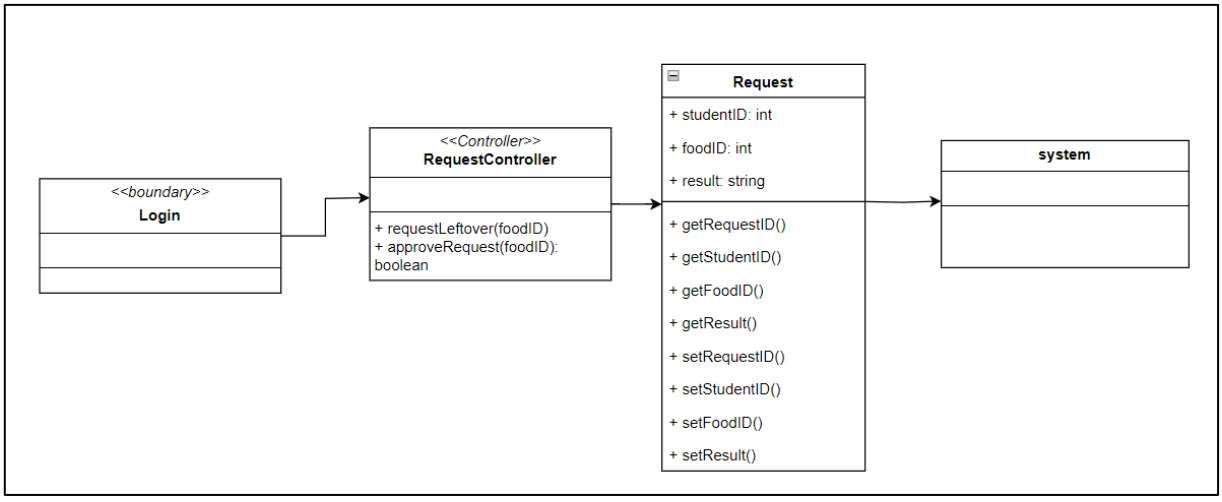
8.7.2 Multilayer Sequence Diagram



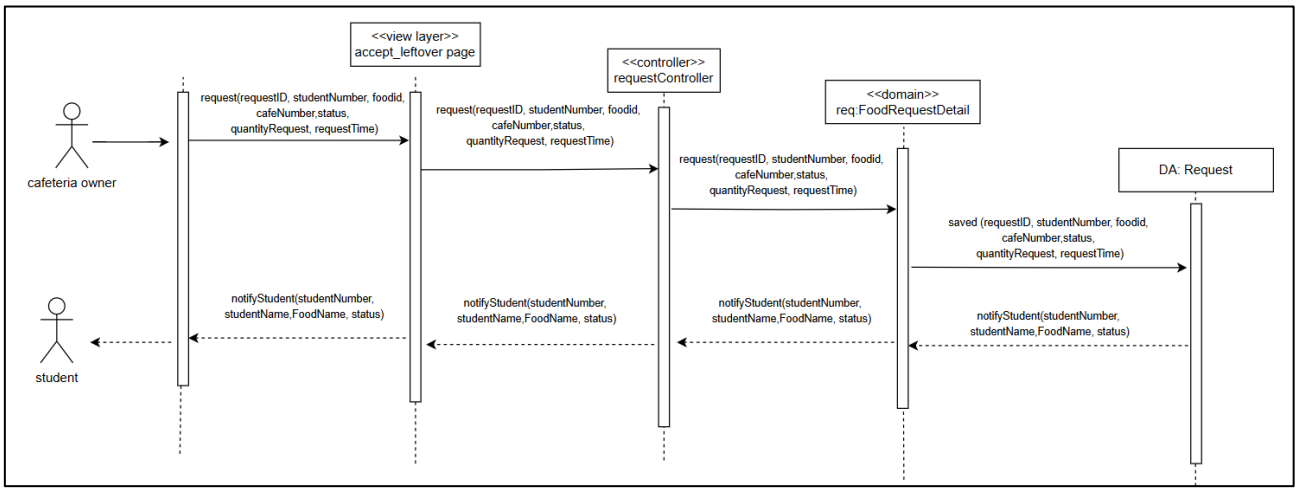
8.8 Accept Request Leftover



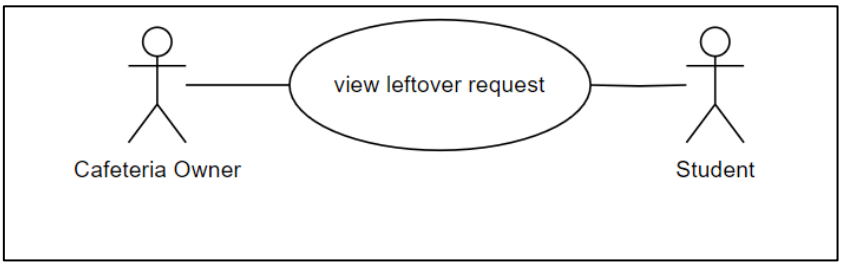
8.8.1 Detail Class Diagram



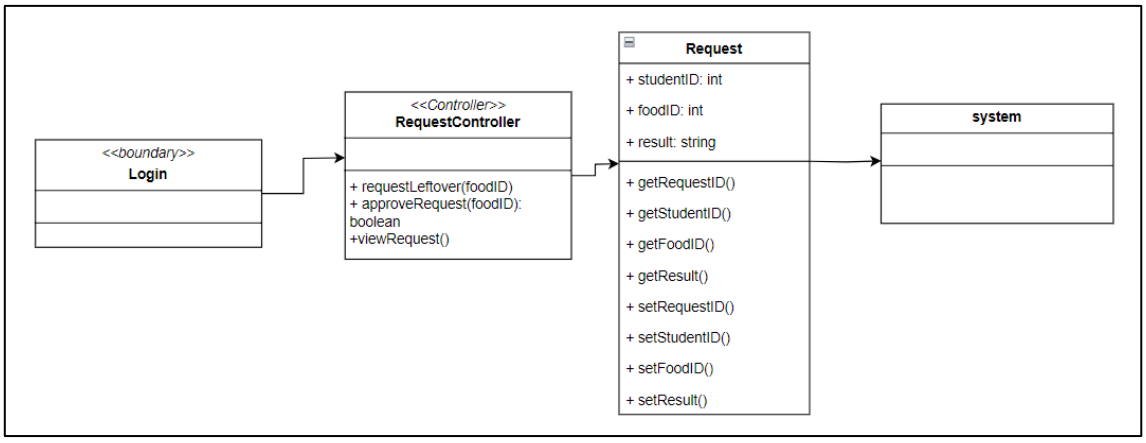
8.8.2 Multilayer Sequence Diagram



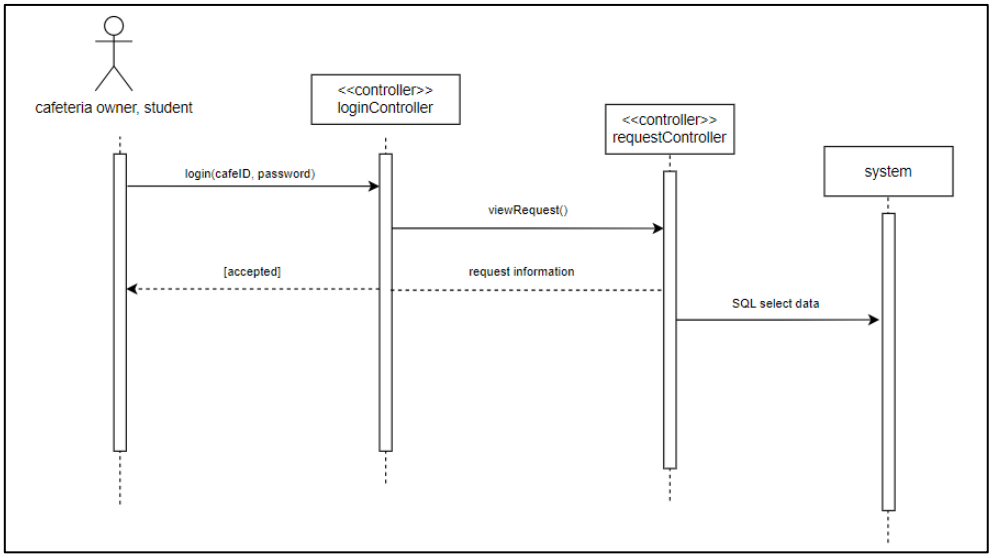
8.9 View Request Leftover



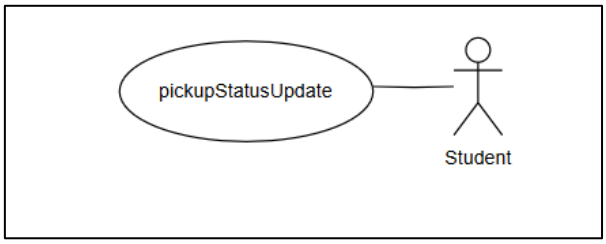
8.9.1 Detail Class Diagram



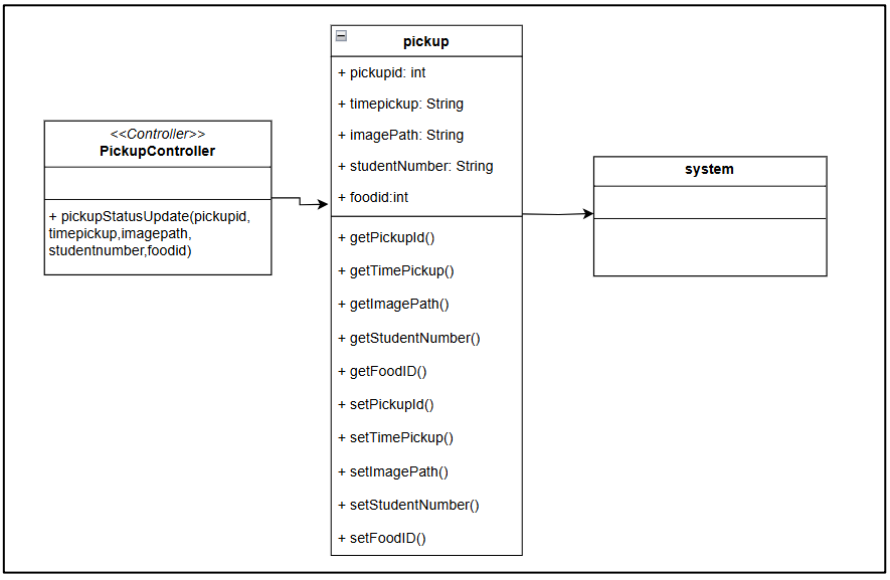
8.9.2 Multilayer Sequence Diagram



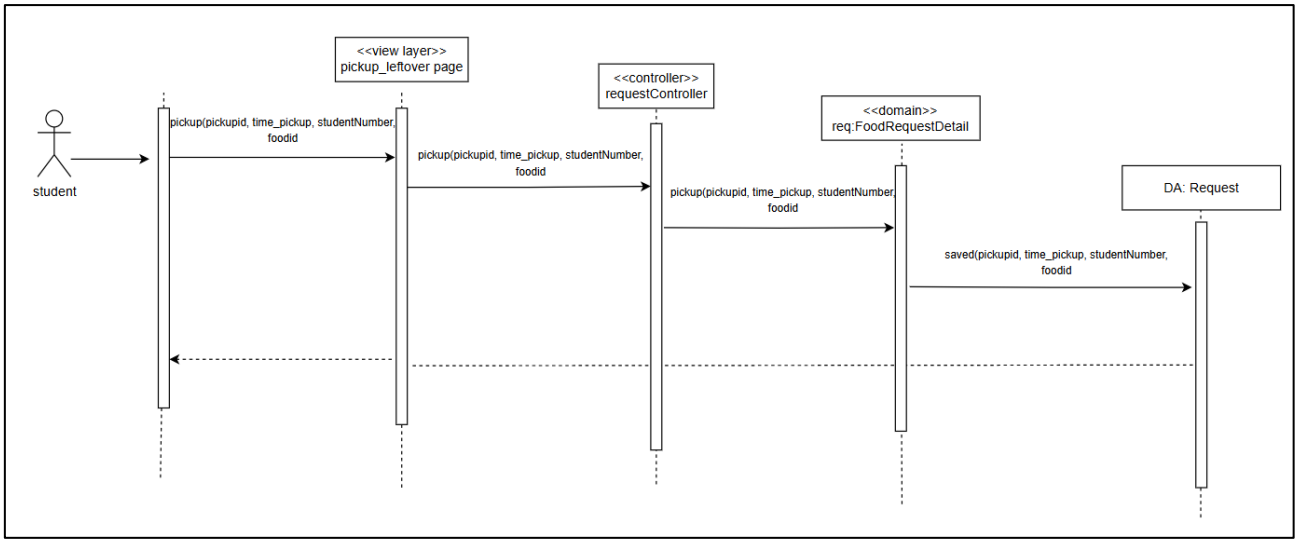
8.10 Pickup Status Update



8.10.1.1 Detail Class Diagram



8.10.2Multilayer Sequence Diagram


<<view layer>>
pickup_leftover page
<<controller>>
requestController
<<domain>>
req:FoodRequestDetail
DA: Request
student
pickup(pickupid, time_pickup, studentNumber, foodid
pickup(pickupid, time_pickup, studentNumber, foodid
pickup(pickupid, time_pickup, studentNumber, foodid
saved(pickupid, time_pickup, studentNumber, foodid